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Thesis for the Degree of Ph.D.

Development of Question-Answering System to Improve Construction Risk Identification in NATM Tunnels

School of Architectural, Civil, Environmental and Energy Engineering,
Major in Civil Engineering
The Graduate School

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December 2023

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ABSTRACT

Tunnel construction projects are prone to many risks due to uncertainties that usually emanate from project complexity and uniqueness, the involvement of multiple stakeholders, huge capital investment, dynamic environments, long completion periods, and exposure to external environments and weather conditions. Thus, identifying and responding proactively to these risks can help mitigate the potential negative impacts on project objectives.

Risk identification is the first and most crucial step in the risk management of construction projects. However, the current risk management practice in construction projects is faced with two problems: (1) lack of formalized representation format for capturing and storing risk data (knowledge) generated from previous projects, hindering knowledge integration, reuse, and sharing among practitioners and computers, (2) labor-intensive and time-consuming procedures which rely heavily on human experience, knowledge, and effort, that may be inadequate for managing complex construction projects such as tunnels, thus negatively influencing project success, performance, and productivity in the construction industry.

Consequently, this study adopted two emerging artificial intelligence (AI) techniques, knowledge graph and GPT-3, to address these problems in order to improve the efficiency and accuracy of the risk identification process in construction projects. The knowledge graph concept was adopted to develop a formal risk knowledge representation for NATM

tunnels with the aid of the Neo4j desktop application. Also, a large language model (LLM), i.e., GPT-3, was adapted to develop an intelligent question-answering system for construction risk (*QAR-System*) to support efficient risk identification in NATM tunnels, thereby reducing manual efforts and time expended on reviewing past project documents during the planning of new construction projects. The proposed QAR-System is appropriate for quick, effective, and intuitive risk identification at the early stage of any construction project, as it can facilitate timely and reliable decision-making to prevent safety accidents and reduce project delays and cost overruns. The contribution of this study to the body of knowledge is in three folds. (1) This study provides the practitioners/researchers with the framework for conceptualizing and representing the risk knowledge acquired from previous projects using the knowledge graph approach to aid reuse and sharing between humans and machines (computers). (2) This research lays the foundation for applying and adopting LLMs such as GPT-3 in the construction domain to advance timely, effective, and reliable decision-making throughout the lifecycle of construction projects. (3) This study demonstrated how AI technologies could facilitate the digital transformation of risk management and construction management practices to enhance informed decision-making, improving project performance and productivity in the construction industry.

CHAPTER 1. INTRODUCTION

1.1 Research Background

The construction industry is one of the principal industries that contribute to global economic growth by implementing various sizes of construction projects (Isah & Kim, 2021). According to (Ribeirinho et al., 2020), the construction industry contributes around 13% to the global GDP, but for two decades now, productivity growth has been increasing by only 1% annually. Also, the Global Construction 2030 report forecasted that the spending on construction and engineering projects would surpass \$212 trillion by 2030 (Mike et al., 2015). However, achieving the desired productivity and project success that corresponds with the value of the capital investment in these projects depends on efficient management of the project objectives (costs, time, scope, and quality), associated risks, and uncertainties that usually influence the project outcome.

Tunnel projects are large-scale construction projects involving the excavation of underground passages. They are critical parts of infrastructural development that connect people and places while driving global economic development. However, the construction of tunnel projects, compared to other infrastructural projects, is very challenging and risky (H. Zhou et al., 2020) due to the complexity and diversity of the surrounding rocks, groundwater, work safety, geological environment, construction method, man-made factors, and several other factors (Lin et al. 2020; Yang et al., 2023). Therefore, it is imperative to identify the potential risks at the preconstruction phase and manage them

effectively throughout the project lifecycle to prevent safety accidents, reduce the probability of risk occurrence during construction, and minimize the adverse impact of the risks on the project objectives (Isah & Kim, 2022, 2023) to ensure project success.

The Project Management Book of Knowledge (PMBOK) defines risk as an uncertain event or condition that, if it occurs, has a positive or a negative effect on project objectives (PMI, 2017). In other words, risk is the effect of uncertainty on the project objectives (ISO, 2018a). Therefore, risk management (RM) is an important measure of enhancing the holistic understanding of the project scope and potential problems that could affect project objectives (Tan, 2006). RM is a “formal process of systematically identifying, analyzing, and responding to risk events throughout the life of a project to obtain an acceptable degree of risk elimination or control”(Jamal & Crandall, 1990). The objective of RM in construction projects is to identify, assess, and manage critical project risks effectively to maximize the chances of project success.

Furthermore, risk identification is the first step and the most critical process of RM in construction projects. It involves documenting the types and sources of significant risks associated with the project so that the project team can prepare appropriate responses to mitigate them (PMI, 2017). Also, identifying risks in the early stage of the project and taking appropriate mitigation and control steps during execution can help abate the delay and cost escalations usually experienced in construction projects (Shelake et al. 2022). The main purpose of risk identification in a construction project is to keep the project team abreast of uncertain situations and potential risks associated with the project (Tan,

2006).

In addition, risk identification and management is a knowledge and experience-based process that generates high data (knowledge) scattered across project documents and stored in various formats. The knowledge and experience generated are useful for managing future projects if captured, structured, and stored in a reusable format for humans and computers. In other words, the lessons learned, knowledge and experiences gained from the previous projects play crucial roles in risk identification, forecasting probability of occurrence, impact prediction, and suggestion of risk consequences and mitigations of new projects (Okudan et al., 2021). However, the traditional RM procedures for construction projects rely heavily upon tacit knowledge stored in the minds of humans (experts) and explicit knowledge stored in non-human repositories of various formats. Besides, there are no formalized formats of representation, capturing and storing this risk-related knowledge to facilitate integration, reuse, and sharing among practitioners. Thus, there is a need to develop formal risk knowledge representation mechanisms to prevent loss and ensure the sharing of valuable risk knowledge for efficient RM in construction projects.

Although several risk identification methods, such as checklist analysis, brainstorming, interviewing, risk workshops, etc., have been established in the literature. Nonetheless, these methods have their inherent limitations, which include: (1) over-dependence on the knowledge of experts that can lead to misidentification or improper management of identified risks if the expertise is flawed or lacking, (2) subjective nature of the process

leading to inconsistencies, (3) resource-intensive and time-consuming procedures that may not account for subtle or unusual risks because of the static and complex nature of construction projects, (4) inadequate utilization of historical data or predictive analytics which can lead to inaccurate risk profiles, and (5) nonavailability of standardized systems for documentation and communication among the stakeholders resulting to possible oversights and misunderstandings (Saka et al., 2023). Hence, the application of advanced computational techniques such as artificial intelligence (AI) has been suggested to mitigate the limitations of the traditional methods (Zhang et al., 2021).

AI is a powerful technology that deals with the ability of machines to execute repetitive tasks that typically require human intelligence, like reasoning, learning, perception, and decision-making. Some of the subfields of AI include machine learning (ML), knowledge-based systems (KBS), computer vision, robotics, and natural language processing (NLP) (Abioye et al., 2021). Studies have highlighted that these AI technologies can be leveraged for dynamic and comprehensive risk identification and assessment by reviewing large project documents and historical data quickly (Saka et al., 2023), thus improving the efficiency of RM, communication among project stakeholders, and decision-making in construction projects.

Moreover, the advancements in AI and NLP-based applications, such as question-answering systems, virtual assistants, chatbots, etc., have become prominent because they allow computers to understand and interact with humans in conversational ways using natural language (NL). These applications leverage large language models (LLMs) for

effective functioning and have been utilized in many fields, such as healthcare, bioinformatics, finance, education, etc. LLMs are advanced machine learning models pre-trained on large datasets to understand and generate human-like language (Zheng & Fischer, 2023). A notable example of the LLMs is OpenAI's GPT series (e.g., GPT-3), which have emerged as powerful tools in NLP tasks due to their in-context learning capabilities. The GPT models can be adapted (through prompting or fine-tuned with a small dataset) to develop an intelligent question-answering system (QAS) for automating information retrieval using NL.

Despite the adaptability of the GPT models in breaking the barrier of developing or adopting advanced AI applications in the construction domain and providing opportunities to improve productivity, only a few studies have adopted GPT in construction literature. Therefore, this study aims to integrate a GPT model with a knowledge graph (KG) to develop intelligent QAS for automating risk information retrieval to mitigate the limitations of the traditional risk identification process in construction projects. This approach involves integrating human-machine intelligence based on the knowledge and experience acquired from previous construction projects to provide a system that can support automatic risk identification in construction projects. It is expected that this study will pave the way for studies on the adoption of advanced AI technologies such as KG and GPT models for risk identification in construction projects, thus improving the efficiency of RM processes, quality of decision-making, increasing productivity, providing a common RM platform for the project team, and ultimately

facilitating the digital transformation of RM management practice in construction projects.

1. 2 Research Objectives

The objectives of this research are as follows:

- i. To develop a mechanism for the formal representation of risk-related knowledge for construction projects to prevent loss of knowledge, enable knowledge sharing between humans and computers, and improve the efficiency of risk identification and RM practice. This objective is achieved by adopting a KG approach to construct a graph database (risk knowledge graph) based on the work breakdown structure (WBS) and risk breakdown structure (RBS) of the project to understand the intrinsic link between construction works, risk sources, and risk factors. The risk knowledge graph (RisKG) is a structured database for developing the question-answering system for construction project risks.
- ii. To develop a question-answering system for construction project risks (QAR-System) based on the structured database (RisKG). This objective is accomplished by adapting OpenAI's GPT-3 model for query generation based on natural language and developing a user interface (UI) to support quick and efficient risk information (knowledge) retrieval using the streamlit Python library.

In a nutshell, this research focuses on developing an intelligent QAS powered by a KG and GPT model that can be used to support the risk identification process in construction projects. While question answering is an NLP task that requires computers to answer questions posed by humans based on the information available in knowledge bases

(Zhang et al., 2021), the KG is an AI technology used for knowledge representation and management (Jiang et al., 2017) and semantic retrieval of information in various domains (Lezhnina et al., 2022). These technologies were leveraged to improve the efficiency of risk identification in construction projects.

1.3 Research Method and Scope

Figure 1 shows the development process of the proposed QAR-System for tunnel construction projects. The process involves the construction of a risk knowledge graph based on the data collected and the development of a question-answering model using the GPT-3 model. First, risk data for tunnel projects was collected. Then, the risk knowledge was modeled by defining entities and relationships of the RisKG based on the data collected. Next, the risk knowledge was extracted as triples from the data collected and saved in an Excel file. Then, RisKG was constructed by storing the risk knowledge in a graph database using the Neo4j desktop application. Second, the Cypher query generation model was developed by training a foundational GPT-3 model using the few-shot learning approach. In other words, the GPT-3 model (i.e., “text-DaVinci-003”) was trained with a few examples of question-query pairs. The query generation model enables the conversion of natural language questions (NLQ) to executable Cypher queries to facilitate the extraction of risk information from the graph database. Finally, the RisKG and query generation model were integrated using the Streamlit Python library to develop the proposed QAR-System.

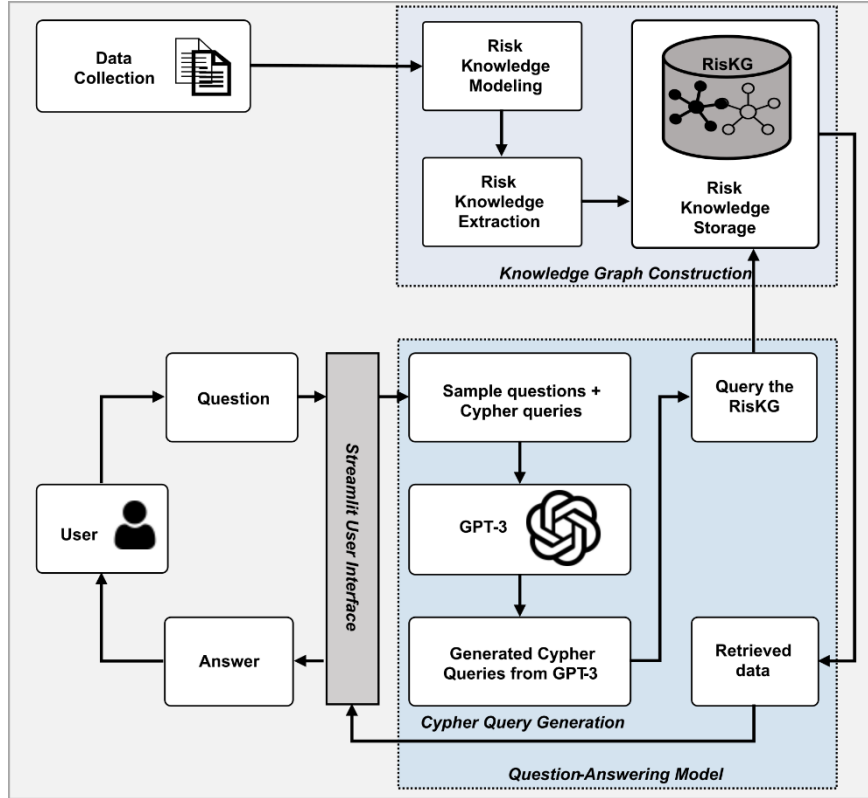


Figure 1. Overview of the QAR-System development process

This study uses the available risk knowledge (explicit and tacit) acquired from previous tunnel projects to identify risks at the preconstruction stage of a new project. At the early stage, limited information on the new project is available to support the project participants in comprehensive and accurate risk identification. Also, at this stage, all potential critical risks that can jeopardize the project objectives (time, cost, and quality) must be identified to ensure accurate risk analysis, control, and mitigation of the identified risks. Therefore, a tunnel construction project was selected to demonstrate the

development of QAR-System. Specifically, the New Austrian Tunnel Method (NATM) tunnel projects were selected because of the non-availability of risk-related information for other tunnel construction projects. However, some major outputs can be referenced, and relevant concepts used here can be borrowed to build intelligent systems for other construction projects that fall outside the scope of this thesis.

Further, the risk-related information used in the model development consists of the project risk sources, risk names, and risk factors associated with the project works. The risk information, such as risk consequences, probabilities, and impacts that can aid risk assessment and control, is outside the scope of this study due to the non-availability of useable data. However, a brief description of how information retrieved from QAR-System can be used for risk assessment was presented in this study. More risk information (risk consequences, probabilities, and impacts) is needed to make the system an integrated RM tool. Ultimately, QAR-System is developed to provide the project manager with automatic risk information retrieval to support efficient risk identification and reliable decision-making at the preconstruction stage of tunnel projects.

1.4 Thesis Organization

This thesis is divided into five chapters, as illustrated in Figure 2. Chapter 1 presents the research background, objectives, an overview of the research method, and the scope of the research. Chapter 2 reviews existing literature on risk management practices focusing on risk identification and knowledge-based risk management (KBRM) for construction

projects. Also, it reviews previous studies on the knowledge graph (KG), question-answering systems, and their applications in the construction domain. In addition, the chapter provides a literature review on applying the GPT models in fields other than the construction domain. The chapter highlights impediments to applying and adapting AI technologies in construction and civil engineering. The chapter concluded with a brief explanation of the popular conversational AI tools powered by LLMs. Chapter 3 presents a detailed explanation of knowledge graph development. In the chapter, a risk knowledge graph (RisKG) for tunnel projects was constructed and evaluated, and the implications of the evaluation results were discussed. In Chapter 4, a brief introduction of question-answering was provided. Then, a detailed explanation of the development process of the proposed QAR-System was presented, including the evaluation of QAR-System and results discussion. Chapter 5 concludes with the study's contributions to the body of knowledge highlighted. Finally, the study's limitations and recommendations for future studies were presented.

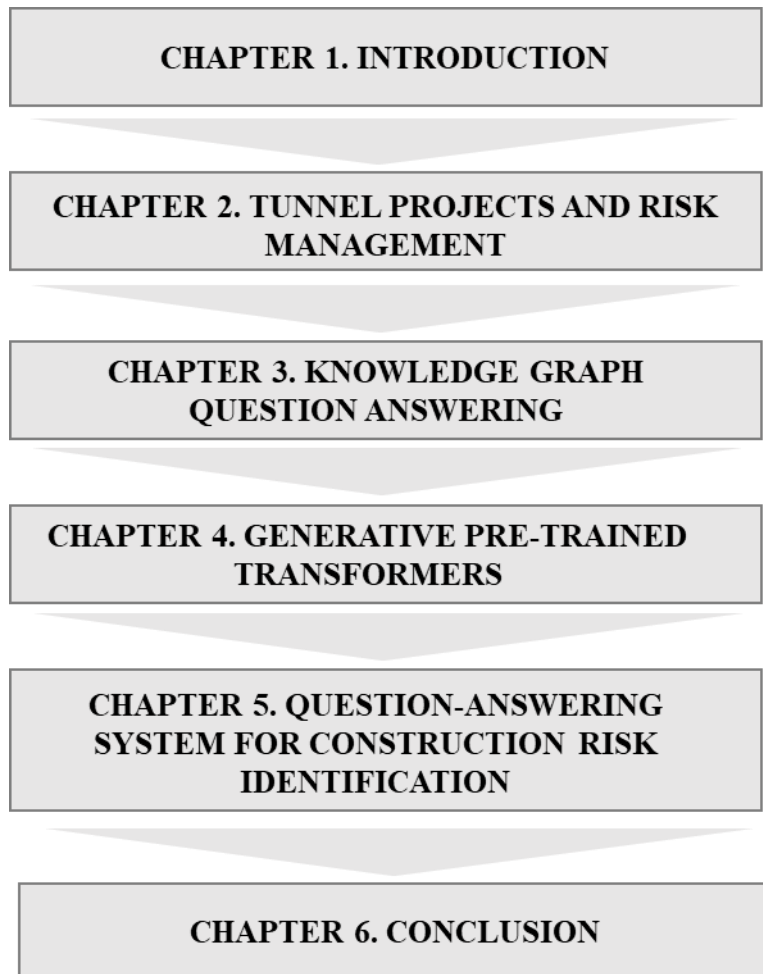


Figure 2. Thesis organization

CHAPTER 2. RISK MANAGEMENT IN TUNNEL PROJECTS

2.1 Introduction

This chapter overviews risk management (RM) in tunnel construction projects. It explains the risk management concept in construction projects, starting with the definitions of risk, an explanation of the risk management processes and frameworks, a literature review of risk identification in tunnel projects, and the existing tools used. The chapter concluded with the introduction of knowledge-based risk management in construction projects.

2.2 Risk Management in Tunnel Projects

2.2.1 Definition

Historically, the word risk was derived from the French word *risqué* and was first used in insurance transactions around 1830 in England (Smith et al., 2014). However, there is no commonly accepted definition for the term risk. Thus, several attempts have been made in the literature to define risk in different ways and contexts. For example, traditionally, it is defined as “an uncertainty that could have a negative/ harmful/ adverse/ unwelcome/bad effect on one or more objectives” (Hillson 2002). From a business perspective, risk is seen as “the probability of occurrence of some uncertain, unpredictable and even undesirable event(s) that would change the prospects for the profitability on a given investment” (Kartam & Kartam, 2001) and “a chance of bad consequences, or exposure to mischance” (IRM, 2002). Also, the Project Risk Management Guide defined risk as any uncertainty that, if it occurs, would affect one or

more project objectives (WSDOT, 2018). According to ISO 31000:2009, risk means the effect (positive and/or negative) of uncertainty on project objectives. Moreso, the Association of Project Management (APM) refers to risk as an uncertain event or set of circumstances which, should it occur, will affect the achievement of objectives” (PRAM, 2018).

In the construction projects context, *risk* is regarded as “an exposure to economic loss or gain arising from involvement in the construction process” and “a consideration in the process of a construction project whose variation results in uncertainty in the final cost, duration, and quality of the project” (Kartam & Kartam, 2001). Ultimately, from all the definitions provided, risk could be viewed from three different perspectives: (1) risk could be all negative (threat), (2) risk could be either positive (opportunity) or negative (threat), and (3) risk could be both opportunity and threat. However, this study viewed risk from a threat perspective and thus adopted the traditional definition of risk, emphasizing the negative impact on predefined project objectives.

Furthermore, uncertainties and risks are unavoidable in tunnel projects as no construction project is risk-free. Tunnel construction is a complex and unique undertaking with inherent risks and uncertainties resulting from complex geotechnical conditions, varied geographical environments, difficult construction methods, etc. These risks cannot be ignored but should be managed to avert disastrous accidents and adverse effects on project objectives, i.e., cost, time, and quality (Isah & Kim, 2022). Therefore, risk management should be incorporated into the delivery of the process of tunnel projects from planning

to completion.

RM is “the process of conducting risk planning, identification, analysis, response planning, response implementation, and monitoring risk on a project” (PMI, 2017). RM refers to “the culture, processes, and structures directed toward the effective management of potential opportunities and adverse effects” (Cooper et al., 2005). The guide to risk management explained that RM is the process of identifying, assessing, and responding to risks and promptly communicating the outcomes of these processes to the appropriate parties (Queensland Treasury, 2020).

In construction projects, RM systematically applies management policies, processes, and procedures to establish the context, identify, analyze, assess, treat, monitor, and communicate risk. According to ISO, construction RM entails coordinated activities to direct and control an organization concerning risk events (ISO, 2018). Merna & Al-Thani (2018) stated that RM is a crucial part of the construction project planning process, which (1) requires acceptance that uncertainty exists, (2) generates a structured response to risk in terms of alternative plans, solutions, and contingencies, (3) is a dynamic process requiring imagination and ingenuity (4) generates a realistic attitude in project staff by preparing them for risk events rather than being taken by surprise when they arrive. The ultimate purpose of RM in construction projects is to identify and manage risks effectively to optimize/increase the chances of project success and to minimize the risks of not achieving the project’s objectives. In the words of Cooper et al. (2005), RM helps project managers set priorities, allocate resources, and implement actions and processes that

reduce the risk of the project not achieving its objectives. Also, an effective RM adds value to construction projects, with many benefits which include but are not limited to facilitating better business and project outcomes through informed decision-making, increasing certainty and reducing overall project risk exposure, providing a framework to avoid sudden surprises, and justifying prudent risk minimization and mitigation measures (Cooper et al., 2005).

2.2.2 Risk Management Process

The generic risk management process has been developed and established in various RM guides (AS/NZS ISO, 2009; Guide, 2018; ISO, 2018b; PMI, 2017, 2019; Queensland Treasury, 2020; WSDOT, 2018). As for the tunnel risk management process, the detailed procedure has been documented by (Eskesen et al., 2004). The risk management process, as established in the project risk management guideline of K-Risk (Korea Risk Expert Council) (K-Risk, 2022), includes risk management planning, risk identification, qualitative risk analysis, quantitative risk analysis, risk response, risk monitoring, and control. The K-Risk RM process aligns with the Washington State Department of Transportation (WSDOT) risk management guide, illustrated in Figure 3.



Figure 3. Risk management process (WSDOT, 2018; K-Risk, 2022)

Risk management planning is a systematic way of deciding how to plan and execute RM activities throughout the project's life. This process aims to maximize the positive outcome of the opportunities and minimize or eliminate the effects of adverse risk events. The process of risk identification determines and documents the risks that can affect the project. In the qualitative risk analysis process, identified risks are assessed for their probabilities of occurrence and impact on project objectives (i.e., cost and time). Quantitative risk analysis involves numerically estimating the probability that a project will meet the planned cost and time. The risk response process is a way of developing options and determining actions to minimize threats or enhance opportunities for the project objectives. This process requires the project manager and team to prepare the best response strategies for each risk and identify the implementation of the strategy. Risk monitoring and control is the final step in RM. It involves tracking the identified risks, monitoring the residual risks, identifying new risks, ensuring the execution of risk plans, and evaluating their effectiveness in minimizing risk. It is important to note that risk

management is not a static process but cyclic and continuous for the project's life.

2.2.3 Risk Identification in Tunnel Projects

RM activities begin at the project's initiation and continue throughout the project's life cycle. However, the first and most important process of RM is risk identification. The remaining RM processes can be performed only when potential risks that will affect the project are identified. Thus, the process must be conducted diligently to ensure that all possible potential risks and their sources are identified to facilitate effective RM and accurate decision-making.

Risk identification is “the process of systematically and continuously identifying potential risks and their consequences on a project using appropriate risk identification tools and techniques, classifying the risks, identifying their root causes, and documenting the characteristics of each risk” (Siraj & Fayek, 2019). This process determines what, how, and why things may happen in a project. In other words, risk identification determines what could affect the project's objectives and how those might happen (Cooper et al., 2005). Therefore, identifying risks in the early stage of the project and taking appropriate mitigation and control steps during execution can help abate the delay and cost escalations usually experienced in tunnel projects (Shelake et al. 2022). The objective of risk identification is to generate a comprehensive list of risks based on those events that might create, enhance, prevent, degrade, accelerate, or delay the achievement of the project's objectives.

In the early 2000s, Working Group 2 (Research) of the International Tunnelling and Underground Space Association (ITA) published the guidelines for risk management of tunnels and underground projects in which they described the process of identification and management of risk factors that cover the entire lifecycle of tunneling and underground projects (Eskesen et al., 2004). Similarly, numerous studies have established different tools and techniques for identifying tunnel project risks. Some of these tools and techniques include but are not limited to checklist analysis, documentation reviews, brainstorming, Delphi techniques, interviewing, root cause analysis, questionnaires, risk workshops, assumption analysis, cause-and-effect diagrams, system or process flow charts, influence diagrams, SWOT analysis, expert judgment, fault tree analysis, decision tree analysis, and failure and effect analysis (Siraj & Fayek, 2019). Nevertheless, (Hillson, 2002) stated that there is no single best method for identifying project risk, but combining these tools and techniques should be employed to achieve better project outcomes. In other words, two or more risk identification tools can be combined and even integrated with other tools for effective and accurate project risk management. For instance, (Xu et al., 2019) demonstrated this by utilizing a combination of risk breakdown structure (RBS), work breakdown structure (WBS), Delphi survey, and fuzzy analytic network process (FANP) for the identification and assessment of risks in a Beijing subway tunnel construction. Moreover, (S. Zhang et al., 2019) proposed a real-time model for identifying potential safety risk factors during metro construction, considering adjacent buildings and structures. The authors established the initial risk factors based on the information

gathered from the web, academic literature, accident news bulletins for metro construction, and expert support. Then, the interval analytic hierarchy process (IAHP) and an extension of the technique for order preference by the similarity to the ideal solution (TOPSIS) methods were used to evaluate and rank the significant potential safety risk factors encountered in real-time metro projects. Correspondingly, (Ou et al., 2021) established a theoretical and technical system for controlling the risks related to tunnel collapse during construction. The authors identified eleven influencing risk factors using the records and descriptions of tunnel collapse accidents in academic literature, public reports, and projects experienced by the authors. Then, the Dempster-Shafer (D-S) fusion rule was used to establish the tunnel collapse risk level.

Recently, (Shelake et al., 2022) combined a questionnaire survey and expert judgment to develop a framework for identifying the risk factors of tunnel projects using the Relative Importance Index (RII). The authors developed an exhaustive tunnel project risk factors list through a comprehensive literature review. Then, experts' opinions were collected through a questionnaire survey to identify fifteen (15) critical risk factors using the RII framework.

Table 1. Tunnel construction risks and risk factors

Author	Risk Category	Risk factor
Eskesen et al., 2004	•Contractual disputes. •Insolvency and institutional problems. •Authorities' interference. •Third-party interference. •Labour disputes. •Accidental occurrences. •Unforeseen adverse conditions. •Inadequate designs, specifications, and programmes. •Failure of major equipment. •Substandard, slow, or out-of-tolerance works.	
Chivata et al., 2013	•Face instability in soft soils when using Slurry Shields. •Face instability in soft soils when using Earth Pressure Balance Shields. •Excessive volume loss leading to settlement in tunnels bored in soft soils. •Excessive deformation, damage, or leakage in concrete linings. •Collapse and large deformations of excavated shafts in soft soils. •Collapse and large deformations of excavated cross passages in soft soils.	
(Xu et al., 2019)	•Management.	-Standard specifications for construction -Safety consciousness -Safety management -Construction coordination
	•Selection of construction equipment.	-Operating standard -Mechanical equipment failure
	•Material.	-Concrete strength -Steel cage welding -Quality of grouting material
	•Construction.	-Instability of a support system -Over excavation -Temporary support -Water inflow -Foundation failure
	•Environment.	-Underground pipeline -Existing building

Author	Risk Category	Risk factor
(Chung et al., 2020)	•Geological	-Discontinuity -Weak ground -Shallow cover depth -Karst topography -Artificial structure -Groundwater/High water pressure -Anisotropic ground -Swelling ground -Ground under great pressure
	•Design	-Numerical problem -Unsuitable excavation and Support design for ground condition -Insufficient geotechnical investigation
	•Construction	-Poor construction management -Excessive excavation
Lin et al., 2020	•Rock mass quality	-Uniaxial compressive strength (UCS) -Rock quality designation -Joint spacing -Joint condition -Groundwater condition
	•Monitoring measurement	-Crown settlement -Peripheral convergence displacement -Ground settlement
	•Safety management of constructors	-Personal protective equipment -Safety supervision and inspection -Safety publicity and education
	•Equipment operation and management	-Construction machinery and equipment -Power supply equipment -Ventilation system -Fire protection equipment
Ou et al., 2021	•Natural condition	-Excavation span -Shape of the tunnel section -Tunnel layouts -Tunnel depth -Rainfall -Negative landform -Earthquake
	•Geological factors	-Uniaxial compressive strength (UCS) -Degree of weathering -Unfavorable geology -Groundwater condition

Author	Risk Category	Risk factor
Ou et al., 2021	•Geological factors	-Attitude of rock -Stress condition -Bias angle
	•Design and construction	-Accuracy of geological survey -Excavation method -Primary support -Stratum reinforcement -Skills of construction workers -Drainage
	•Organization and management	-Monitoring -On-site management -On-site supervision
Yang et al., 2023	•Human factor	-Working ability -Security awareness -Professional quality -Health status -Mental state
	•Material factor	-Equipment advancement -Equipment operation status -Safety devices -Quality of construction materials -Materials storage
	•Geological exploration design	-Geological Exploration accuracy -Drilling design -Route design -Construction drawing design -Implementing the design intent
	•Technical management	-Safety education train -Program change -Construction organization -Quality inspection -Monitoring management -Monitoring analysis -Geologic analysis -Detail design -Technical tests
	•Safety management	-System construction -System implementation -Information receiving -Emergency plan system -On-site environmental management
	•Natural environment	-Water supply -Regional structural -Climatic conditions -Geologic environment
	•Tunnel construction accident	-Objective hazard sources -Subjective hazard sources

Similarly, (Y. Yang et al., 2023) used a literature review of tunnel construction accidents and expert experience (opinion) using questionnaires for the identification of risk factors that influence tunnel accidents and then established a structural equation model to investigate the relationship between the risk factors, thus improving safety awareness in tunnel projects. Table 1 highlights some risks and risk factors identified in previous studies for tunnel construction.

Furthermore, the literature reviewed on tunnel risk identification revealed that (1) researchers have perpetually adopted the traditional risk identification techniques for tunnel project risks, and (2) risk identification is a knowledge-intensive process hinged upon both explicit knowledge (risk repositories) and tacit knowledge (i.e., knowledge stored in the minds of individuals, e.g., experts) (Isah and Kim 2023; Okudan et al. 2021). In other words, it is a knowledge discovery process that requires creative thinking, imagination, and leveraging project team experience and knowledge (Siraj & Fayek, 2019). (Chivatá et al., 2013) pointed out that the knowledge gathered from previous construction projects is extremely useful for managing new projects. Thus, to effectively mitigate and manage these risks, studies have proposed the adoption of an integrated human-machine intelligence approach like the knowledge-based risk management (KBRM) method, which entails using the generated risk-related knowledge during RM of past projects to facilitate efficient and reliable decision-making in new projects (Okudan et al., 2021; Serpella et al., 2014; Tah & Carr, 2000; Zoysa & Alan D. Russell, 2003).

2.3 Knowledge-Based Risk Management

Knowledge is the bedrock of human intelligence Yuan et al. (2019). Knowledge is defined as the information stored in the mind of humans about facts, experiences, concepts, interpretations, ideas, observations, judgments, expert insights, etc., articulated and presented in graphics, textual or symbolic forms (Serpella et al., 2014). According to Okudan et al. (2021), knowledge is “a fluid mix of framed experience, values, contextual information, and expert insight”. Knowledge is divided into explicit and tacit knowledge. Explicit knowledge is objective knowledge that is accessible and shareable, while tacit knowledge is subjective knowledge that is difficult to articulate but can be acquired through experience (Maravilhas & Martins, 2019). In the risk identification process of a new project, the interaction between explicit and tacit knowledge usually generates new risk-related knowledge useful for managing future projects. Hence, appropriate mechanisms are needed to capture, share, and manage the knowledge generated to maximize the benefits.

Knowledge management (KM) involves capturing, storing, and sharing knowledge using the appropriate mechanisms. According to Tan (2006), KM is the process of “acquiring, storing, retrieving, applying, generating and reviewing an organization’s knowledge assets in a controlled manner. Also, KM is described as “the process of using cautious steps to acquire, design, manage, and share knowledge in an organization to achieve performance improvements like reduced cost of work, fast-track work, and use of best practices” (Santoro et al., 2018). Therefore, risk-related knowledge acquired from the

present project can be utilized to improve future project performance using the appropriate KBRM tools and techniques. KBRM is a systematic approach to capturing and organizing risk-related knowledge acquired from past projects and reusing it in the RM of new projects to aid reliable decision-making that will lead to successful project delivery.

Numerous existing construction literature has demonstrated and documented the application of KBRM. For example, Tah and Carr (2001) introduced a generic process and underlying information model using a common language for risk descriptions and remedial actions. The study developed a fuzzy knowledge representation model to support quantitative risk analysis and a user-friendly knowledge-based system that controls the entire RM process. Further, Yuan et al. (2019) combined a knowledge base (prevention through design, PtD) with building information modeling (BIM) to develop an automatic plugin-in for checking safety risks in buildings. The PtD knowledge base was constructed using the safety regulations documents and best practices. Also, the study utilized Revit API and Visual Studio to develop the automatic plug-in. Using a case study and questionnaire survey, the authors validated the proposed PtD tool. The results show that the PtD is effective and efficient for automatically detecting safety risks at the design stage, thus saving the designer's effort and time. Similarly, Dadashi Haji et al. (2022) integrated a knowledge base with BIM to develop an add-on to assess the impact of safety-leading indicators on construction projects. The knowledge-based safety-leading indicator was developed using the extracted information from subject matter experts,

documents, and best practices. Nevertheless, similar to Yuan et al. (2019), the authors utilized Navisworks API and Visual Studio framework to create the add-on for determining the impact of safety-leading indicators on construction projects based on the knowledge base. The tool's efficiency was verified through an expert survey, and the results indicated that the add-on could support an automatic impact assessment of safety-leading indicators and aid a 38% reduction in construction site accidents. Moreover, to improve the KBRM practice, Okudan et al. (2021) used an AI approach (i.e., case-based reasoning (CBR)) to develop a KBRM tool (CBRisk) to support effective RM in construction projects. The authors explained that the proposed CBRisk is a web-based tool that can facilitate the cyclic RM process based on an effective case retrieval method. Although researchers have made concerted efforts to develop practical KBRM tools, however, the research area is still developing. Thus, more studies are required to enable the development of practical KBRM tools leveraging emerging AI technologies, such as knowledge graphs, to improve RM practice in construction projects. In other words, it is important to adopt appropriate knowledge representation techniques that can facilitate the accurate capturing and storage of risk-related knowledge (explicit and tacit) and support quick knowledge retrieval and sharing between project participants during the RM planning of a new project.

CHAPTER 3. KNOWLEDGE GRAPH QUESTION ANSWERING

3.1 Introduction

Knowledge graph question answering (KGQA) is a research field rooted in the intersection of natural language processing (NLP), semantic web technologies, and AI, which is focused on providing a concise answer to user queries using knowledge graph technology. This section provides a brief explanation of the concept of knowledge graph and question answering.

3.2 Knowledge Graph

3.2.1 Concept of knowledge graph

Graph database (GDB) is a departure from conventional relational database system (RDBMS) that has been around since the 60s. GDBs have been the fastest-growing database technology for the past decade, introducing a new way of modeling, traversing, and using interconnected data. A GDB is an efficient tool for modeling data (entities) when the relationships between the entities are of great interest. Further, GDB utilizes interlinked data structures, which are built from nodes, relationships, and properties, making it a sophisticated technology that supports semantically rich queries of various scales while enabling users to gain intuitive insight into data models (Webber & Van Bruggen, 2020). It consists of nodes and relationships, where the nodes represent entities such as a person, product, place, or patient diagnosis, while the relationships link nodes together. Figure 4 shows a conceptual diagram of a GDB consisting of living things taxonomy.

A knowledge graph (KG) is a collection of real-world entities (data) connected through semantic relations, which supports advanced reasoning to discover unseen conceptual connections and facilitate informed decision-making. KG is an interconnected dataset enriched with semantics to enable reasoning about the underlying data and can be used confidently for making complex decisions (Maya Natarajan, 2021).

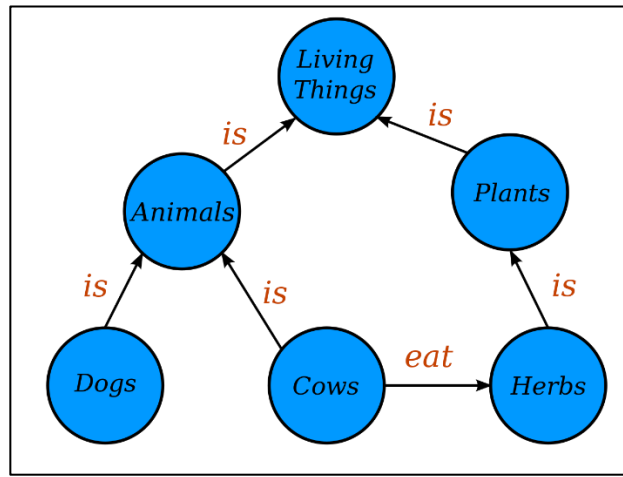


Figure 4. Conceptual diagram (Wikipedia)

According to (Ma, 2021), KG refers to “a group of nodes connected by edges (relationships) where the nodes represent entities in the real world and edges for the relationships between those entities”. In the business context, KG combines business data and knowledge to provide a more complete and integrated experience with the organization’s data. The functional definition of KG is given as a set of *triples*, with each triple intuitively representing a *fact*. Figure 5 shows a triple, i.e., 3-tuple (h,r,t) where h represents a head entity, t represents a tail entity, and r expresses a relationship between

the two entities. Simply put, KG is the connection between two things, e.g., person and company, person and product, etc.

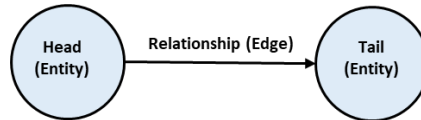


Figure 5. An illustration of a triple (3-tuple)

According to (Zdonik et al., 2019), KG, otherwise known as the knowledge base, has many advantages over the relational database, which include but are not limited to:

- supports explicit and implicit knowledge representation, aiding semantic metadata to describe data sources uniformly,
- provides the underpinnings and architecture in which data can be shared effectively throughout an organization,
- simple and easy to read for machine learning and data analytics programs.
- fast, flexible, and low-cost approach to enterprise data management.

In a nutshell, KG is a knowledge representation and AI technique for domain conceptualization and data management (Abu-Salih, 2021) that allows the machine to conduct reasoning and inference over the graph to answer questions (queries) in more detail way while enabling humans to learn from the graph data model. For example, Figure 8 shows a sample KG of a company, *ABC Inc.* Figure 6 shows the relationship between ABC Inc (company), CEO (person), PM 1 (person), PM 2 (person), Project 1

(object) and Project 2 (object).

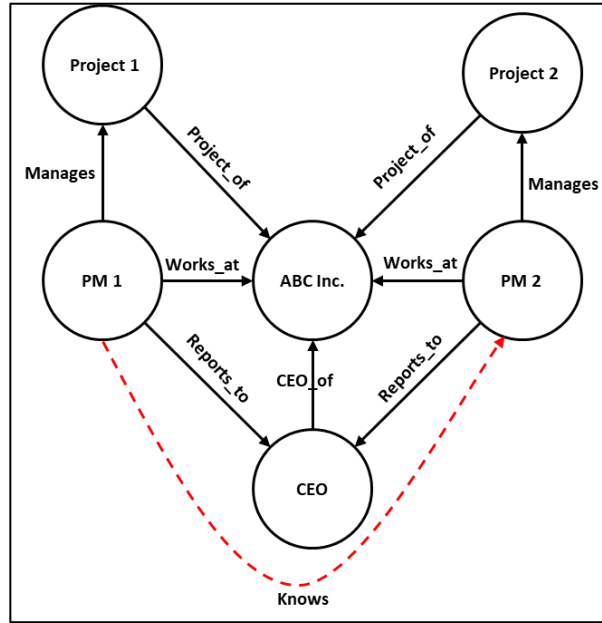


Figure 6. Sample of a company KG

As depicted in the figure, the relationships between all the entities are clear. However, the relationship between PM 1 and PM 2, which cannot be seen in a relational database, is revealed with the red dashed arrow. The simple interpretation is that if PM 1 and 2 both work at ABC Inc. and report to the CEO, it can be inferred that both PM 1 and PM 2 know each other. This type of inference is possible only with a KG.

3.2.2 Brief History of KG

In the early 1950s, knowledge representation and knowledge-based systems were used for reasoning and complex problem-solving (S. Ji et al., 2022). However, the concept of

KG is a semantic network, and its history can be dated back to the early 1970s. The term “knowledge graph” was first coined by an Austrian Emeritus Professor of linguistics, Edgar W. Schneider. Subsequently, in 1985, Wordnet was developed, capturing semantic relationships between words and their respective meanings. In 2005, a KG capturing geographic names, locations, and associated entities was developed by Marc Work. Similarly, in 2007, two graph-based knowledge repositories for general-purpose knowledge, DBpedia and Freebase, were established (Wikipedia). However, in 2012, KG became popular after Google developed its KG and used it to support intelligent search engines.

Generally, KG is categorized into two: open domain (generic) and domain-specific KG. The generic KGs are publicly available KGs developed based on general knowledge available on the web, some of which include FreeBase, NELL, DBPedia, YAGO, etc. In contrast, the domain-specific KG focuses on “an explicit conceptualization of a high-level subject-matter domain and its specific subdomains represented in terms of semantically interrelated entities and relations” (Abu-Salih, 2021). Domain-specific KGs are developed based on domain knowledge to address specific and complex domain problems. Example of such includes but is not limited to Medical KG (L. Li et al., 2020), Hepatocellular Carcinoma KG(KGHC) (N. Li et al., 2020), Food KG (Hausmann et al., 2019), K12 EduKG (P. Chen et al., 2018), etc.

3.2.3 Construction of KG

KG is a structured knowledge base that can be constructed from unstructured and semi-

structured data, such as texts, tables, etc. According to (Abu-Salih, 2021), the construction of KG can be categorized mainly into three, including the knowledge extraction level, the knowledge base type, and the incorporated construction techniques. Figure 7 shows the taxonomy of KG construction adopted from (Abu-Salih, 2021). Nonetheless, this study is focused on only knowledge base categories and is explained as follows.

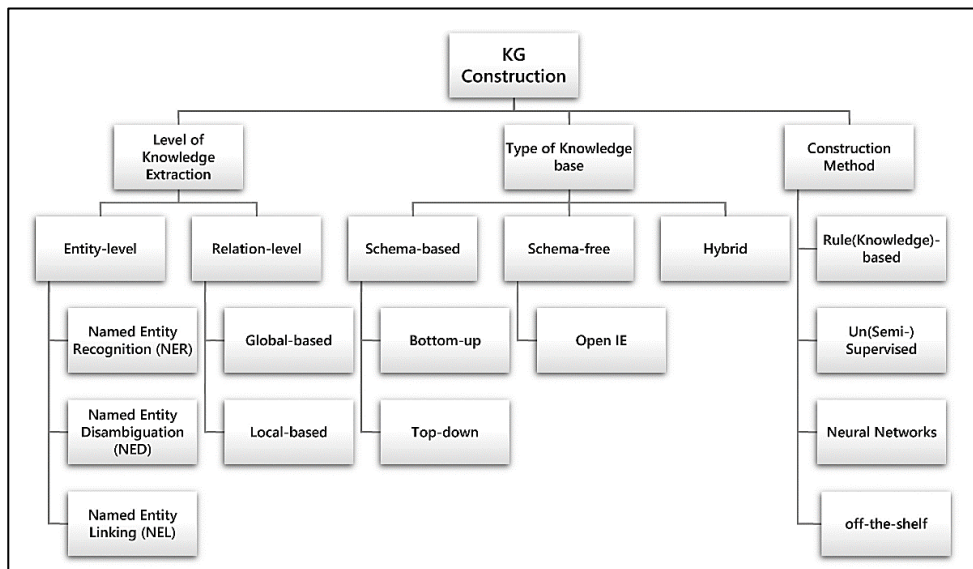


Figure 7. KG construction taxonomy (Abu-Salih, 2021)

The construction method for the knowledge base type is further subdivided into schema-based, schema-free, and hybrid methods. The schema-based method is based on the predefined schema and can be grouped into two based on data sources and ontology, namely, (1) bottom-up method means building the KG ontology from scratch and incorporating the ontology structure into the KG (e.g., Wikipedia was established using a predefined ontology model i.e., DBpedia) which is appropriate for a domain-specific KG,

and (2) top-down method is a data-driven approach which involves building the ontology KG by inferring from an underlying structured data (e.g. YAGO) or constructed using a structured database (Abu-Salih, 2021). In contrast, the schema-free technique is an open information extraction approach that relies upon the open data from the web with no consideration for creating a unified ontology. The hybrid approach is a technique that relies on knowledge from predefined ontology and information gathered from the web (e.g., KnowledgeVault and NELL).

Further, KGs can be constructed either manually or in an automated way. The manual method ensures high-quality and unique KG but requires qualified domain experts and knowledge engineers. Also, the manual approach relies on crowdsourcing efforts, which are labor-intensive and may be prone to human errors and biases. The automatic approach uses machine learning (ML) to extract entities and relationships to construct the KG. The ML approach is fast but unsuitable for a domain-specific KG, especially when there is no ML expert to develop the pipeline for constructing the KG because no one-fits-all ML pipelines are available. Ultimately, each method has advantages and disadvantages, which should be carefully considered for the appropriate selection of construction techniques.

3.2.4 Evaluation of KG

In the era of big data, there is a proliferation of many KGs that pose questions concerning the quality of embedded data (knowledge), i.e., entities and relations where they correctly represent and convey the intended real-world concepts. Thus, it is important to ascertain the completeness and correctness of the constructed KG using the appropriate evaluation

methods to determine whether it fits the intended purpose, use, and application. Some available evaluation measures include Hits@N and Mean Reciprocal Rank (MRR), Accuracy, Precision, Recall, and F- score, summarized in Table 2.

Table 2. Summary of KG evaluation metrics (Abu-Salih, 2021)

Metric	Description	Formal definition	
Hits@N	Indicates the number of elements in the ranking vector retrieved from the model is positioned in the top (N) locations.	$Hits@N = \begin{cases} \sum_{i=1}^{ Q } 1, & \text{if } rank(s, p, o)_i \leq N \\ 0, & \text{otherwise} \end{cases}$	where $rank(s, p, o)_i$ refers to the rank of a positive element i against a list of negative elements, T is a set of test triples and (s, p, o) is a triple $\in T$.
MRR	is a function that computes the mean of the reciprocal of elements embodied in a vector of rankings. It is used as a measure to evaluate the system performance against the retrieved elements.	$MRR = \frac{1}{ T } \sum_{i=1}^{ Q } \frac{1}{rank(s, p, o)_i}$	
MR	refers to the mean rank of the correct test facts/triples embodied in a vector of rankings (i.e., the average of the predicted ranks).	$MR = \frac{1}{ T } \sum_{i=1}^{ Q } rank(s, p, o)_i$	
Accuracy	specifies the accuracy of the incorporated embedding model in making a correct prediction. Accuracy is the ratio obtained between the accurate predictions (i.e., $TP + TN$) and the overall inferred predictions ($FN + TP + FP + TN$).	$Accuracy = \frac{TP + TN}{FN + TP + FP + TN}$	True-Positives (TP): refer to the true facts that are classified by the model as true statements. False-Positives (FP): refer to the false facts that are classified incorrectly as true statements. True-Negatives (TN): refer to the false facts that are classified correctly as false statements. False-Negatives (FN): refer to the true facts that are classified incorrectly as false statements.
Precision	refers to the proportion of those facts that were classified accurately as positive and they are actually positive.	$Precision = \frac{TP}{TP + FP}$	
Recall	recall indicates the proportion of true positive facts were correctly classified as positives facts.	$Recall = \frac{TP}{TP + FN}$	
F- measure	(a.k.a. f -measure/ f -score) is a harmonic measure used to provide a trade-off between precision and recall	$F - score = \frac{Precision \cdot Recall}{Precision + Recall}$	

3.2.5 Application of KG

KG is a flexible and powerful approach to knowledge representation and utilization in various domains. Consequently, its applications are numerous diverse, and encompass many industries. Figure 8 illustrates some major areas where KGs are being applied.

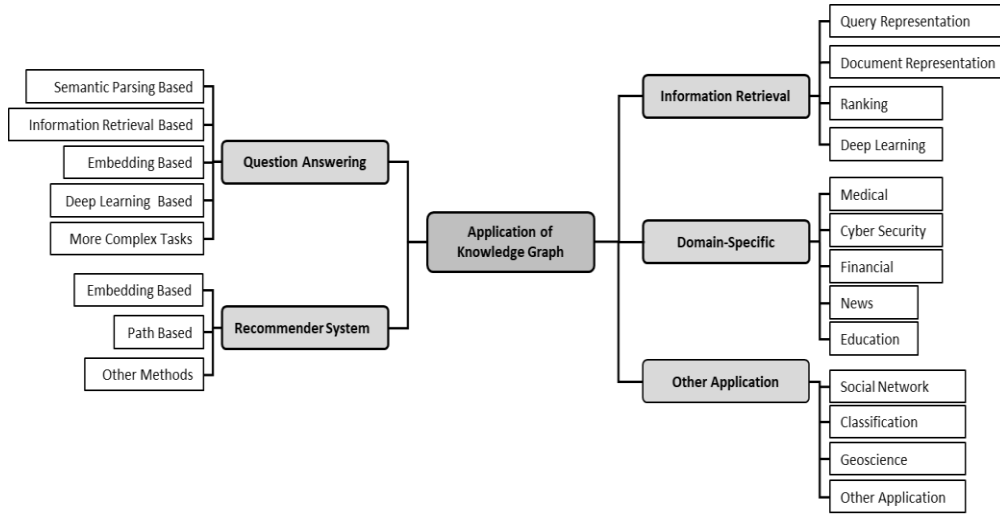


Figure 8. Key applications of knowledge graphs (Zou, 2020)

Furthermore, since KG was officially used by Google in 2012 to aid intelligent search engines, its application has grown in various industries for semantic search, intelligent question answering, personalized recommendation, intelligence analysis, fraud detection, and other fields (Abu-Salih, 2021; L. Li et al., 2020). For example, the medical field has used KG for the improvement of health delivery through the development of various intelligent systems used for supporting, enriching, and accelerating clinical decision-making (Santos et al., 2020), discovering associations between risk factors and breast cancer (Daowd et al., 2021), inferring possible diseases and recommending medical orders and assessing and preventing the risk of prostate cancer (PCa) associated with a

particular lifestyle (F. Chen et al., 2022).

Furthermore, in tourism, KG was adopted in the development of question-answering systems (L. Yang et al., 2020), touristic recommender systems for personalized attractions recommendations (Lu et al., 2016), and the best accommodation prices (Kärle et al., 2018). Moreover, the information communication technology (ICT) sector has used KG to solve the communication gap between Internet of Things (IoT) devices by developing a multilayer IoT middleware for IoT devices based on IoT KG (Xie et al., 2021). Likewise, KG has been utilized in finance to support AI-based systems for detecting online fraud (Wang et al. 2020) and predicting stock prices based on financial news (Liu et al. 2019). In the same way, KG has been used in traffic and transportation management to support the development of an urban traffic prediction model based on a deep Spatio-temporal convolution neural network (CNN) model (G. Zhou & Chen, 2019). Equally, J. Liu et al. (2021) proposed a KG-based approach to explore and understand railway operational accidents to reveal the potential accident rules and knowledge while contributing to preventing accidents.

Correspondingly, in construction research, few studies have proposed several ontology-based methodologies (a similar concept to KG) for safety risk identification (Xing et al., 2019; Zhang et al. 2020), job hazard risk identification (Fang et al., 2020) and decision support systems (Zaouga et al. 2021) for construction projects. Other construction-related studies have applied the KG technique for traffic management (G. Zhou & Chen, 2019), accident control (J. Liu et al., 2021), effective risk management, and reliable decision-

making in construction projects (Isah & Kim, 2023). However, there is a need to develop more domain-specific KGs for the subdomains of the construction engineering and management domain to conceptualize and test the potential applicability of KG in facilitating the development of different AI-based systems that can be used to enhance efficiency, productivity, and project performance. Therefore, this study aims to investigate the applicability of KG in aiding the development of an intelligent question-answering system for construction risks (QAR-System) using risk KG (RiskKG) to facilitate automatic risk information retrieval to support a comprehensive and dynamic risk identification in construction projects.

3.3 Question Answering

3.3.1 Introduction

Question-answering (QA) is a classic natural language processing (NLP) task. In the era of big, the internet is flooded with a sea of information. Search engines are useful tools that assist us in sailing through the information but with limited capabilities and applications. However, search engines can only present the user with a ranked list of relevant documents extracted from the web of documents so that the user examines each document in order to get the desired information. This method of information retrieval is time-consuming and labor-intensive. Therefore, an automatic answer retrieval system is crucial to save time and effort for information retrieval. Question-answering (QA) systems were introduced to mitigate these problems. QA is concerned with developing systems that can provide answers automatically to questions asked by humans in natural

language.

QA research is an intersection of natural language processing (NLP), information retrieval (IR), knowledge representation, machine learning (ML), and semantic search (Zope et al., 2022). Also, QA is an AI concept that enables interaction between humans and machines (computers) using natural language. Contrary to a search engine concept, a question-answering system (QAS) provides concise answers to the user's questions instead of returning a list of relevant documents containing the answers. Besides, (Ojokoh Adebisi, 2019) stated that QAS automatically provides solutions to queries expressed in natural language. Ultimately, the QAS aims to offer users succinct answers to questions posed by the users in natural language.

3.3.2 History of Question-Answering Systems

The early QASs developed belonged to the restricted or closed domain category; the prominent ones were BASEBALL and LUNAR (Mishra & Jain, 2016). Green et al. (1961) introduced BASEBALL, which answered questions and provided information about the American baseball leagues for a particular season. Similarly, Woods (1975) proposed that LUNAR could answer questions about the geological analysis of rock samples from the Apollo lunar missions. In 1971, LUNAR was demonstrated at a lunar science conference and responded correctly to 90% of the questions posed by users with knowledge about the system development. BASEBALL and LUNAR were based on a core database or knowledge base hand-crafted by domain experts. They have exceptional language abilities similar to the first chatterbot programs (i.e., ELIZA and DOCTOR) (Wikipedia).

Moreover, in the 1980s, QA systems with text comprehension capabilities were developed. An example of such a system was the Unix Consultant (UC) developed by *Rober Wilensky*. UC was developed based on a comprehensive, hand-written knowledge base of its domain. The system can answer questions on the Unix operating system and accommodate various user questions. Further, in the 1990s, there was a significant advancement in the development methods and techniques of QASs, leading to improved systems. For example, in 1993, Boris Katz introduced the first web-based QAS, START, and has been running since then. Correspondingly, competitions such as the Text Retrieval Conference (TREC), Stanford Question Answering Dataset (SQuAD), etc., were launched to advance the field of question further answering.

3.3.3 Question-Answering System Architecture

As illustrated in Figure 9, a typical QAS consists of three major modules: question processing, document processing, and answering processing.

The main function of the question (query) processing module is to analyze and process the questions by creating representations of the requested information. The system first analyzes the questions to appropriately represent the required information to answer the user's questions. Then, the question type is classified to ensure that relevant data/information is correctly extracted for answer verification. The common question types are *what*, *why*, *who*, *how*, *when*, *where*, etc. Once the question has been analyzed and classified, the question is then reformulated to facilitate question phrasing and

transformed into executable queries for information retrieval in the document processing module.

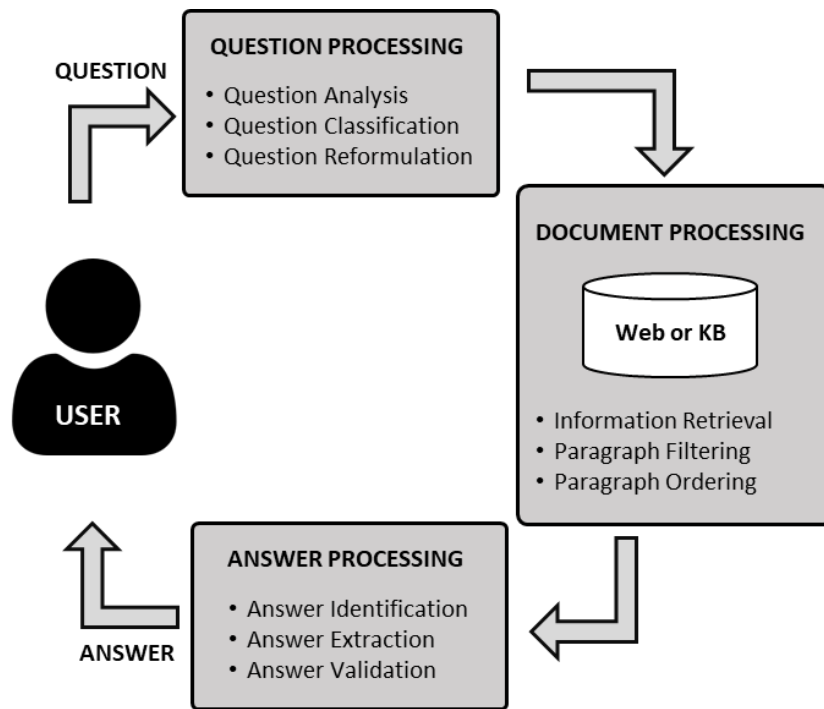


Figure 9. A typical question-answering components

The objective of the document processing module is to create a set of candidate-ranked paragraphs that contain the answers to the user's questions. This module is also known as the paragraph indexing module, where a ranked list of relevant documents is retrieved based on the reformulated question from the preceding module. Specifically, this module retrieves ranked documents relevant to the question posed. Then, the retrieved documents are filtered to reduce the number of candidate documents to the most relevant ones. Next,

the candidate paragraphs are ordered based on the degree of plausibility of the correct answers contained in the candidate paragraphs. In addition, this module uses one or more information retrieval systems for effective information-gathering from document corpora. This module ensures that only the most relevant and reasonable number of paragraphs are passed to the answer processing module.

The first step in the answer processing module is to identify correct answers based on the answer type determined in the question processing module. The parsers for named entity recognition and part-of-speech tagger are used to complete this step. The second step involves extracting the correct answer from the candidate's answers in the paragraphs. Also, parsers are employed in this step based on a set heuristic to extract only relevant words or phrases that answer the user's question. Next, answer validation is carried out, and it involves checking the correctness of the answer before presenting it to the user. Usually, lexical resources such as *WordNet* validate that a candidate's answer is from the correct answer type. Further explanation of each module of the system can be found in (Allam et al., 2012). Finally, it is important to note that the architecture presented here consists of the generic components of a simple QAS. Thus, the QAS architecture can vary depending on the type and intended use of the system. The next sub-section discusses the classification of QASs.

3.3.4 Classification of Question-Answering Systems

Question-answering research intends to develop approaches to answering a wide range of question types, including fact, list, definition, how, why, hypothetical semantically

constrained, and cross-lingual questions (Wikipedia). Many existing studies have classified question-answering systems depending on the point of view and the criteria. For instance, (Mishra & Jain, 2016) presented eight criteria for the classification of QASs, including (1) application domains for which QASs are developed, (2) types of questions asked by the users, (3) types of analyses performed on users' questions and source documents, (4) types of data consulted in data sources, (5) characteristics of data sources, (6) types of representations used for questions and their matching functions, (7) types of techniques used for retrieving answers, and (8) forms of answers generated by QASs. Also, (Ojokoh & Adebisi, 2019) classified QAS based on domain, question type, data source, knowledge base, multiple information sources, forms of answers generated, language paradigm, and approaches adopted. This study adopted the classification presented in (Pandya & Bhatt, 2021), which is summarized in Figure 10.

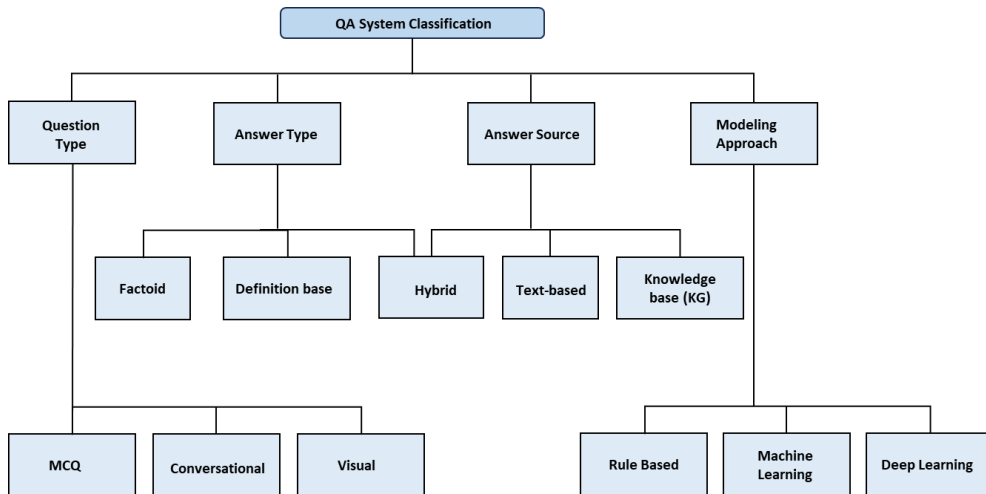


Figure 10. Classification of QAS

As shown in Figure 10, the broad classification of the QAS is based on (1) the question type, (2) the answer type, (3) the answer source, and (4) the modeling approach.

The classification based on the question type is divided into multiple choice questions (MCQ), conversational, and visual QA. The MCQ QA can answer the questions with multiple options and is challenged with eliminating the wrong option. In conversational QA, the conservation history plays an important role in directing the user to answer the question, which cannot be achieved by looking at the context of the question alone. The expected answer from the system can be a factoid, a definition, or complex questions. The last of this category is visual question-answering (VQA), which can answer questions from visual content such as pictures and videos. VQA is developed based on NLP and computer vision techniques. Usually, convolution neural networks (CNN) and recurrent neural networks (RNN) are used to map the images and questions features to common feature space. VQA can answer factoid, non-factoid, yes/no, etc., question types.

Another classification of the QASs is based on the type of answer expected. Thus, QASs are based on answer type, including factoid, definition base, and hybrid. The factoid type answers questions about simple facts with a few words (e.g., What is the capital of South Korea? Answer is Soeul). This system provides knowledge-based answers using inference that require aggregation from multiple information sources. The definition-based QA extracts short or long answers from a knowledge base to respond to user questions. For example, Q: *What is a chatbot?* A: *a computer program designed to simulate conversation with human users, especially over the internet.* Some techniques

that can be used to develop this type of QAS include but are limited to TF-IDF, BLSTM, BERT, GPT, etc. As for the hybrid QA, questions such as “What is the second largest planet in the solar system?”. This question can be answered in both a factoid and a non-factoid way, depending on the type of knowledge base/graph adopted in the system. Multiple query graphs are first generated for the question if a knowledge graph is deployed in the system. Then, semantic similarities between the question and each query graph are determined using a deep neural network, and the best candidate graph that can predict the correct answers is selected.

The category of QAS based on data sources includes text-based, knowledge-based, and hybrid. Text-based, sometimes called information retrieval-based (IR-based) QA, relies on the huge amount of information that can be accessed on the web as text format or ontologies. This QAS answers users’ questions by retrieving text snippets from the document corpora or Web. CNN, Bi-LSTM, Bi-directional GRU, and other deep neural networks can be adopted to improve text matching and retrieval. The knowledge base QAS answers questions by mapping the question (query) over a predefined ontology such as DBpedia, Freebase, etc. The data source of the knowledge base QAS can be simple or complex in structure, requiring complex logical queries (e.g., SQL or CQL). The methods employed for the mapping include rule-based, supervised, and semi-supervised methods. The hybrid QAS combines the features of text-based and knowledge-based QA to provide answers to user’s queries.

The classification based on the modeling approach includes rule-based, machine learning,

and deep learning methods. The rule-based QASs involve designing a hand-crafted logical rule for obtaining the correct answers based on NLP tasks such as tokenization, POS tagging, semantic class tagging, entity recognition, etc. In the machine learning approach model, such support vector machine (SVM), decision tree (DT), naïve Bayes (NB), etc., are used for sentence parsing and question classification to get the most accurate results. The advancement in the NLP field has given rise to the development of powerful deep learning models like RNN, BERT, GPT, etc., that have been used to improve the performance of QASs. The survey on QAS classification has been documented in (Allam et al., 2012; Mishra & Jain, 2016; Ojokoh & Adebisi, 2019; Pandya & Bhatt, 2021; Zope et al., 2022).

3.3.5 Knowledge-based QAS

As explained in the previous section, the knowledge base QAS (KBQAS) is developed based on structured knowledge bases or ontologies that are either opened (DBpedia, Freebase) or closed domain knowledge base. The tasks performed include question analysis, phrase mapping, disambiguation, query construction/generation, and querying distributed knowledge, as shown in Figure 11.

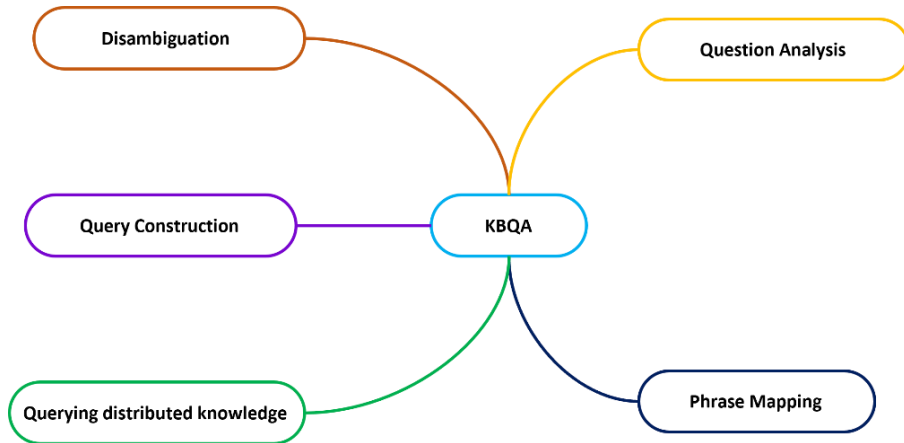


Figure 11. Task performed in KGQA

Question analysis involves interpreting and understanding the user's question to facilitate the retrieval of the correct information. Phrase mapping is the process of translating the key components of the user's natural language question into the corresponding entities and relationships found in the knowledge graph or structured database. Disambiguation is a crucial task in KBQA, which involves resolving ambiguities that arise when a word or phrase is associated with multiple meanings or entities in the knowledge base. Query generation involves converting the user's question in natural language to a logical or formal query that can be executed over the knowledge base to retrieve the answer. In addition, when the answer is retrieved from multiple knowledge sources, a query-distributed knowledge approach is used to integrate the knowledge bases and query this data effectively.

3.3.6 Evaluation of QAS

Evaluating a question-answering system is vital to assess how well the system meets the user's needs in providing accurate and relevant answers. The evaluation metrics can be categorized into quantitative and qualitative measures. However, the performance of the evaluation metrics depends on the type of QAS built and the component/part/module of the system to be evaluated. The commonly used evaluation metrics include Accuracy, Mean Reciprocal Rank (MRR), Precision, Recall, F1-Score, and Average Precision (AP). The mathematical expression for each evaluation metric can be found in Chapter 3 (section 3.2.4).

3.3.7 Application of QAS

QASs are utilized in various applications to improve information retrieval and user experience by providing succinct answers to user queries (questions). QAS is used in standard question-answering systems (i.e., web base QA), dialogue systems (i.e., chatbots like Siri and chatGPT), community QA (e.g., stack overflow), multimedia QA (e.g., mid-journey, DALL.E 2), reading comprehension QA, etc.

Further, QAS are powerful tools that can be integrated into various fields and industries to enhance information accessibility and decision-making processes. Therefore, researchers have made several efforts across different domains to develop many QASs, including but not limited to medical, finance, defense, tourism, legal, safety hazard management, and manufacturing. KBQAS in construction project management was

described in a previous study (Cheng et al., 2002). The proposed KBQAS framework employed ifcXML for knowledge representation, and the XML query engine was utilized for information query. The authors evaluated the system using selected questions relating to the residential building project, and the result obtained demonstrated reasonable accuracy.

Moreover, Chi Minh et al. (2021) developed a Vietnamese tourism QAS that can be used to search tourist attraction locations in Vietnam with detailed information about the place. The QAS used a deep learning model to generate simple NL answers for simple questions, while a knowledge graph was used to generate an NL answering list. The authors reported an 83% F1 score and 87% precision based on the test set, indicating high performance. In addition, Gupta et al. (2021) presented a template-based QAS for answering questions about Indian missiles. The study used a Missile KG consisting of 177 entities and 400 relationships and a query-answering model to develop QAS. The highest accuracy obtained in the performance evaluation of the QAS is 0.9786 for precision, 0.9898 for recall, and 0.484 for the F1-score. Further, another KBQAS based on a knowledge graph (Marie) was developed by X. Zhou et al. (2022) for answering questions and automatic information retrieval in the chemistry domain. Recently, Martinez-Gil (2023) presented a survey of QASs used in the legal domain. The author pointed out that adopting QAS will aid the automatic retrieval of legal information in seconds using NLQs, thus saving resources such as cost and time.

Moreover, a typical KBQAS architecture consists of the knowledge base, question pre-

processing, question classification, query generation, and answer generation modules (Pereira et al., 2022; Wu et al., 2019). Previous studies have introduced several techniques to improve each component of KBQAS. However, in recent studies, deep learning models have been employed to enhance the performance of KBQASs (Stroh & Mathur, 2016; C. Zhang, Lai, et al., 2021; Zhong et al., 2020). Specifically, recent studies have introduced deep learning models with transformer layers (i.e., large language models such as BERT, GPT, etc.) in KBQASs to improve their performances (Abbas et al., 2023; Kim et al., 2022; Qiu et al., 2023; Tian et al., 2023). However, in the construction domain, only a few existing studies have adopted the GPT models in the development of QAS or similar other applications. Hence, this study proposes adapting OpenAI's GPT-3 model for query generation in QAR-System. The GPT-3 model was selected because it supports query generation for structured query language (SQL), SPARQL, Cypher query language (CQL), etc., based on natural language prompts.

CHAPTER 4. GENERATIVE PRE-TRAINED TRANSFORMERS

4.1 Introduction

This chapter explains the generative pre-trained transformer model, its general, and its application in the construction domain. A brief description of the most popular conversational AI tools powered by large language models, i.e., chatGPT, Bing chat, and HuggingChat, was given. The chapter concluded by explaining the hallucination phenomenon and how to overcome it when adapting large language models for specialized use.

4.2 Overview

Generative artificial intelligence (GenAI) is artificial intelligence (AI) capable of generating texts, images, videos, etc., using generative models. The generative models are transformer-based deep-learning neural networks that can accept natural language as input. A good example of GenAI is the generative pre-trained transformer (GPT). GPT is a large language model (LLM) designed based on deep neural networks that utilize a transformer architecture. LLMs are trained with a large amount of data using self-supervised and semi-supervised learning. Figure 12 illustrates the architecture of a transformer and GPT.

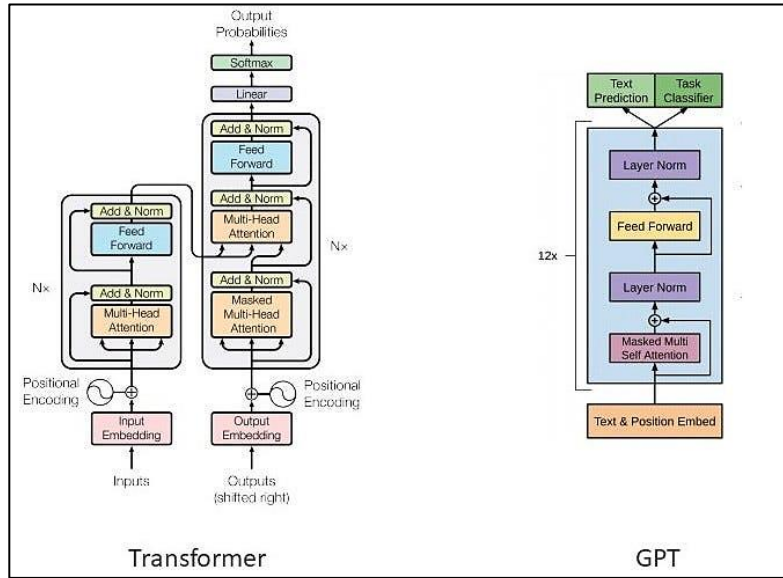


Figure 12. Architecture of Transformer and GPT model

The transformer networks, feedforward neural networks, and attention processes constitute the building blocks of GPTs. The transformer-based neural architecture layer of GPT uses attention techniques to concentrate on a particular area of the text input. Then, the model can pick up on the statistical patterns in natural language traceable to its architecture without requiring explicit hand-crafted grammar rules or syntax. The main function of the transformer is to create a coherent and fluent output, while the attention mechanism allows the model to focus on a particular portion of the input text.

GPT is a category of NLP directed toward understanding, interpreting, and generating natural language. The research area of generative pretraining has been established in machine learning applications (Deng, 2014; Hinton et al., 2012). However, the first transformer architecture was developed in 2017 at Google (Vaswani et al., 2017). Since

then, there has been the emergence of different LLMs, such as Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) and XLNet (Z. Yang et al., 2019), which were pre-trained transformers but without generative capabilities.

In 2018, OpenAI introduced its first GPT foundational model with generative capability (Alec et al., 2018). The GPT foundation model is an AI model trained on extensive and broad-scale data that can be adapted for various tasks. This model had 117 million parameters and was a huge step toward natural language understanding and generation. In 2019, the second version of GPT (GPT-2) was released by OpenAI. GPT-2 was trained on 10 billion tokens and had 1.5 billion parameters. It attracted global attention as it can generate coherent and compelling text passages. GPT-3 was introduced in 2020 and had more training parameters and capabilities than the previous two. GPT-3 had 175 billion training parameters and was trained on 499 billion tokens. The current version, GPT-4, was released in 2023 and has more advanced features than the previous versions (OpenAI, 2023). Table 3 shows the progression of GPT model development from GPT-1 to GPT-4.

Table 3. Progression of GPT development from GPT-1 to GPT-3 (Saka et al., 2023)

Date	Milestone
11th June, 2018	OpenAI introduced GPT-1, the first model in the GPT series.
14th Feb., 2019	OpenAI announced the release of GPT-2.
28th May, 2020	OpenAI published the initial GPT-3 preprint paper on arXiv.
11th June, 2020	OpenAI launched a private 1beta for the GPT-3 API.
22nd Sep, 2020	OpenAI licensed GPT-3 to Microsoft.
18th Nov, 2021	The GPT-3 API was opened to the public.
27th Jan 2022	OpenAI released InstructGPT as text-DaVinci-002, later renamed GPT-3.5.
28th July, 2022	OpenAI published a paper on exploring data-optimal models with FIM.
1st Sep, 2022	OpenAI reduced the pricing of the GPT-3 model by 66% for the DaVinci model.
21st Sep, 2022	OpenAI announced Whisper (speech recognition).
28th Nov, 2022	OpenAI expanded GPT-3.5 to text-davinci-003 with improved language generation capabilities.
30th Nov, 2022	OpenAI announced ChatGPT.
14th Mar 2023	OpenAI released GPT-4, the latest and highly anticipated addition to the GPT series.

4.3 Applications of GPT

Some of the capabilities of GPT models include but are not limited to text/code completion, chat completion, language understanding, image generation, etc. As a result, they have been successfully utilized in healthcare, bioinformatics, education, business, and commerce for various applications such as electronic records analysis, medical diagnosis, genome analysis, proteomics, flexible learning, real-time feedback, fraud detection, financial analysis, forecasting, chatbots, virtual assistants, etc., (Li et al., 2023, Zhang et al., 2023, Firat, 2023 Zong & Krishnamachari, 2022).

Furthermore, one of the main advantages of the GPT models is their ability to generate language that is coherent, fluent, and nearly indistinguishable from texts produced by humans due to being trained on vast amounts of unstructured text data (Saka et al., 2023).

Also, the quality and accuracy of the text generation accomplished for tasks such as language modeling, text generation, and question-answering have improved exponentially. These tasks are accomplished by fine-tuning the GPT model or prompt engineering. For example, the GPT-3 model can be fine-tuned with small data or prompt with NL to perform domain-specific tasks such as text categorization, sentiment analysis, reading comprehension, code generation, chatbot completion, etc.

Fine-tuning is a method by which GPT models can be adapted for a specialized task or domain. It involves further training GPT on a specific small dataset to customize its knowledge or skills in a particular domain. On the other hand, prompt engineering entails structuring text that can be interpreted and understood by a GenAI model. In other words, prompt engineering is formulating prompts to maximize the effectiveness, relevance, and accuracy of GPT models. A prompt is natural language text describing what an AI should perform (Wikipedia). Prompt guides and optimize the responses from LLM like GPT-3 or GPT-4. This study adopted prompt engineering to aid the conversion of natural language questions to logical queries, i.e., Cypher query, to facilitate risk information extraction from the risk knowledge graph. The current areas of applications of GPT models in the construction industry are explained in the next subsection.

4.4 Application of GPT Models in the Construction Domain

Despite the great capabilities and possibilities of GPT models overcoming the impediments of adopting AI-based solutions in the construction industry. However, only some studies in construction literature have applied the GPT models as they are still new

to the industry, unlike other LLMs like BERT. For example, in construction project scheduling, (Prieto et al., 2023) demonstrated the applicability of the GPT-3.5 model (ChatGPT) in automating construction scheduling based on natural language prompts. ChatGPT was used in the study to generate a simple construction project schedule using natural language prompts. The pool of experts evaluated the construction schedule generated, and the results revealed that ChatGPT can assist in generating a coherent schedule automatically, therefore reducing time expended on traditional scheduling methods. Similarly, (Amer et al., 2021) adapted the GPT 3.5 model to develop a model to aid the automatic matching of look-ahead plans to master schedules of a construction project in order to generate lists of tasks based on natural language prompts. The validation result (i.e., the precision of 76.5%) obtained shows that the technique can assist the project planners in automatically matching the look-ahead plan with the master schedule during project execution.

Furthermore, to resolve the problem of excessive time and cost, inaccuracy, inefficiency, and instability associated with the manual compliance checking process during the execution of building construction projects, (X. Liu et al., 2023) propose automatic compliance checking (ACC) using the GPT model. The authors adapted the GPT-3.5 model through prompt engineering to generate compliance results based on building design specifications, corresponding codes, and task-specific natural language prompts. Also, (Zheng & Fischer, 2023) introduced building information modeling (BIM)-GPT to enhance the efficiency of information retrieval from building information models (BIMs).

Specifically, the authors integrated a prompt-based virtual assistant (VA) with BIM and GPT-3.5-turbo to support NL-based information retrieval. The BIM-GPT framework was validated using a hospital building project and achieved accuracy rates of 83.5% and 99.5% for NL queries with no data and 2% data incorporated in the prompts, respectively.

Moreover, (Saka et al., 2023) conducted a review and highlighted thirty-eight (38) possible areas of application GPT models covering the entire lifecycle of a construction project from pre-design, design, construction, operation, and maintenance to demolition stages. In addition, the authors explained six other areas that are not specific to a particular phase of the construction project lifecycle that can benefit from the GPT models, including knowledge management and training, customer services, stakeholder communication, business intelligence, and conversational AI/chatbots. The study concluded by demonstrating the use of GPT-3.5-turbo in developing a material selection and optimization model for building projects. Table 4 below summarizes the current applications of GPT models in the construction domain.

Table 4. Current applications of GPT models (Saka et al., 2023)

S/N	Study	Purpose	GPT model utilized
1	Amer et al. (2021)	Automated linking of short-term and long-term schedules in construction	GPT-2
2	Liu et al. (2023)	Automated compliance checking in AEC	GPT-3.5
3	Prieto et al. (2023)	Construction tasks scheduling	GPT-3.5
4	You et al. (2023)	Automated sequence planning of construction tasks in robot-enabled construction assembly	GPT-4
5	Zheng & Fischer (2023)	Information retrieval from BIM using natural language	GPT-3.5-turbo
6	Saka et al., (2023)	Review and case study on GPT model application in the Construction industry	GPT-3.5-turbo

Although the existing studies have demonstrated the adaptability of various GPT models in solving scheduling problems, compliance checking, material selection/optimization, and information retrieval. However, the existing studies focused mainly on building construction projects tackling specific problems in the project. There is no existing literature that has demonstrated the adaptability of GPT or other LLMs in construction project risk management. Therefore, this current research differs from the existing studies as it focuses on addressing the inefficiency associated with manual risk identification practice in complex civil engineering projects such as Tunnels. Specifically, this study attempts to adapt the GPT-3 model (text-daVinci-003) through prompting to generate Cypher queries for an intelligent question-answering system to retrieve succinct risk information from the KG using natural language.

In a nutshell, this study intends to explore the adaptability of the GPT-3 model to support automatic risk identification for construction projects at the preconstruction stage. Thus, this research contributes to the body of knowledge by offering a practical workflow for researchers and practitioners seeking to leverage KG and GPT models in comprehensive risk identification and management for construction projects to facilitate reliable decision-making. Moreover, it is expected that this study will birth an intelligent tool (i.e., QAR-System) that can assist the project manager in automated and timely retrieval of risk information to enhance efficient identification and management of construction project risk at the preconstruction stage while facilitating informed decision-making that will lead to project success. Finally, this study is expected to mitigate the challenges of adapting AI technologies for managing repetitive tasks in construction projects to save time and cost, improve efficiency and productivity, optimize overall project performance, and increase the chances of achieving project objectives.

4.5 Conversational AI Tools Powered by LLMs

The proposed QAR-System is an example of a conversational AI tool powered by an LLM. Thus, the following sections introduce some popular existing LLM-powered conversational tools.

4.5.1 ChatGPT

ChatGPT is an LLM-based chatbot developed by OpenAI and a sister model to *InstructGPT* that was trained on billions of tokens to follow instructions in a prompt and

produce a detailed response. ChatGPT, since its release in November 2022, has received widespread attention and interest globally, resulting in research harnessing its potential in various fields (Y. Liu et al., 2023). The free version of ChatGPT released was powered by GPT-3.5, enabling users to ask questions in any textual format, style, and language. Recently, OpenAI released a paid version called “*ChatGPT Plus*,” powered by GPT-4. The GPT-4 model is an improved version of GPT 3.5 and is currently OpenAI’s most advanced LLM that exhibits high capabilities in understanding and generating natural language. Also, it is a multi-modal model that can process text and image inputs seamlessly and produce text outputs. Figure 13 shows the user interface (UI) of ChatGPT.

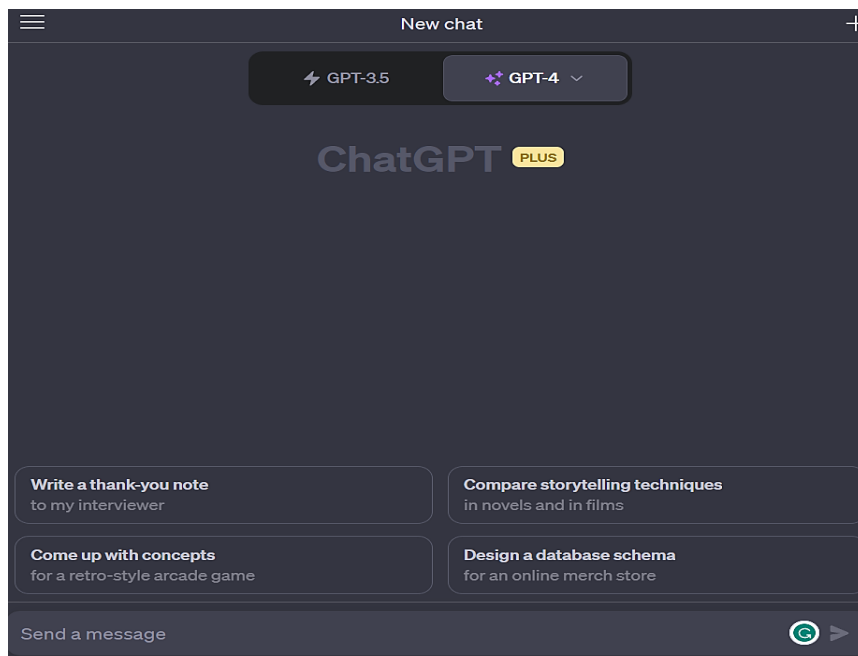


Figure 13. UI of ChatGPT Plus

4.5.2 Bing Chat

Bing Chat is a chatbot incorporated into the Microsoft Bing search engine. BingChat is powered by OpenAI's GPT-4 model. It has general and rich information on many subjects/topics and can answer general questions on various issues, including the risks of tunnel construction projects. The UI of Bing Chat is presented in Figure 14.

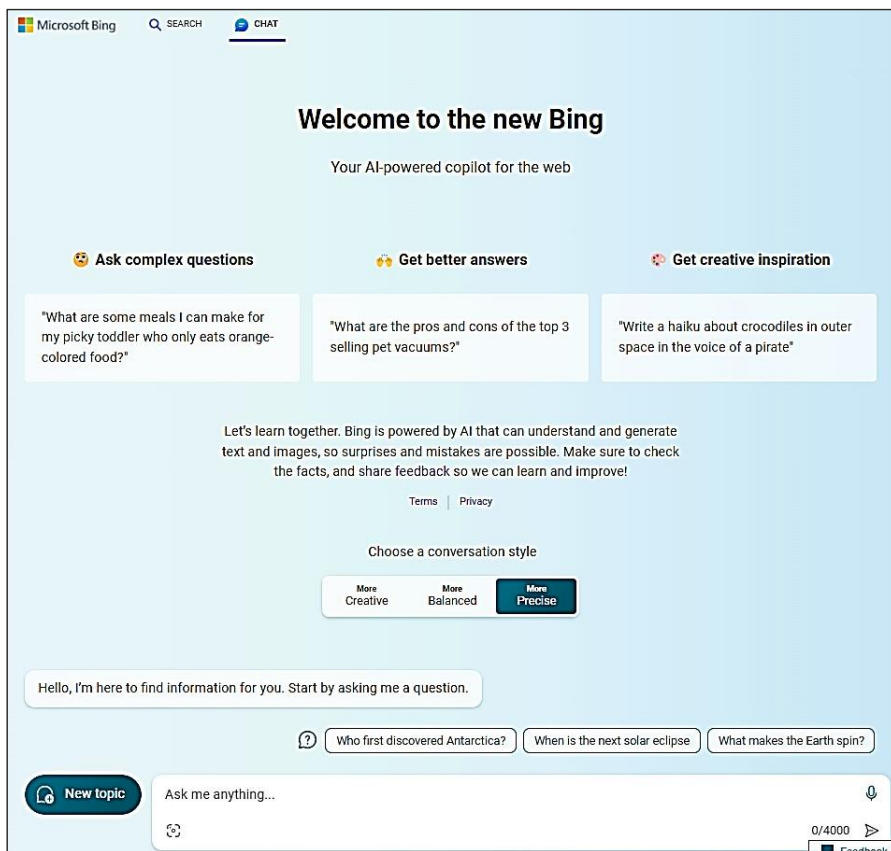


Figure 14. UI of Bing Chat

4.5.3 HuggingChat

HuggingChat is another conversational AI tool similar to OpenAI's ChatGPT, but open source hosted by HuggingFace. HuggingChat is powered by three different LLMs developed by Meta AI (Facebook), with the *Large Language Model Meta AI-2 (Llama-2)* having the greatest capability to perform various tasks. The Llama-2 model is an open-source LLM pre-trained with 2 trillion tokens, 70 billion parameters, and 4096 maximum token content length. Llama-2 can be used to develop Chatbots, code generation, QA systems, etc. Thus, the Llama-2 model was selected for the validation process of QASRisk. Figure 15 shows the UI of HuggingChat.

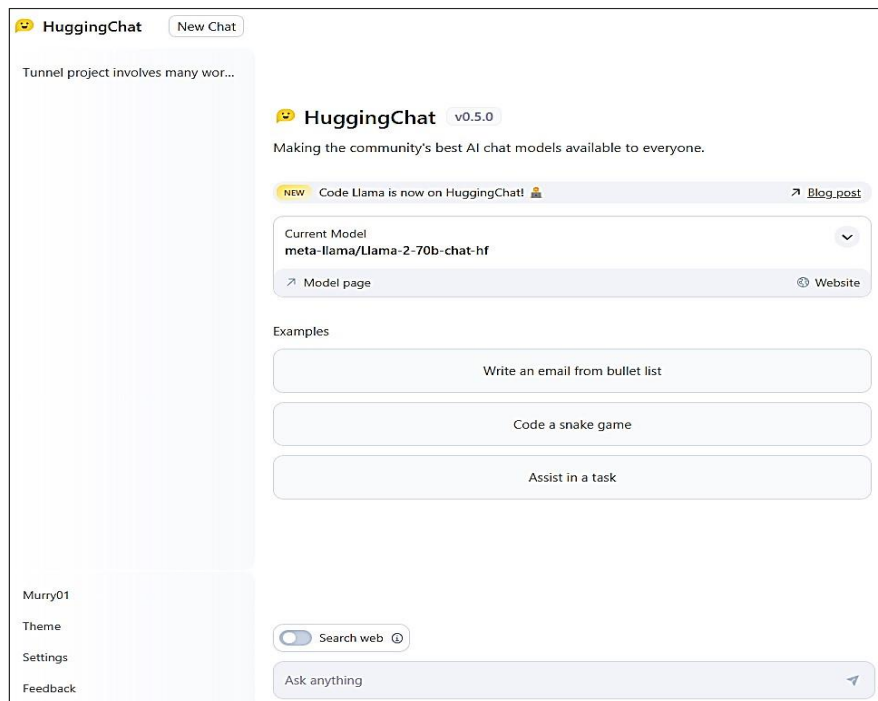


Figure 15. UI Hugging Chat

4.6 Hallucination in LLMs

As powerful as the LLMs are, they still have several limitations. One of the major limitations of LLMs is hallucination. Hallucination is the phenomenon of LLMs “generating unfaithful or nonsensical text” (Z. Ji et al., 2022). For instance, OpenAI acknowledges this limitation in the technical report of all its GPT models, including the newly released GPT-4 (OpenAI, 2023). Specially OpenAI states that “ChatGPT sometimes writes plausible-sounding but incorrect or nonsensical answers” and that great care should be taken when using LLM outputs. HuggingFace stated a similar limitation about HuggingChat. To address this problem, studies have suggested various ways of mitigating hallucinated texts in LLMs. For example, (Z. Ji et al., 2022) suggested manually building a faithful and clean dataset for training the LLMs. In other words, constructing structured databases can help resolve hallucinated texts, enabling the LLMs to respond concisely and accurately to users. This was demonstrated in this study using a knowledge graph (i.e., RiskG) to power the proposed QAR-System to ensure accurate and consistent response.

CHAPTER 5. QUESTION-ANSWERING SYSTEM FOR CONSTRUCTION RISK IDENTIFICATION

5.1 Introduction

This chapter gives a detailed explanation of the construction and evaluation risk knowledge graph (RisKG) for construction risk identification. The implications of the evaluation results were discussed. Further, this chapter explained the detailed process of developing the QAR-System. The chapter concluded with a discussion of the validation results of QAR-System.

5.2 Construction of Risk Knowledge Graph (RisKG)

The risk identification process in construction projects leverages experts' knowledge, stakeholders' experiences, and information gathered from various explicit sources. Thus, organizing risk-related knowledge gathered through experience from previous projects is crucial to developing a robust knowledge-based system (KBS). In other words, formalizing the knowledge acquired through experience can help develop a formidable KBS. Presently, there are no formalized risk knowledge bases in the construction domain that can support automatic risk identification and management for construction projects. Consequently, this section presents the methodology that utilizes KG to formalize and conceptualize risk-related knowledge for construction projects to aid the development of an intelligent KBRM tool. Specifically, the procedure for the construction of a risk knowledge graph (RisKG) that can support the proposed QAR-System is explained in

detail in this section.

5.2.1 Methodology

The risk knowledge graph (RisKG) construction is a fundamental step toward developing the proposed QAR-System because it uses the RisKG as the knowledge base to answer user questions precisely. Usually, domain-specific KG, like RisKG, is constructed using a top-down approach (Abu-Salih, 2021). Thus, the top-down approach was adopted in this study, and the process of constructing the RisKG is depicted in Figure 16, followed by a detailed explanation of the steps.

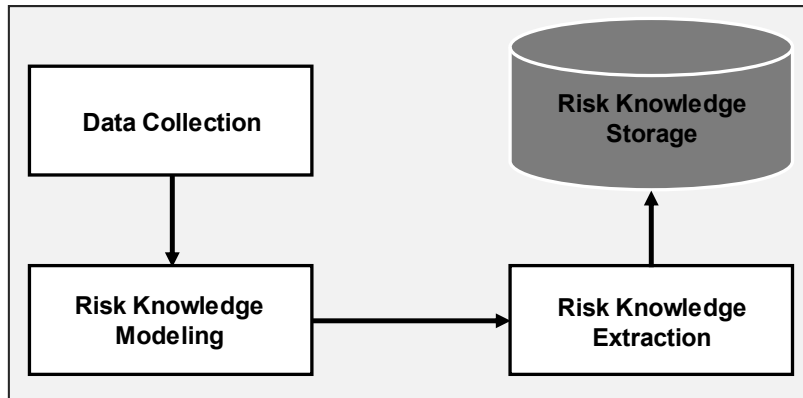


Figure 16. Process of constructing RisKG

5.2.1.1 Data Collection

Data collection plays a significant role in developing a robust KG and functional QAS. Thus, tunnel construction project data were collected for the RisKG construction through a literature review and experts' workshop in this study. Specifically, the New Austria Tunnel Method (NATM) tunnel was used as a case study. First, a comprehensive list of

potential tunnel project risks and risk factors was collected through a literature review of published articles on risk management of tunnel projects. Over 200 risks and about 450 risk factors related to NATM were identified through the literature review.

Second, an expert workshop was conducted to verify and validate the collected risk data. Five experts with 10-25 years of experience in tunnel design and construction and risk management were invited to the risk identification workshop. The experts reviewed the data collected. After extensive discussion and deliberation, 142 risks and 369 risk factors were agreed upon for developing the risk knowledge graph. Also, the experts classified 142 risks and 369 risk factors under internal risks for the tunnel project, while the remaining were grouped under external risks. Since this focused on the risks under management control, only internal risks were considered for the RisKG.

Figure 17 shows the risk breakdown structure (RBS) adopted at the workshop by experts. Four construction projects, including the tunnel, bridge, highways, and buildings, are shown in the figure focusing on the tunnel project. Six internal risk categories were identified, including design, construction, technical, safety, management, and environmental risks. On the other hand, four categories of external risks, economic, physical, political, and technological change risks, were identified, as shown in Figure 17.

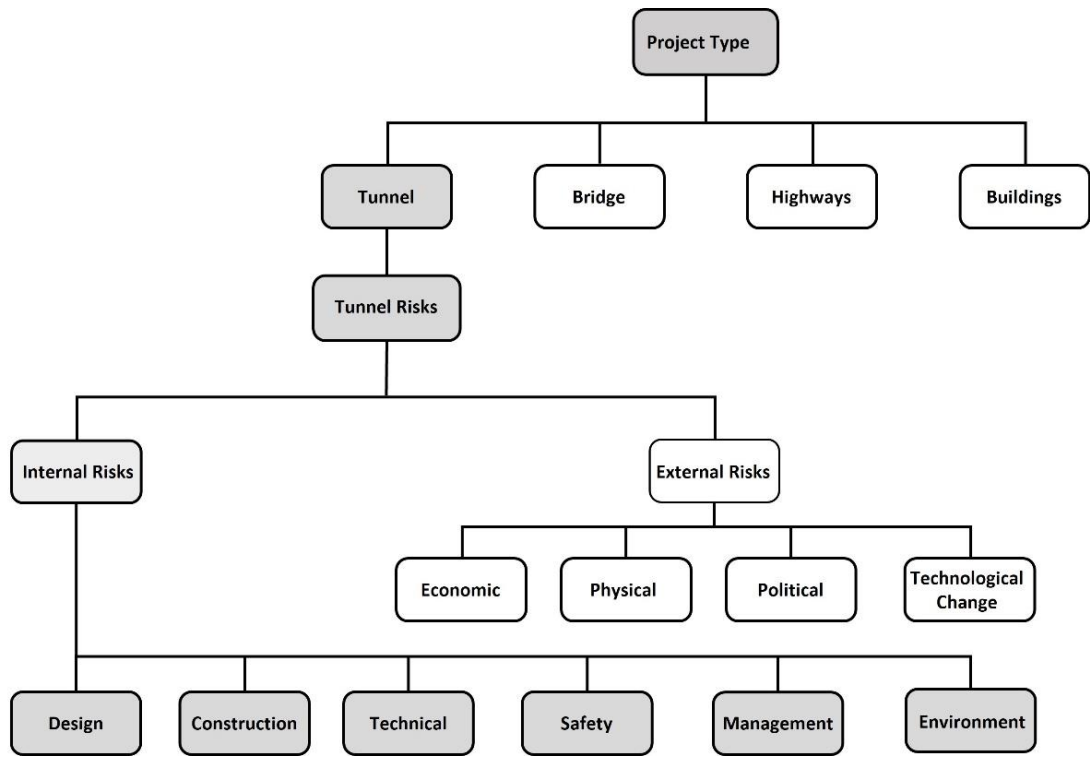


Figure 17. Risk breakdown structure (RBS) for NATM Tunnel

Furthermore, the experts determined the work breakdown structure (WBS) for the NATM tunnel and linked the risk categories, risks, and risk factors to each work package in the WBS. In summary, WBS comprises fifteen (15) works associated with tunnel construction, six (6) risk categories, 142 risks, and 369 risk factors were established. The established WBS-RBS for the NATM tunnel construction is shown in Figure 18, corresponding to the risk data verified by the experts at the workshop. Table 5 shows the excerpt of the data collected, while the complete data collected is shown in Appendix B.

Table 5. An excerpt of data for the NATM tunnel

Project	Work type (WBS)	Risk Category	Risk	Risk factor
1.0 Tunnel	1.1 Shotcrete Work	1.1.1 Construction	1.1.1.1 Arch formation difficulty	1.1.1.1.1 Long pouring time
			1.1.1.2 Cost increase due to excessive use of shotcrete material	1.1.1.2.1 Occurrence of over burrow
			1.1.1.3 Insufficient secondary support	1.1.1.3.1 Over pouring of shotcrete
			1.1.1.4 Defective shotcrete attachment	The thickness of one casting is thick (within 100mm) Foreign matter exists on the surface of the 1st shotcrete
				1.1.1.4.1 Insufficient cleaning of the pouring surface before construction (failure to remove floating stones and pumice stones, etc.)
			1.1.1.5 Collapse	1.1.1.5.1 Shotcrete pouring delay
		1.1.1.6 Environmental problems		1.1.1.6.1 Insufficient rebound material treatment
				1.1.1.6.2 Environmental problems arise due to inability to separate waste from shotcrete sealing
		1.1.2 Safety	1.1.2.1 Occupational accidents of workers	1.1.2.1.1 dust penetration into human body
				1.1.2.1.2 Overloaded connections due to clogged shotcrete nozzles
		1.1.3 Technical	1.1.3.1 Defective shotcrete attachment	1.1.3.1.1 Lack of steel fiber content and length
				1.1.3.1.2. Failure to establish appropriate drainage measures for the water generating section
			1.1.3.2 Insufficient shotcrete strength	1.1.3.1.3 Poor quality of quick payment
				1.1.3.2.1 Poor aggregate granularity
			1.1.3.3 Increased shotcrete disposal cost	1.1.3.3.1 Increased rebound
				1.1.3.3.2 Defective shotcrete material (sand with a lot of soil, problems with quick fixing, etc.)
				1.1.3.3.3 Non-compliance with shotcrete placement angle

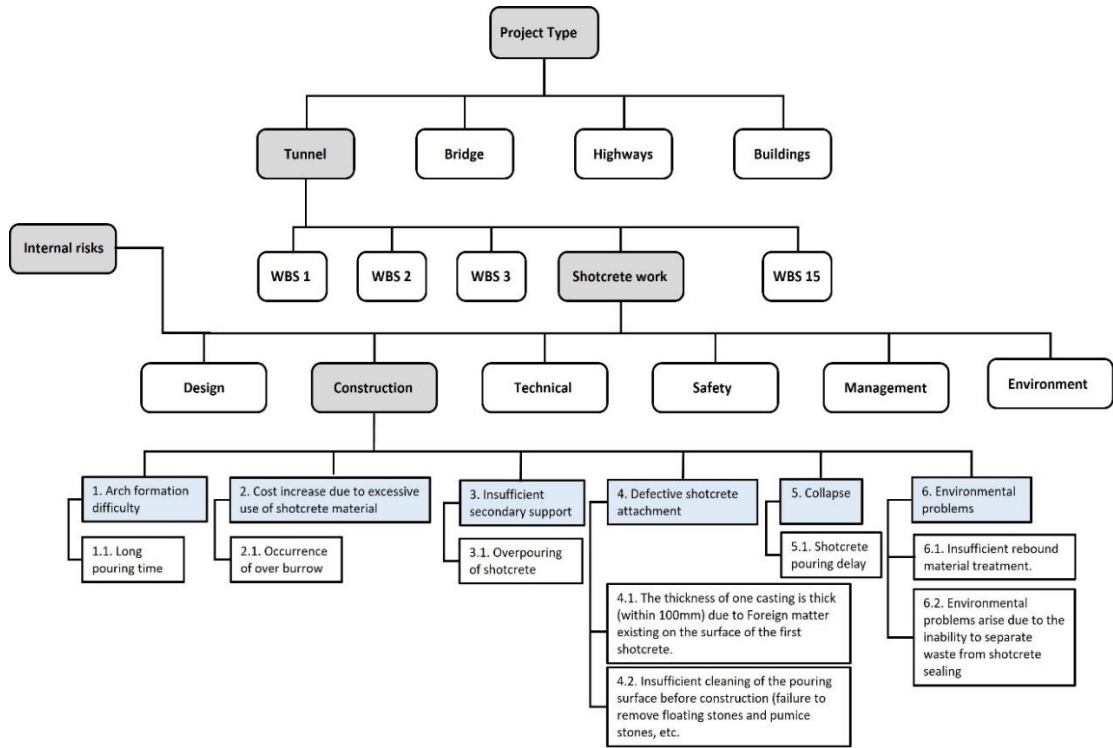


Figure 18. WBS-RBS for NATM tunnel

5.2.1.2 Risk Knowledge Modeling

Risk knowledge modeling is important in constructing KG because it establishes the model and methods for machine reasoning, the data structure for efficient computing, and the interface between machine and human understanding. In this step, the ontology and data model are defined. The ontology definition involves identifying the graph entities, the relationships, and their attributes. Thus, based on the data collected in section 5.2.1.1, the RiskKG entities, attributes, and relationships were defined. The nodes defined include

project name (Tunnel), work packed (WBS), risk, and risk factors. Also, the relationships defined include `has_work_type`, `has_risk`, and `has_riskfactor`. The attributes corresponding to the entities identified are name and riskcategory. Further, the graph data was modeled based on the predefined RisKG ontology.

Data modeling is the process of determining the structure of the database to ensure the usability, scalability, expansion, and accurate performance of the RisKG. The output of this process is the ontology model for the RisKG depicted in Figure 19. The RisKG ontology consists of four entities (nodes) and three edges (relationships) between the nodes.

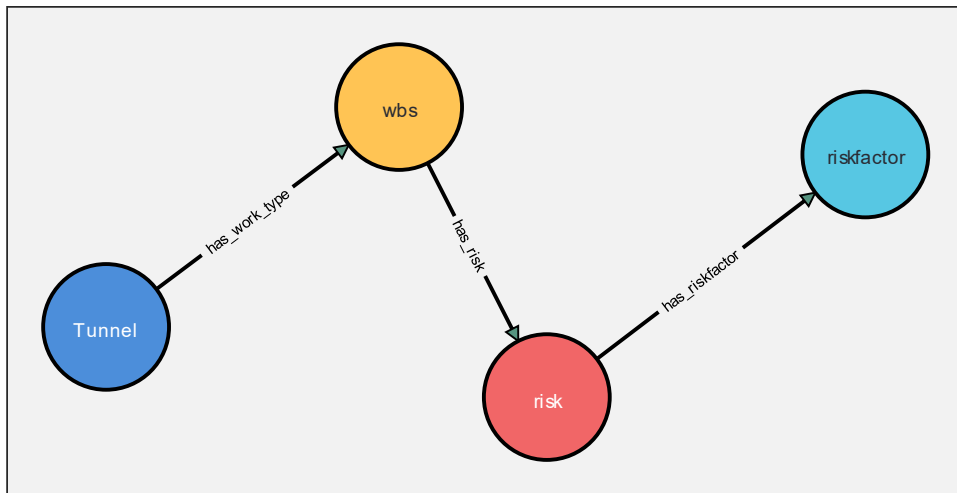


Figure 19. RisKG ontology model

5.2.1.3 Risk Knowledge Extraction

Risk knowledge extraction is another critical step after risk knowledge modeling. This step ensures data homogeneity and prevents ambiguity, especially if the data is extracted

from multiple sources before storing them in a graph database. Risk knowledge extraction involves identifying the RisKG element from the data collected manually or automatically. However, this study achieved this step through manual means to ensure high-quality, rich, and unique KG. As shown in Table 6, the total number of entities, edges, and attributes extracted for the construction of the RisKG includes 527, 526, and six (6), respectively.

Table 6. Entities and relationships for the RisKG

Category	Entity and Relationship Type	Number	Total
Entity	Project name (NATM: Tunnel)	1	527
	Work Type (wbs)	15	
	Risk (risk)	142	
	Risk factor (riskfactor)	369	
Relationship	has_work_type	15	526
	has_risk	142	
	has_riskfactor	369	
Attribute	riskcategory	6	6

5.2.1.4 Risk Knowledge Storage

After determining and extracting the risk knowledge, storage of the risk knowledge is the final step in the construction of RisKG. In this step, all the entities, relationships, and attributes, as determined in the preceding sections, were organized in a triple form and then stored in the Neo4j database by implementing the Cypher script (code) developed for this study on the Neo4j desktop application. Neo4j is an open-source graph database that stores structured graph data using Cypher programming. Although there are several

graph databases (such as ArangoDB, Amazon Neptune, Dgraph DataStax, etc.) that can be utilized to store the extracted risk knowledge. However, Neo4j was selected for the present study because it is an open-source graph database that is lightweight, high-performance, and can support intelligent application development and advanced ML pipelines.

Finally, the RisKG was constructed by running the Cypher code on the Neo4j desktop application. Figure 20 and 21 shows the excerpt of Cypher code and the resultant RisKG. The complete code for the RisKG can be found in the GitHub link (github.com/Murry01/RisKG) on the Neo4j desktop application.

```

1 //create new file
2 CREATE(p:Tunnel{ name:"TUNNEL" }), //check sequence of node creation
3
4 //create wbs node
5 (Ex:wbs{ name:"Excavation" }),
6 (St:wbs{ name:"Support Treatment" }),
7 (Sw:wbs{ name:"Shotcrete Work" }),
8 (Kj:wbs{ name:"Formwork" }),
9 (Rb:wbs{ name:"Rock bolting" }),
10 (L:wbs{ name:"Lining" }),
11 (Wp:wbs{ name:"Water Proofing" }),
12 (Dr:wbs{ name:"Drain" }),
13 (Ca:wbs{ name:"Common area" }),
14 (Gang:wbs{ name:"Gang Gates and Retaining Walls" }),
15 (Sp:wbs{ name:"Shaft part Work" }),
16 (Af:wbs{ name:"Auxilliary Facilities" }),
17 (Inst:wbs{ name:"Instrumentation" }),
18 (Am:wbs{ name:"Steel Pipe Multi-stage Grouting" }),
19 (Gib:wbs{ name:"Geotechnical Investigation" }),
20
21 //create wbs relationship
22 //create relationship between project and work type
23 (p)-[:has_work_type]->(Ex),
24 (p)-[:has_work_type]->(St),
25 (p)-[:has_work_type]->(Sw),
26 (p)-[:has_work_type]->(Kj),
27 (p)-[:has_work_type]->(Rb),
28 (p)-[:has_work_type]->(L),
29 (p)-[:has_work_type]->(Wp),
30 (p)-[:has_work_type]->(Dr),
31 (p)-[:has_work_type]->(Ca),
32 (p)-[:has_work_type]->(Gang),
33 (p)-[:has_work_type]->(Sp),
34 (p)-[:has_work_type]->(Af),
35 (p)-[:has_work_type]->(Inst),
36 (p)-[:has_work_type]->(Am),
37 (p)-[:has_work_type]->(Gib),
38

```

Figure 20. Part of the RisKG Cypher Code

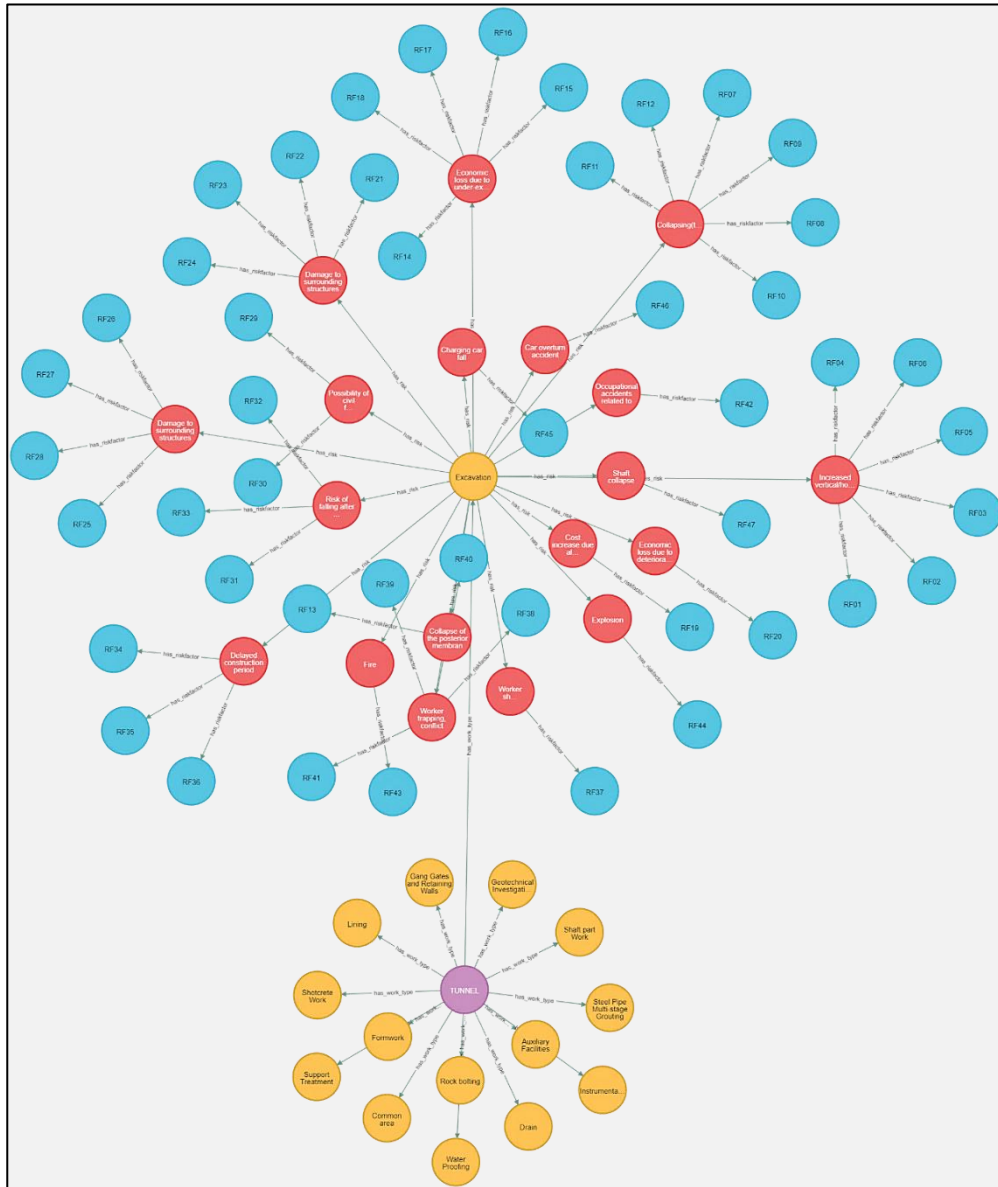


Figure 21. An excerpt of the constructed RiskKG

5.2.2 Evaluation of RiskG

The evaluation of a KG involves assessing the completeness, correctness, and quality of

the knowledge (i.e., entities and relations) embedded in the graph to ascertain whether the real-world concepts (entities) and relations are correctly represented in the graph database (Abu-Salih, 2021). Thus, verifying the completeness and correctness of the constructed RisKG is crucial to determining its fitness for this study's proposed application. Usually, various evaluation metrics are employed to measure a domain-specific KG's completeness and correctness. Some of the metrics include but are not limited to Hits@N, Mean Reciprocal Rank (MRR), Mean Rank (MR), Accuracy, Precision, Recall, F-measure, Use-case-specific metrics, etc. (Zaveri et al., 2016). However, in this study, the use-case-specific metric was adopted to evaluate the completeness and correctness of the RisKG, and the results obtained were compared with manually extracted data in section 5.2.1.3.

The RisKG was evaluated using three specific use cases based on the Cypher queries in Table 7. Firstly, the completeness of all the entities and relationships in KG was assessed using Cypher query (1) in Table 7. The result revealed 527 nodes (entities) and 526 relations (edges), corresponding to the manually extracted data in Table 6 and the screenshot in Figure 21. Secondly, we retrieved all the work packages (work type) associated with the project (i.e., Tunnel) using the Cypher query (2) of Table 7. The result is shown in Figure 22.

Table 7. Use cases Cypher query

Use case	Cypher query
1	Match (n) return(n)
2	Match (n:Tunnel)-[:has_work_type]->(m:wbs) return n,m
3	match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk) return n,m

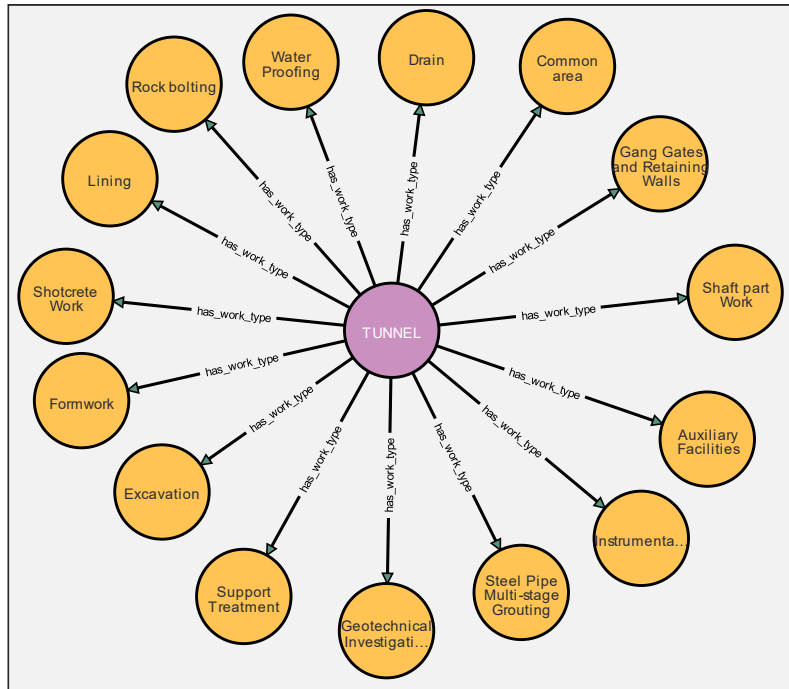


Figure 22. Result of the second use case

From the figure, the purple node represents the project (i.e., Tunnel), and the yellow nodes represent the list of fifteen (15) works of the tunnel project. Also, the arrows represent the relationship between the project and the work packages. The list of the works retrieved includes “*Geotechnical Investigation, Excavation, Support Treatment, Rock bolting,*

Formwork, Shotcrete Work, Lining, Water Proofing, Drain, Common area, Gang Gates and Retaining Walls, Shaft part Work, Steel Pipe Multi-Stage Grouting, Instrumentation, and Auxiliary Facilities works”.

Lastly, the risks associated with the work package: Geotechnical Investigation was retrieved using the Cypher query (3) in Table 7. The result is shown in Figure 23, where the yellow node represents the work type, i.e., “*Geotechnical Investigation*”, and the red nodes represent the corresponding risks associated with the works. The risks identified include *construction cost increase and construction delay due to ground grade (pattern) change, increased construction cost and delayed construction due to an additional ground investigation, drop in groundwater level, increased construction cost, and delayed construction due to soft ground and ground disturbance*”. Overall, the results confirmed the completeness and correctness of the constructed RisKG, indicating its fitness for application in the proposed QAR-System.

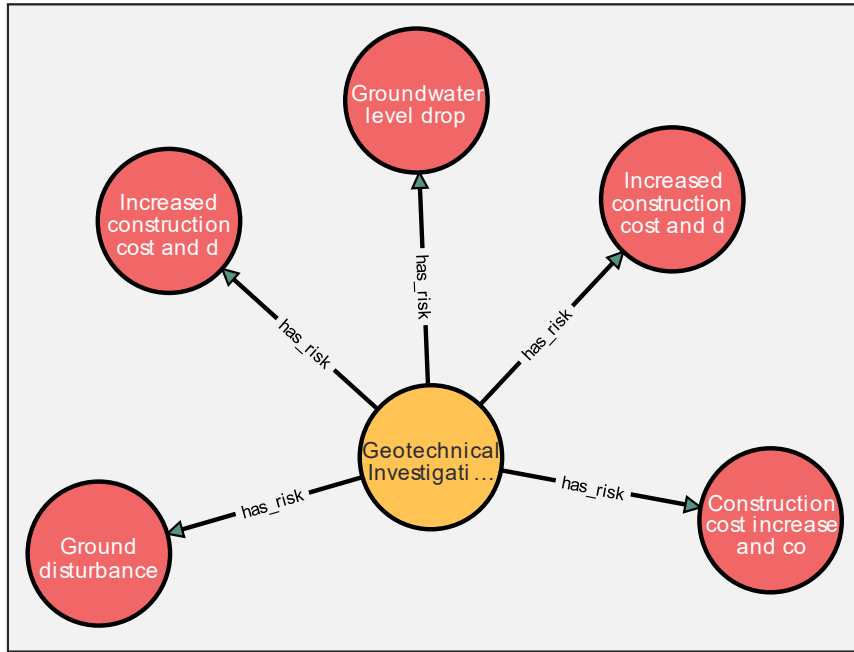


Figure 23. Result of the third use case showing the risks identified

5.2.3 Results Discussion

The results obtained from the specific use cases conform with data extracted manually, indicating the completeness and correctness of the constructed RisKG. Also, the practical implication of the result is that construction project risk information can be represented using KG to aid quick information retrieval and the development of intelligent KBRM tools such as the proposed QAR-System. Thus, allowing for effective storage, sharing, reuse, and communication of project risk knowledge. Also, the RisKG will assist the project manager/team in searching and retrieving risk information within seconds to facilitate efficient risk identification and reliable decision-making that will lead to actualizing project objectives in construction projects.

5.3 Development of the Question-Answering System for Risk Identification

The current risk management practice involves a substantial manual effort to retrieve risk-related information from past project documents. It becomes very difficult, hectic, and time-consuming when the project complexity and size increases, especially during the risk identification process of such projects. Hence, developing a faster method for reviewing and retrieving important risk-related information is essential for effective and dynamic risk identification in construction projects. Question-answering is an AI technique that can help in the fast and succinct retrieval of risk information using natural language. Accordingly, this chapter explains the detailed development and evaluation processes of the proposed QAR-System for risk identification in construction projects.

5.3.1 Methodology

The proposed QAR-System for construction projects is a knowledge-based question-answering system (KBQAS) that leverages the KG technique and OpenAI's generative pre-trained transformer model (i.e., GPT-3). Typically, the constructed RisKG in the previous section consists of linked data that cannot be easily accessed by users who do not have specialized knowledge in Cypher query language (CQL), etc., coupled with an in-depth understanding of content structure in the underlying KG (Liang et al., 2021). As a result, a question-answering system over the RisKG was developed to enable users to easily access the KG in order to benefit optimally from information stored in the RisKG. To achieve this, OpenAI's GPT-3 model was used for Cypher query generation based on

NL to aid the timely retrieval of risk-related information from the constructed RisKG. This approach is an automated and accelerated approach for accurately retrieving risk information to enhance efficient and comprehensive risk identification at the preconstruction stage of a new project, thus eliminating the laborious effort and time usually spent on risk documentation reviews and expert meetings. An overview of the development process of the proposed QAR-System is shown in Figure 1 of Chapter 1. Data collection and knowledge base development are important steps in developing a functional KBQAS. These steps have been elucidated in detail in section 5.3 of this work. The development of a query generation model and its integration with the constructed RisKG to establish the proposed QAR-System are explained in the following subsections.

5.3.1.1 Cypher Query Generation

After establishing the graph database (RisKG), the next step is *Query Generation*, which involves constructing executable logical statements (queries) over the constructed RisKG. This important task entails generating triple graph patterns to retrieve information (answer) from the RisKG based on the user’s question. In other words, query generation enables a set of queries (in the form of triples) to be constructed based on the user’s natural language questions (NLQs) to retrieve the stored in the KG elements, i.e., entities (nodes), attributes, and edges (relationships). For example, the question: “*How many risks are in the project?*” has a corresponding triple pattern given as: “*match(n:risk) return count(n) As response*”.

This study adapted OpenAi’s GPT-3 (“*text-DaVinci-003*”) model through prompt

engineering to develop a query generation model (query generator) that can be used to access the RisKG using NL input. Prompt engineering involves designing and refining input text or “prompts” to guide an LLM such as ChatGPT in generating specific and relevant responses. In other words, prompting means telling an AI agent what to do and how it should be done using human language. In the case of the proposed QAR-System, the GPT-3 model was adapted to translate or covert NLQs to executable Cypher queries over the RisKG to automatically retrieve risk-related information that can facilitate efficient risk identification in construction projects.

As shown in Figure 24, the GPT-3 model consists of several layers of transformer blocks with a normalized layer and two activation functions (linear and SoftMax) connected to the output layer. Each transformer block in the architecture acts like a single model with its number of trainable parameters, making the GPT-3 large model consisting of a family of models. The GPT-3 model is a powerful LLM consisting of between 154 and 175 billion parameters, which has been deployed in many applications, including conversation AI, QA systems, etc.

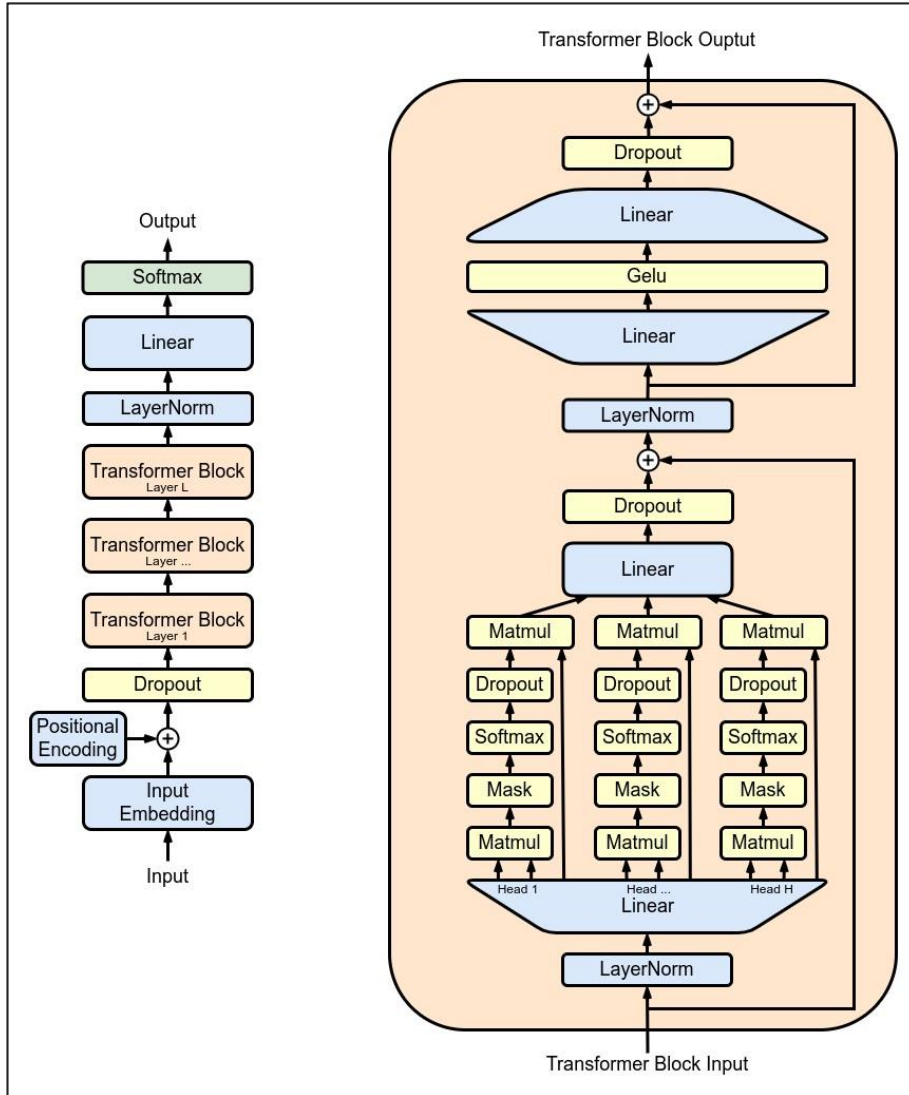


Figure 24. Architecture of GPT-3 model (Wikipedia)

Furthermore, several prompting approaches, such as zero-shot, few-shot, role, chain-of-thought (CoT), etc., can be used to train the GPT-3 models for specific tasks. However,

the few-shot prompting was adopted in this study because the approach is simple and does not require special coding or programming language knowledge for the implementation. Few-shot prompting involves providing examples of tasks and expected solutions to the LLM in NL for training. For example, in the case of the QAR-System, a few questions and the corresponding Cypher querying examples (question-query pairs) were used to train the GPT-3 model to adapt the model for Cypher query generation. Specifically, the GPT-3 model was trained with only fourteen (14) pairs of NL question-Cypher query (training set) shown in Table 8, following the training procedure explained by (Sixing 2022; Tomaz, 2023). After completing the training, new user questions were posed to the model, and the corresponding new Cypher queries were generated, as shown in Figure 25.

Table 8. Question-query pair training set

S/N	Natural Language Questions	Cypher query
1	# What is the list of works in the tunnel project?	match(nTunnel){:has_work_type}>(m:wbs) return m.name As response
2	# What are the risks associated with the tunnel project?	match(m:risk) return m.name + " - " + " Risk Category: " + m.riskcategory As response limit 20
3	# What are the risk factors associated with the tunnel project?	match(m:riskfactor) return m.Riskfactor As response limit 20
4	# List risks associated with the "Geotechnical Investigation" of the tunnel project	match(m:wbs{name:"Geotechnical Investigation"}){:has_risk}>(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response
5	# What are the risk factors of "Geotechnical Investigation" of the tunnel project?	match(m:wbs{name:"Geotechnical Investigation"}){:has_risk}>(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response
6	# List risks associated with the "Shaft part Work" of the tunnel project	match(m:wbs{name:"Shaft part Work"}){:has_risk}>(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response
7	# What are the risk factors of "Shaft part Work" of the tunnel project?	match(m:wbs{name:"Shaft part Work"}){:has_risk}>(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response
8	# List risks associated with the "Lining" of the tunnel project	match(m:wbs{name:"Lining"}){:has_risk}>(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response
9	# What are the risk factors of "Lining" of the tunnel project?	match(m:wbs{name:"Lining"}){:has_risk}>(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response
10	# List risks associated with the "Excavation" of the tunnel project	match(m:wbs{name:"Excavation"}){:has_risk}>(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response
11	# What are the risk factors of "Excavation" of the tunnel project?	match(m:wbs{name:"Excavation"}){:has_risk}>(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response
12	# How many risks are in the project?	match(m:risk) return count(n) As response
13	# How many risk factors are in the project?	match(m:riskfactor) return count(n) As response
14	# List the risk category associated with the tunnel project	match(m:risk) return distinct n.riskcategory As response

```

# List the works in the tunnel project?
match(n:Tunnel)-[:has_work_type]->(m:wbs) return m.name As response

# What are the risks associated with the tunnel project?
match(n:risk) return n.name + " - " + " Risk Category: " + n.riskcategory as response limit 20

# What are the risk factors associated with the tunnel project?
match(n:riskfactor) return n.Riskfactor As response limit 20

# List risks associated with the Geotechnical Investigation of the tunnel project.
match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response

# What are the risk factors of Geotechnical Investigation of the tunnel project?
match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response

# What are the risk factors of Excavation of the tunnel project?
match(n:wbs{name:"Excavation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response

# List risks associated with the Excavation of the tunnel project.
match(n:wbs{name:"Excavation"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response

```

Figure 25. Query generation using GPT-3 model

Figure 25 above shows parts of the training samples in the red box, consisting of the NL questions and the corresponding Cypher queries. Also, as shown in the green box of Figure 25, the new user inputs (questions) starting with hashtags and the generated Cypher queries by the GPT-3 model based on the user inputs are highlighted in green. Further, part of the Python implementation code for the query generation is shown in Figure 26, while the flow chart for the query generation is depicted in Figures 27 and 28.

```

15
16 # Generating Cypher Query from Users' Input.
17 def generate_response(prompt, cypher=True):
18     if cypher:
19         completions = openai.Completion.create(
20             engine="text-davinci-003",
21             prompt=examples + "\n#" + prompt,
22             max_tokens=1000,
23             n=1,
24             stop=None,
25             temperature=0.5,
26         )
27         cypher_query = completions.choices[0].text
28         message = read_query(cypher_query)
29         return message, cypher_query
30

```

Figure 26. Python implementation code for query generation

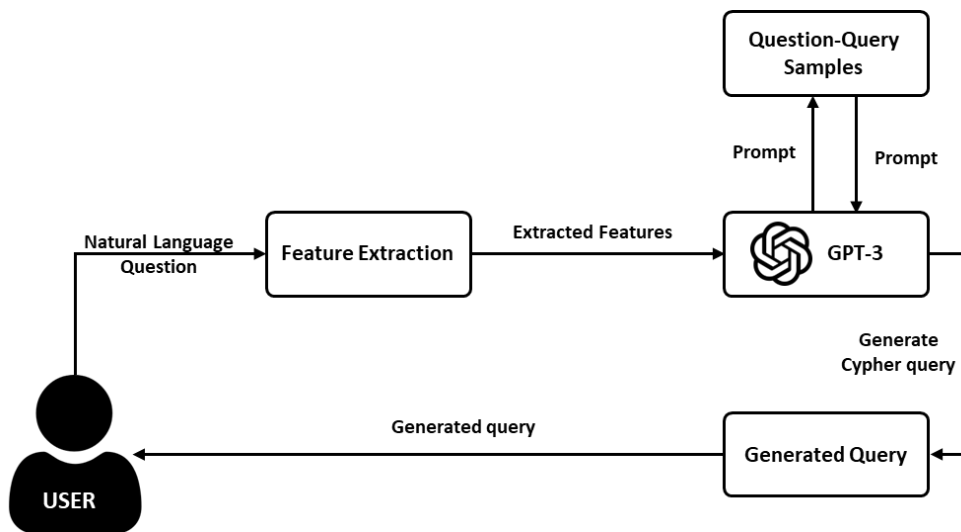


Figure 27. Query generation flowchart

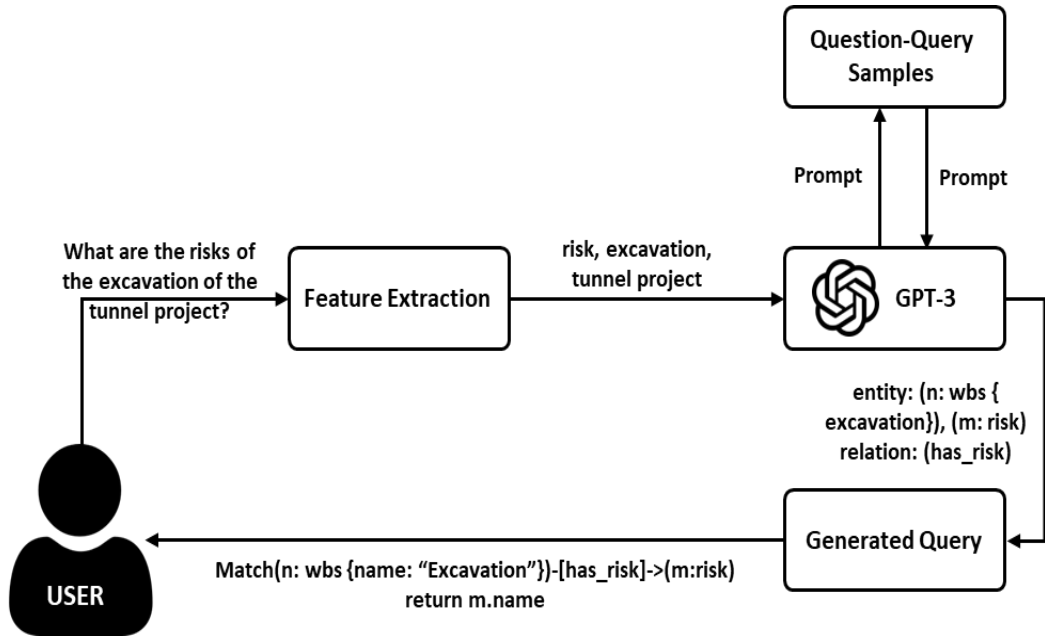


Figure 28. Query generation example

5.3.1.2 Integration of Query Generator with RisKG

In the previous step, i.e., query generation, the GPT-3 model was trained to adapt it for Cypher query generation based on NL questions. The next step is integrating the query generator with the constructed RisKG using Python programming language and developing a User Interface (UI) for the proposed QAR-System using the Streamlit Python library. A Python script for integrating the query generator and RisKG was developed to accomplish this. Then, the UI was developed using Streamlit to enable users to interact with the QAR-System through NL easily. Streamlit is a Python library that provides a faster way to build and share data on web applications and local networks. The

source code for accomplishing this task can be accessed via github.com/Murry01/QASRisk. The home screenshot and UI of the proposed QAR-System are shown in Figure 29 and Figure 30, respectively, while the system's architecture illustrating the workflow is shown in Figure 31.

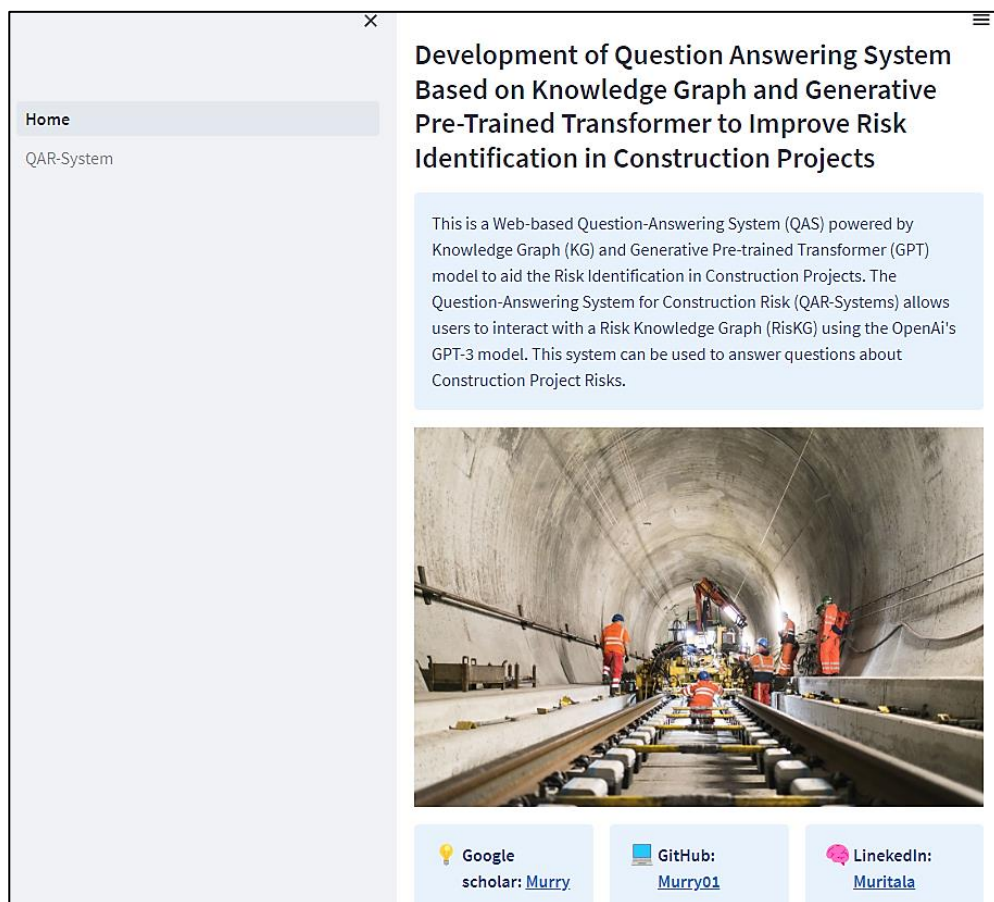


Figure 29. Home Screenshot

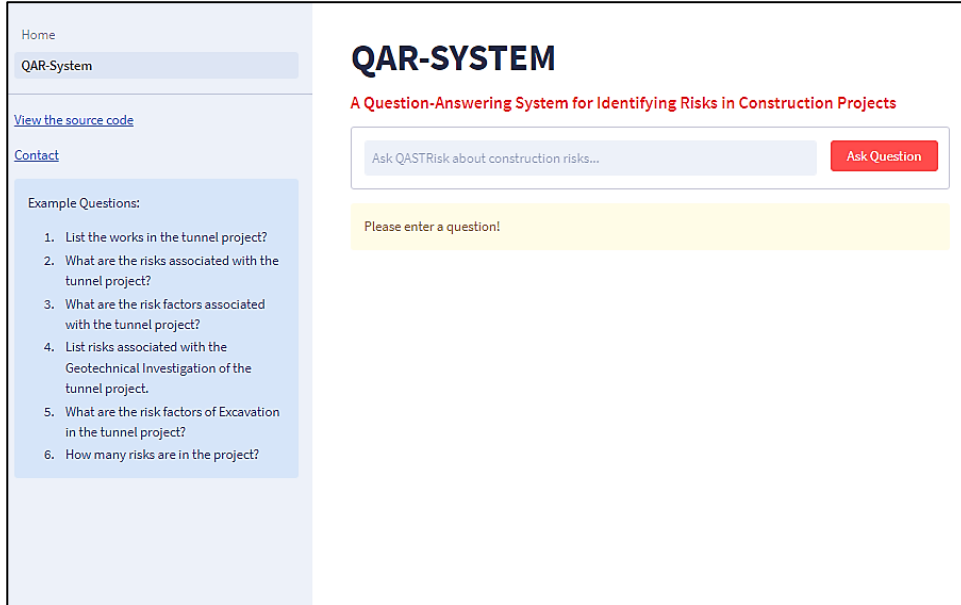


Figure 30. User Interface of the proposed QAR- System

As depicted in Figure 29, the home screen shows the thesis title and a brief description of the QAS developed, while Figure 30 depicts the user interface of the QAR-System UI, showing the input textbox and a sidebar consisting of sample questions that can be posed to QAR-System. Further, as illustrated in Figure 31, the user inputs a question via the UI, and then the question is sent to OpenAi's GPT-3 model as a prompt request to translate the question from an NL question to a Cypher query. Next, the GPT-3 model returns the Cypher query, which is then used to retrieve the risk-related information from the RiskKG. After that, the retrieved risk information is displayed to the user via the UI in NL. The demo video can be accessed via github.com/Murry01/QASRisk.

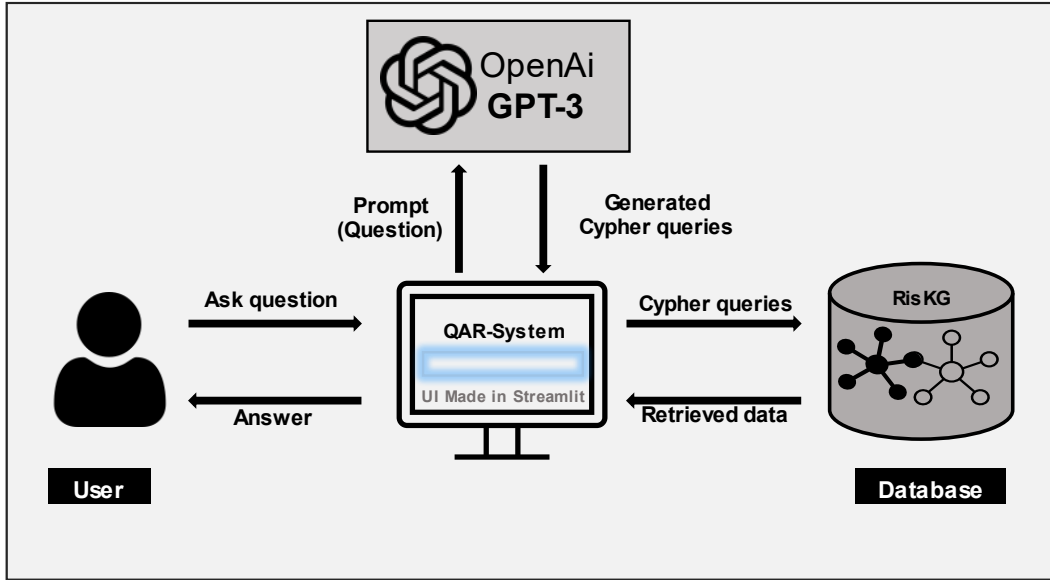


Figure 31. Architecture of QAR-System

Moreover, Figure 31 shows a flow chart elaborating on how QAR-System generates answers to users' questions. When the user enters a question in natural language, the GPT-3 model does the feature extraction by identifying the entities and relations in the question that will facilitate accurate answer retrieval from the RiskKG. Suppose the question does not contain relevant features (entities and relationships) related to the information in the RiskKG ontology. In that case, a warning message is delayed on the UI, telling the user to enter appropriate questions. Suppose the question contains the relevant entities and relationships of the RiskKG. In that case, GPT-3 compares the extracted features with the question-query samples to generate an appropriate query containing the extracted features, i.e., entities and relationships corresponding to the user's question. The generated query

by GPT-3 is then executed on the RiskKG to retrieve the answer to the user's question automatically.

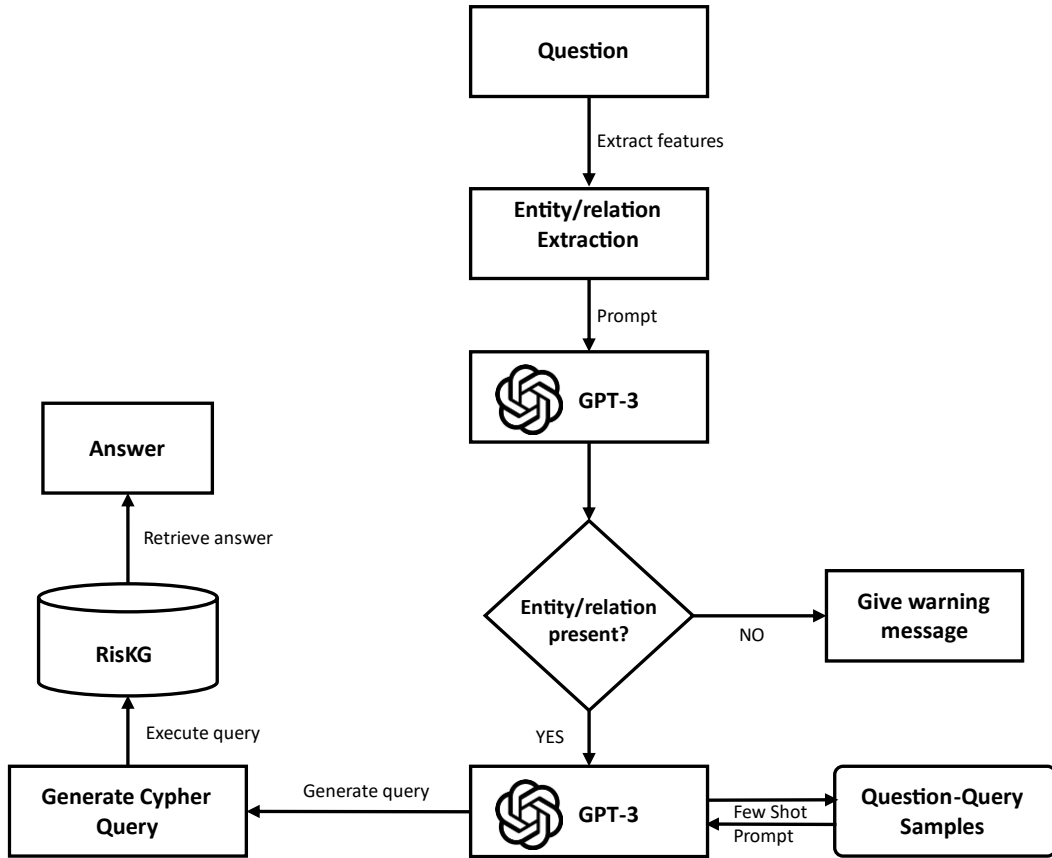


Figure 32. Flowchart of QAR-System

5.3.2 Evaluation of QAR-System

The proposed QAR-System was evaluated using accuracy measures and compared with similar LLM-powered conversational tools based on NL prompts. The evaluation process is elucidated in the following subsections.

5.3.2.1 QAR-System Evaluation Based on Accuracy Measure

Although several evaluation metrics for evaluating the accuracy of QA systems have been proposed in the literature (Rodrigo & Peñas, 2017). However, the final performance of a QA system depends on the system's feature to be evaluated and the metric selected because each metric evaluates a separate set of features (Martinez-Gil, 2023). This study evaluated the query generation model (query generator) of the proposed QAR-System. This is of utmost importance because the performance of QAR-System depends on the ability of the query generator to generate correct and accurate Cypher queries that can be executed over the RisKG to retrieve succinct answers based on questions posed by users in NL. Therefore, a binary approach (for checking correct versus incorrect answers) using the *Accuracy* measure was adopted for the performance evaluation of QAR-System. Accuracy is a simple and intuitive metric representing the proportion of correctly answered questions (Rodrigo & Peñas, 2017). Besides, it is an easy-to-understand evaluation metric for a factoid QAS (such as QAR-System) that restricts the number of answers to one per question. Thus, it can be computed using Eqn (1), where n_{ac} is the number of questions answered correctly, and n is the total number of questions answered.

$$accuracy = \frac{n_{ac}}{n} \quad (1)$$

To evaluate the proposed QAR-System, five (5) questions (validation set) were selected randomly from the previous training sample questions, and 50 new questions (test set) outside of the training set based on the RisKG ontology were created to test the system.

First, QAR-System was asked the questions selected from the training sample (validation set) questions. Thus, the accuracy of the QAR-System for the validation set questions was computed using Eqn (1), and the result is presented in Table 9. From the results in Table 9, all five (5) questions for the validation set were answered correctly, with zero incorrect answers and an accuracy of 1, indicating 100% precision. Second, the newly created 50 questions (test set) were posed to the QAR-System, and the number of correct and incorrect answers was noted. Accordingly, the accuracy was computed, and the result is shown in Table 9.

Table 9. Evaluation results of QAR-System

	Number of Questions (n)	Number of questions answered correctly (n_{ac})	Number of questions answered incorrectly	Accuracy
Validation set	5	5	0	1
Test set	50	48	2	0.96

From the results, out of the 50 test questions, 48 were answered correctly, while two returned incorrect answers with an accuracy of 0.96 (i.e., 96% precision). Overall, the accuracy results confirmed the capability of QAR-System to generate correct queries to aid quick and succinct risk information retrieval, thus facilitating efficient and accurate risk identification in construction projects while fostering reliable decision-making.

5.3.2.2 Accuracy Result Discussion

The purpose of developing the proposed QAR-System is to improve the efficiency of the

risk identification process at the preconstruction stage of a new construction project by eliminating the manual effort and time expended on reviewing risk documentation of previous projects and expert workshops. Automating and digitizing the risk identification process through an integrated human-machine intelligence leveraging emerging AI technologies is daunting. Therefore, achieving a system performance of 100% is not realistic. Nonetheless, the performance will increase continually through a concerted effort to continuously improve the system's development process and access to better training data.

Based on the evaluation results, two incorrect answers were recorded from the test set questions, as shown in Table 9. One question with a punctuation mark (e.g., full stop) omission resulted in an error message. This error was resolved by using appropriate punctuation marks at the end of the question and training the query generation model with a different mix of sentences and punctuation marks. The second incorrect answer resulted from a typographical error in the question, which generated a query with entities and relations that do not match RisKG ontology. This problem was resolved by inputting typographical error-free questions. Hence, if neither of the above two situations occurs in the questions, the accuracy measure of QAR-System will be 100%.

5.3.2.3 QAR-System Evaluation Through Comparison with LLM-based Tools

QAR-System was further validated through comparison with three similar LLM-based conversational tools to ensure a sufficient confidence level. The objective is to assess the framework's effectiveness in risk information retrieval. The process involves comparing

the risk information retrieved from the proposed QAR-System with ChatGPT Plus Bing Chat, and HuggingChat.

To validate the efficacy of both QAR-System, ten (10) new NL questions consisting of the words “What”, “List”, “How” and “Count” were created and executed on QAR-System, ChatGPT Plus, Bing Chat, and HuggingChat. The validation results are shown in Table 10, while Figures 33, 34, 35, and 36 show the screenshots of answers returned by the four models for question Q6. As seen in Table 10, the proposed QAR-System model responded to all ten questions with concise and complete answers. The ChatGPT Plus responded to questions Q1, Q2, Q3, Q4, Q5, and Q9 with generic answers, while Q8 with generic and partial answers. Also, ChatGPT Plus responded to questions Q6 and Q10 with generic and partial answers compared to the QAR-System. Further, BingChat responded to Q1, Q2, Q3, Q5, Q7, and Q8 with generic and partial answers, while Q4 and Q9 answers were generic. Like the QAS-GPT model, BingChat returned generic and inaccurate answers to Q6 and Q10. As for HuggingChat, only the Q1 answer was generic and partial, while the Q6 and Q10 answers were the same as in ChatGPT Plus and BingChat, i.e., generic and inaccurate. However, HuggingChat responded to Q2, Q3, Q4, Q5, Q7, Q8 and Q9 with generic answers.

Generic answers mean answers returned based on the general knowledge acquired by the LLMs through pre-training. Conversely, specific answers are based on knowledge extracted by the LLM from a structured knowledge base. Complete answers mean the answers returned by the proposed QAR-System model based on the RisKG, while partial

answers are the incomplete answers returned by ChatGPT Plus, BingChat, and HuggingChat compared to QAR-System. Accurate answers imply the correct answers returned by the QAR-System based on the risk information in the RisKG, while inaccurate answers are those returned by the remaining models compared with the proposed QAR-System.

Table 10. Validation results of QAR-System

S/N	Question	QAR-System	ChatGPT-4	Bing Chat	Hugging Chat
Q1	What are the works involved in the tunnel project?	S, C, A	G	G, P	G, P
Q2	List the risk categories in the tunnel project.	S, C, A	G	G, P	G
Q3	List all the risks under the construction risk category.	S, C, A	G	G, P	G
Q4	What are the risks under the design risk category?	S, C, A	G	G	G
Q5	List all the risks under the technical risk category.	S, C, A	G	G, P	G
Q6	How many risks are associated with the construction risk category?	S, C, A	G, IA	G, IA	G, IA
Q7	What are the risks associated with the safety risk category?	S, C, A	G	G, P	G
Q8	List the risks associated with Management risk category of the tunnel project.	S, C, A	G, P	G, P	G
Q9	What are the risks associated with the Environment risk category of the tunnel project?	S, C, A	G	G	G
Q10	Count the risks associated with Environment risk category of the tunnel project.	S, C, A	G, IA	G, IA	G, IA

Note - G: Generic, S: Specific, C: Complete, P: Partial, A: Accurate, IA: Inaccurate

Figure 33 shows the response of QAR-System to question Q6: “*How many risks are associated with the construction risk category?*” The total number of risks stored in the data for construction risks is 63. Thus, the QAR-System returned 63 as the answer to question Q6. In Figure 34, the answer returned by ChatGPT Plus is 23, which is the number of generic risks it identified. ChatGPT Plus was able to count the number it identified under the construction risk category. However, the answer is inaccurate compared to the answer returned by QAR-System. Further, in Figures 35 and 36, Bing Chat and HuggingChat returned seven (7) and nine (9), respectively as answers to question Q6. Both Bing Chat and HuggingChat answers were inaccurate compared to QAR-System.

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

Generated Cypher Query

```
match (n:risk[riskcategory:"Construction"]) return count(n) as response
```

1. 63

Figure 33. QAR-System response to question Q6

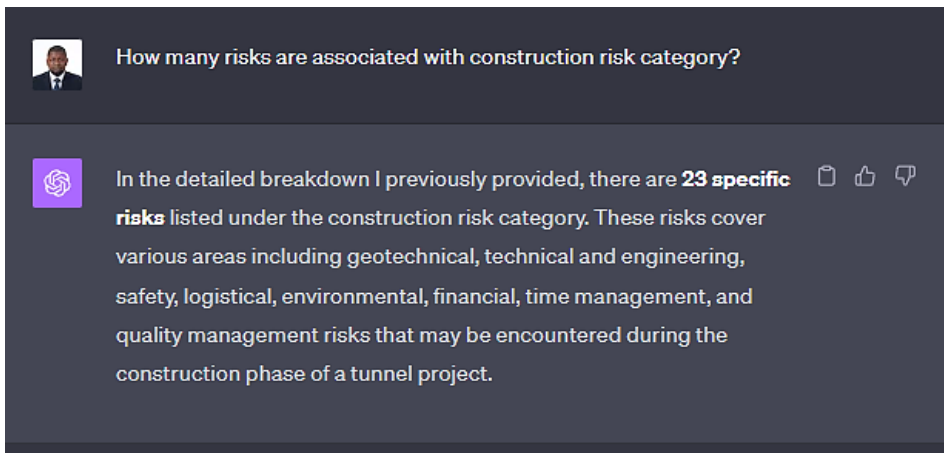


Figure 34. ChatGPT Plus response to question Q6

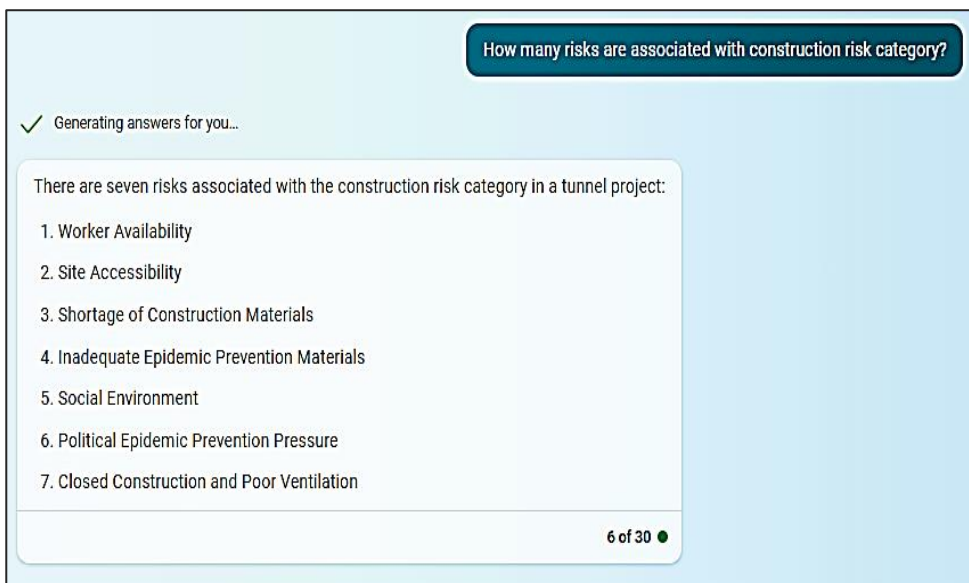


Figure 35. Bing Chat response to question Q6

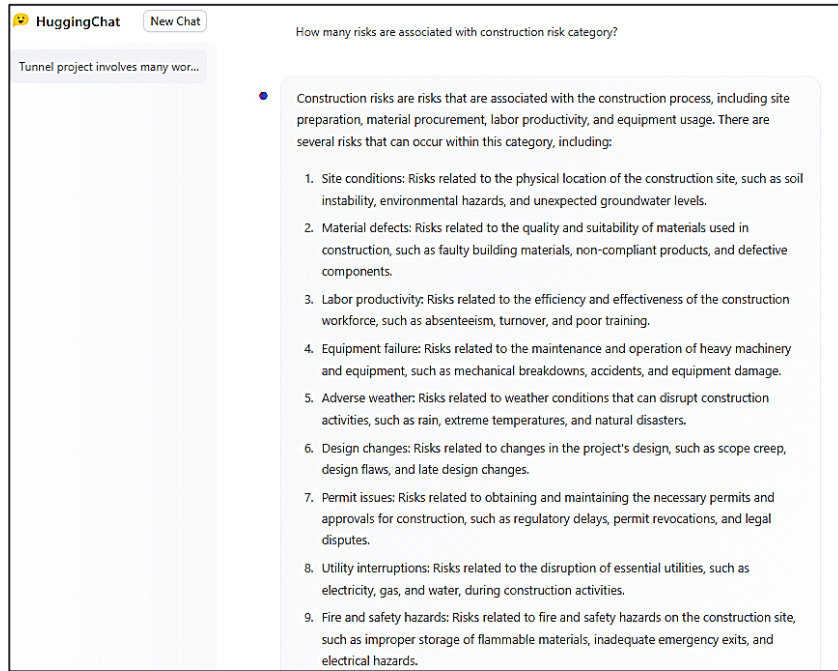


Figure 36. HuggingChat response to question Q6

5.3.2.4 Discussion of QAR-System Validation Results

From the results, it can be observed that the results from QAR-System were consistent compared with ChatGPT Plus, BingChat, and HuggingChat due to incorporating a structured knowledge database (RisKG). ChatGPT Plus and BingChat are powered by the GPT-4 model, the most advanced LLM of OpenAI with the highest capabilities in understanding and generating natural language. Similarly, HuggingChat is built upon Facebook's highest-performing LLM (Llama-2) regarding natural language generation. However, the responses of the three Chatbots might look plausible and correct at first glance but are indeed inaccurate compared to the result obtained from QAR-System. The inaccurate answers resulted from the problem of hallucination, which all LLMs exhibit,

as earlier explained in section 4.6. Therefore, humans should verify results obtained using the popular LLM-based Chatbots before being used as the basis for crucial decision-making.

In contrast, the proposed QAR-System results show that a dedicated and structured database can resolve hallucination problems and improve the accuracy of the results generated by LLMs. In addition, it indicates that adapting LLMs for specific tasks in a particular domain requires structured data for optimal performance and accurate results. Moreso, the accurate answers obtained from QAR-System confirm the adaptability of LLMs for domain-specific tasks through training on a small dataset. This addresses the problem of large dataset requirements for training other ML models hindering the adoption of AI technology in the construction industry.

5.3.3 Summary

The proposed QAR-System is related to NLP, knowledge-based question-answering (KBQA), and semantic web technologies research but differs from existing works in a few respects. For example, (Gupta et al., 2021) adopted a template-based approach in their QAS, where the questions-queries template was manually predefined with syntactic-semantic mapping patterns. The main issue with this approach is the rigidity, where the user is restricted to only the template's question type and sentence pattern. In contrast, the approach presented in this thesis is flexible and only requires training an OpenAI's GPT-3 model with a few samples of question-query pairs (i.e., few-shot prompting) to allow

automatic conversion of the NLQs into an executable query (Cypher). This allows the users to ask questions of different patterns without restriction to the questions-queries used during the model training or a predefined question-query template. Also, the average performance of the system (KG4ASTRA) developed in (Gupta et al., 2021) is 0.4902 in terms of F-score, while the performance QAR-System is 0.96 using the accuracy measure. QAR-system accuracy is higher than that of KG4ASTRA, indicating better performance. Furthermore, (Abujabal et al., 2017) in their study tried to resolve the problems of hand-crafted templates for QAS using automatically learned utterance templates based on standard dependency parse trees. QUINT is more like a semi-automatic system that uses a template to support automatic semantic answer typing. This implies that QUINT still needs hand-crafted syntactic rules (dependency parser) to learn how to generate answers. However, QAR-System uses the GPT-3 model that understands syntactic and semantic rules and does not require a predefined template to know how to generate executable queries over the knowledge base. In summary, the proposed QAR-System powered by the GPT-3 model can understand grammatical rules and automatically perform NLP tasks such as tokenization, name entity recognition (NER), name entity disambiguation (NED), POS tagging, dependency parsing, etc., through prompt engineering. Therefore, the proposed QAR-System provides more robust support to the user regarding question-answering capabilities and achieves the highest accuracy compared to these existing studies.

CHAPTER 6. CONCLUSION

6.1 Conclusion

Risks are inherent in all construction projects, including NATM tunnels. These risks can arise in a project due to the complexity involved and the diversity of the surrounding rocks, groundwater, work safety, geological environment, construction method, man-made factors, and many other factors. Identifying the critical risks and planning the appropriate risk responses at the preconstruction stage can help prevent the potential negative impact of the risk on project objectives.

Two major issues are associated with the current risk identification practices in construction projects. First, there is lack of a formalized and standardized representation format for capturing and storing the risk-related knowledge generated from previous projects that hinders integration, reuse, and sharing among practitioners and computers. Second, there are limitations in the traditional procedures and tools used for risk identification in construction projects. These limitations include but are not limited to over-reliance on the knowledge of experts that may lead to the misidentification of critical risks and improper management of identified risks if the expertise is flawed, resource-intensive, and time-consuming procedures that may not account for subtle or unusual risks because of the static and complex nature of construction projects, subjective nature of the process leading to inconsistencies, etc. Hence, to address these problems, this research developed an intelligent question-answering system powered by two emerging AI

technologies, knowledge graph (KG) and generative pre-trained transform (GPT), to support effective risk identification in construction projects, i.e., NATM tunnels.

To address the problem of the lack of a formalized format for risk knowledge representation in construction projects, a risk knowledge graph (RisKG) was constructed and established with the aid of neo4j based on NATM tunnels construction risk data collected through literature review and expert support. Then, the constructed RisKG was evaluated through the use cases to ascertain the completeness, correctness, and quality of knowledge embedded in the graph. The result revealed that the RisKG consists of 527 nodes (entities) and 526 relations (edges), which correspond to the manually extracted data, indicating the completeness and correctness of the RisKG. The constructed RisKG was then used as the knowledge base for developing the QAR-System for construction projects.

Also, a question-answering system for construction project risks (QAR-System) was developed to improve the traditional laborious and time-consuming risk identification procedures using the constructed RisKG and OpenAI's GPT-3 model. This approach attempts to automate risk identification, assessment, and management for construction projects leveraging powerful LLM. After establishing the RisKG, the GPT-3 model was adapted through prompt engineering to develop a query generator that converts natural language questions into Cypher queries, which can automatically retrieve risk information from the RisKG to support risk identification in NATM tunnel projects. The RisKG and query generator were integrated into the user interface (UI) using the streamlit

Python application to develop the QAR-System.

The proposed QAR-System's ability to retrieve concise risk information from the graph database (RisKG) was evaluated using *Accuracy Metrics*, and a 96% accuracy was achieved. Further, the QAR system was validated through comparison with the state-of-the-art conversational AI models powered by LLMs, namely ChatGPT Plus, Bing Chat, and HuggingChat. This is to verify the confidence level of the proposed QAR-System in automatic risk information retrieval using natural language. The comparison result confirmed the conciseness and succinctness of QAR-System over ChatGPT Plus, Bing Chat, and HuggingChat in retrieving the project risk information. In a nutshell, this study has successfully demonstrated the application and adaptability of emerging AI tools (KG and GPT-3) for improving efficiency in risk identification of construction projects. Finally, the proposed QAR-System can aid comprehensive, dynamic, and effective risk identification while enhancing the overall accuracy of the risk profile of construction projects.

6.2 Contributions

This study proposes a QAR-System to tackle the problems associated with risk management practices, specifically on risk knowledge storage and retrieval in construction project risk identification. The proposed QAR-System is appropriate for quick, effective, and intuitive risk identification at the early stage of any construction project, as it can facilitate timely and reliable decision-making to prevent safety accidents and reduce project delays and cost overruns. The main contributions of this study to the

body of knowledge are as follows.

- i. This study provides a framework for structuring and storing risk knowledge acquired from previous construction projects using knowledge graph concepts, thereby setting a formalized and standardized format for sharing risk knowledge among humans and computers.
- ii. The research provides the practitioners with an integrated human-machine intelligence (IHMI) framework for the development of intelligent systems that can help reduce manual effort and time spent on the risk identification process using an LLM, i.e., GPT-3 trained with a small dataset, thus improving the efficiency of risk management practice in the construction domain. Also, this study lays the foundation for applying LLMs in the construction domain to advance timely efficiency and reliable decision-making throughout the lifecycle of construction projects.
- iii. The proposed QAR-System could also facilitate the digital transformation of risk management and construction management practices leveraging AI technology, thereby enhancing the decision-making that can improve project performance and productivity in the construction industry.

6.3 Limitations and Recommendations

The adoption of AI techniques in the construction industry is still growing due to the unique characteristics of construction projects rooted in the differences in project size, needs, methods, environment, and involvement of many stakeholders. Consequently,

impediments and limitations were encountered during the development of the proposed QAR-System, as highlighted below.

- The knowledge base (RisKG) used for developing the QAR-System consists of only NATM tunnel construction risk data, limiting its application for risk identification of other tunnel types and construction projects such as highways, bridges, buildings, etc. A future study focusing on constructing a robust risk knowledge graph to capture different types of tunnels and other projects is recommended. This will facilitate the development of robust and intelligent risk identification systems for construction projects.
- The data type utilized in developing the current QAR-System is limited to semi-structured data, i.e., tabular data format. Future research should incorporate unstructured data (text), as project risk management documents are usually stored in various formats. A hybrid system consisting of unstructured, semi-structured, and structured databases will help improve the efficiency and accuracy of the system in facilitating automatic risk management in the tunnel and other construction projects.
- The proposed QAR-System capability is limited to the risk identification process. Risk management processes, such as risk assessment, analysis, and response processes, should be integrated into the system. Further research is required for an integrated risk management system for NATM tunnels and other construction

projects. This can be achieved by incorporating project risk assessment, analysis, and risk response details in the risk knowledge graph to facilitate holistic risk management of the project while enabling informed, accurate, and data-driven decision-making, leading to successful project delivery.

- The ontology model (schema) of the RisKG is simple and supports only factoid questions; therefore, the construction of a more complex ontology that can support complex question-answering for complex decision-making should be considered in future studies.
- Finally, the inherent limitations of GPT models, such as hallucination, black-box phenomenon, data confidentiality, trust, acceptability, etc., are major barriers to their adoption/application in the construction industry. Thus, more studies demonstrating the safe and ethical use of the GPT models in the construction industry are required, as only a few construction-related studies are available presently.

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NATM 터널시공 리스크 식별을 향상시키기 위한 질의-응답시스템의 개발

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(초록)

터널 프로젝트는 일반적으로 프로젝트 복잡성과 고유성, 다수의 이해 관계자의 참여, 막대한 자본 투자, 경의 변화, 긴 프로젝트 기간, 외부 환경 및 기상 조건 등과 같은 이유로 많은 리스크에 노출되기 쉽다. 따라서 이러한 리스크를 사전에 파악하고 대응하면 프로젝트 목표에 대한 잠재적인 부정적 영향을 완화하는 데 도움이 될 수 있다.

건설 프로젝트의 리스크 관리에 있어서 리스크 식별은 첫 번째이자 가장 중요한 단계지만 건설 프로젝트의 리스크 관리 실무는 현재 (1) 과거 수행한 프로젝트에서 발생하거나 식별되었던 리스크를 관리하기 위한 구조화된 형식 체제 미비로 인해 전산화와 재사용 및 공유 미흡 (2) 터널과 같은 복잡한 건설 프로젝트를 관리하기 위해 인력 위주의 노동집약적 방식으로 인하여 프로젝트 성과와 생산성에 부정적인 영향이 발생한다.

따라서 본 연구는 건설 프로젝트에서 리스크 식별 프로세스의 효율성과 정확성을 향상시키기 지식 그래프와 GPT-3 모델을 활용하였다. 지식 그래프 개념은 Neo4j를 활용하여 NATM 터널에 대한 일반 리스크 지식 그래프를 개발하였다. 또한 대형 언어 모델의 일종인 GPT-3를 활용하여 건설 리스크에 대한 지능형 질문 응답 시스템(QAR-System)을 개발하여 NATM터널에서 발생하는 리스크에 대하여 효율적으로 식별할 수 있도록 하였다. 이를 통해 신규 건설 프로젝트 계획 시 과거 프로젝트 문서 검토에 소요되는 수작업과 시간을 단축할 수 있도록 하였다.

본 연구에서 개발한 QAR-System은 터널 프로젝트의 초기 단계에 활용하여 의

사결정 단계에 신뢰성 있는 참고자료를 제공하여 안전사고를 예방하고, 프로젝트 지연 및 비용 초과를 효과적으로 관리할 수 있을 것으로 기대된다.

본 연구의 기여하는 바는 크게 세 가지다. (1) 본 연구는 지식 그래프 방법론을 활용하여 과거 프로젝트에서 수집된 리스크 지식을 개념화하고 체계적 전산화를 통한 재사용 및 공유를 지원하는 프레임워크를 제공한다. (2) 본 연구는 GPT-3와 같은 LLM 기술을 건설영역에 적용하여 건설 프로젝트의 생애주기 전반에 걸쳐 적시에 효과적이고 신뢰할 수 있는 의사결정을 제공할 수 있는 기반을 마련한다. (3) 본 연구는 AI기술을 활용하여 리스크 관리 및 건설관리에 정보기반 의사결정 기반을 제시하고 건설 프로젝트의 성과와 생산성을 향상시킨다.

APPENDIX A: Abbreviations

The following abbreviations are used in this thesis:

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
CQL	Cypher Query Language
GDP	Gross Domestic Product
GPT	Generative Pre-trained Transformer
IR	Information Retrieval
IHMI	Integrated Human-Machine Intelligence
KBQAS	Knowledge-Based Question-Answering System
KBRM	Knowledge-Based Risk Management
KBS	Knowledge-based System
KG	Knowledge Graph
KM	Knowledge Management
LLM	Large Language Model
ML	Machine Learning
NATM	New Austrian Tunnel Method
NL	Natural Language
NLP	Natural Language Processing
NLQ	Natural Language Question
PMBOK	Project Management Book of Knowledge
QA	Question Answering
QAR-System	Question-Answering System for Construction Risk
QAS	Question-Answering System
RBS	Risk Breakdown Structure
RisKG	Risk Knowledge Graph
RM	Risk Management
SQL	Structured Query Language
UI	User Interface
WBS	Work Breakdown Structure

APPENDIX B: Risk Data of Tunnel Construction Projects

Work Type (WBS)	Risk Category	Risk	Risk Factor
1. Excavation (NATM tunnel)	1.1. Construction	1.1.1. Increased vertical (horizontal) strain	1.1.1.1. Heaven (Tunnel Ceiling) Deformation
			1.1.1.2. Adjacent to the crusher
			1.1.1.3. Sidewall deformation
			1.1.1.4. Excessive displacement
			1.1.1.5. Tunnel subsidence
			1.1.1.6. Excessive internal displacement
		1.1.2. Collapse (Tunnel collapse, subsidence)	1.1.2.1. Crushing zone, progressive over burrowing
			1.1.2.2. Insufficient coping with rapid changes in membrane lipids
			1.1.2.3. Unsecured bearing capacity of existing bedrock
			1.1.2.4. Unsecured safety of support materials (shotcrete, rock bolts, steel supports)
			1.1.2.5. Non-compliance with design excavation site
			1.1.2.6. Appearance of abnormal geologic zone and occurrence of water
		1.1.3. Collapse of the posterior membrane	1.1.3.1. Insufficient rear wall lock bolt and support work
		1.1.4. Economic loss due to under-excavation and over-excavation	1.1.4.1. Excessive use of explosives
			1.1.4.2. Survey error
			1.1.4.3. Failure to comply with the proper amount of gunpowder
			1.1.4.4. Non-compliance with design excavation site
			1.1.4.5. Joint, fault zone impact
		1.1.5. Cost increase due to drilling alignment error	1.1.5.1. Move the location of the survey datum point (insufficient confirmation)
		1.1.6. Economic loss due to deterioration of blasting efficiency	1.1.6.1. Insufficient coloring work, many joints and wide gaps

Work Type (WBS)	Risk Category	Risk	Risk Factor
1. Excavation (NATM tunnel)	1.2. Management	1.2.1. Damage to surrounding structures (roads, buildings, underground pipes, utility poles, etc.)	1.2.1.1. Too much charge
			1.2.1.2. Vibration value exceeded during blasting
			1.2.1.3. Inadequate investigation of underground facilities
			1.2.1.4. Excess displacement during excavation
		1.2.2. Damage to surrounding structures (roads, buildings, underground pipes, ground poles, etc.), sinkholes	1.2.2.1. Groundwater level drop
			1.2.2.2. Groundwater runoff during excavation
			1.2.2.3. Lowering of the groundwater level in urban low-depth tunnels
			1.2.2.4. Insufficient groundwater layer reinforcement, non-treatment of water during drilling
		1.2.3. Possibility of civil complaints for damage to fixed-temperature facilities such as neighboring houses and barns, delay in construction	1.2.3.1. Blasting noise and vibration effects
			1.2.3.2. Too much charge
		1.2.4. Risk of falling after excavation and before support installation	1.2.4.1. Lack of design and judgment according to rock quality (blasting pattern, reinforcement method, etc.)
			1.2.4.2. Risk of falling until the first shotcrete is poured
			1.2.4.3. Insufficient clean-up of dead ends
		1.2.5. Delayed construction period	1.2.5.1. Calculation of appropriate time considering excavation pattern and recent blasting permission time
			1.2.5.2. (Types 4 to 6 are between 7:00 and 19:00 when two blasts are performed per day, and types 1 and 3 are finished between 7:00 and 11:00 the next day when 2 blasts are performed per day. increase permit time)
			1.2.5.3. Rocky non-uniform section
			1.2.5.4. Excavation method change and rock determination
	1.3. Safety	1.3.1. Worker electric shock	1.3.1.1. Jumbo Drill Temporary Wire Leakage Current

Work Type (WBS)	Risk Category	Risk	Risk Factor
1. Excavation (NATM tunnel)	1.3. Safety	1.3.2. Worker being trapped	1.3.2.1. Charging car malfunction: unskilled operator
			1.3.2.2. Pumice stone removal
			1.3.2.3. Close-up work
			1.3.2.4. Poor visibility due to dust after blasting
		1.3.3. Occupational accidents related to workers' diseases	1.3.3.1. Poor working conditions in the pit due to gas and dust after blasting
		1.3.4. Fire	1.3.4.1. Concern about fire due to blasting during tunnel excavation
		1.3.5. Explosion	1.3.5.1. Risk of explosion from striking residual unexploded gunpowder
	1.4. Technology	1.3.6. Charging car conduction	1.3.6.1. Ground weakening (soft ground)
		1.3.7. Car overturn accident	1.3.7.1. Steep slope of sand pit material transport route
2. Support treatment	2.1. Management	2.1.1. Delay due to earthwork flow imbalance	2.1.1.1. Disruption in the selection of adjacent fill area or sand pit.
			2.1.1.2. Additional quantity due to over-excavation
			2.1.1.3. When rock filling is used, additional work types for rock filling are generated.
		2.1.2. Damaged dump truck cover and rockfall	2.1.2.1. Overloading/Unloading
		2.1.3. Dump overturning in case of prop failure	2.1.3.1. Loading box jam due to excessive load size,
			2.1.3.2. Poor ground conditions at the landing site
			2.1.3.3. Traffic damage due to speeding and overloading
	2.2. Safety	2.2.1. Safety accident	2.2.1.1. Fall due to poor loading and overloading during propulsion transportation using buckets
			2.2.1.2. Fall due to non-execution of workplace control during bucket movement and rotation using a crane
			2.2.1.3. Fire due to overheating of dump truck and loading equipment in the tunnel

Work Type (WBS)	Risk Category	Risk	Risk Factor
2. Support treatment	2.2. Safety	2.2.1. Safety accident	2.2.1.4. Fall due to wire rope breakage during lifting with a crane
			2.2.1.5. Overturning and narrowing of running vehicles due to lack of illumination and poor zoning of movement lines
			2.2.1.6. Accidents occur when moving workers with upper supports
			2.2.1.7. Equipment narrowing during loading and unloading
			2.2.1.8. Unsecured safety due to increased driver fatigue
			2.2.1.9. In the case of a railway tunnel with a small cross-sectional width, the dump truck has to come out backward when handling the support, so there is a risk of colliding and colliding with workers.
			2.2.1.10. Collision with other equipment and workers due to speeding in tunnels and construction roads
			2.2.1.11. Risk of collision and suffocation with nearby workers due to narrow space during loading work
	2.3. Construction	2.3.1. Waste environmental issues	2.3.1.1. Difficulties in securing a site for a temporary stockyard in the complex
			2.3.1.2. Mixed with shotcrete repellent material and taken out
			2.3.1.3. Shotcrete is mixed with the blasting stone during buckling, causing environmental issues.
			2.3.1.4. Noise and dust generated due to damage caused by the support vehicle
		2.3.2. Increase in economic loss and cost (division, waste disposal)	2.3.2.1. Consideration of equipment combination considering transport distance and support size
			2.3.2.2. Addition of tent installation cost for separating shotcrete classified as waste
			2.3.2.3. Excavation rock transport required
			2.3.2.4. Cycle time increase due to unsecured stocking
			2.3.2.5. Due to the shotcrete rebound, it is mixed in the holding force when excavating the lower half
			2.3.2.6. Increase in noise, vibration, and cycle time due to breakdown

Work Type (WBS)	Risk Category	Risk	Risk Factor
2. Support treatment	2.3. Construction	2.3.3. Additional yard storage permits and increased yard storage costs	2.3.3.1. In the case of a long tunnel, a separate storage yard is required
3. Formwork	3.1. Construction	3.1.1. Formwork transformation	3.1.1.1. Stress concentration on the right leg
			3.1.1.2. Formwork transverse hook not fastened
			3.1.1.3. Delayed installation of steel girders, insufficient maintenance of the arch shape due to severe irregularities in the original ground
		3.1.2. Tunnel deformation	3.1.2.1. Occurrence of the section where steel support is not installed due to excessive excavation site
		3.1.3. Lower support effect	3.1.3.1. Construction is difficult by water
			3.1.3.2. Structural instability due to poor bolting connection between upper and lower halves
		3.1.4. Decreased support effect (separation behavior)	3.1.4.1. H-beam support member shotcrete casting creates a gap on the back side
		3.1.5. Steel support beam deformation and damage	3.1.5.1. Poor support member connection
		3.1.6. Unexploded strong beam linear mismatch	3.1.6.1. Insufficient shotcrete thickness around formwork (decreased durability)
		3.1.7. Support deflection	3.1.8.1. Too much quick-setting shotcrete
			3.1.8.2. Decreased ground support capacity
	3.2. Safety	3.2.1. Formwork Falling Accident	3.2.1.1. Material lifting rope peeling off
		3.2.2. Worker safety accident	3.2.2.1. Accident caused by a worker falling while working on support installation.
			3.2.2.2. Worker collision due to fall of support material while transporting support material
		3.2.3. Strong support conduction occurs	3.2.3.1. Insufficient support and excessive weight
		3.2.4. Strong support for overturning and subsidence	3.2.4.1. Poor support
	3.3. Technology	3.3.1. Decrease in durability	3.3.1.1. Difficulty in securing internal holes and reinforcing bar coating due to support material deformation or curvature error

Work Type (WBS)	Risk Category	Risk	Risk Factor
3. Formwork	3.3. Technology	3.3.2. Strong support movement or warping occurs	3.3.2.1. Installation error occurs when tunnel auxiliary construction method (pipe loop, pore pulling) is performed in parallel
	3.4. Design	3.4.1. Delay due to scope change	3.4.1.1. Produce new types when changing types according to cancer determination during excavation
		3.4.2. Sunk costs incurred	3.4.2.1. Pre-prepared materials are not required when changing the type due to cancer determination during excavation
		3.4.3. Suspension of construction due to the need for reinforced steel support	3.4.3.1. Discrepancy between design and actual ground condition
4. Shotcrete Work	4.1. Construction	4.1.1. Arch formation difficulty	4.1.1.1. Long pouring time
		4.1.2. Cost increases due to excessive use of shotcrete material	4.1.2.1. Occurrence of over burrow
		4.1.3. Insufficient secondary support	4.1.3.1. Over-pouring of shotcrete
		4.1.4. Defective shotcrete attachment	4.1.4.1. Increased thickness of the first layer of casted shotcrete due to the presence of foreign matter on the surface.
			4.1.4.2. Insufficient cleaning of the pouring surface before construction (failure to remove floating stones and pumice stones, etc.).
		4.1.5. Collapse	4.1.5.1. Shotcrete pouring delay
	4.2. Safety	4.2.1. Occupational accidents of workers	4.1.6.1. Insufficient rebound material treatment
			4.1.6.2. Environmental problems arise due to the inability to separate waste from shotcrete sealing
			4.2.1.1. Dust penetration into the human body
	4.3. Technology	4.3.1. Defective shotcrete attachment	4.2.1.2. Overloaded connections due to clogged shotcrete nozzles
			4.3.1.1. Lack of steel fiber content and length
			4.3.1.2. Failure to establish appropriate drainage measures for the water-generating section
			4.3.1.3. Poor quality of quick payment

Work Type (WBS)	Risk Category	Risk	Risk Factor
4. Shotcrete Work	4.3. Technology	4.3.2. Insufficient shotcrete strength.	4.3.2.1. Poor aggregate granularity.
		4.3.3. Increased shotcrete disposal cost.	4.3.3.1. Increased rebound.
			4.3.3.2. Defective shotcrete material (sand with much soil, problems with quick fixing, etc.).
			4.3.3.3. Non-compliance with shotcrete placement angle.
5. Rock bolt	5.1. Construction	5.1.1. Rock bolt installation is not possible.	5.1.1.1. Excessive outflow of groundwater.
			5.1.1.2. Collapse of the drilled holes in poor ground.
		5.1.2. Fastening loosening due to excessive stress caused by blasting vibration, etc.	5.1.2.1. Fastening with a force exceeding the yield strength.
		5.1.3. Lower support effect	5.1.3.1. Flowing down of the fixing material from the ceiling rock bolt
			5.1.3.2. Insufficient rock bolt preload due to blasting without sufficient curing time for rock bolt grouting.
		5.1.4. Increase in construction cost	5.1.4.1. Excessive injection of mortar in cracks.
		5.1.5. Lack of support	5.1.5.1. Poor construction
	5.2. Management	5.2.1. Damaged waterproof sheet	5.2.1.1. Damage to waterproofing sheets due to failure to remove wire for construction verification
	5.3. Safety	5.3.1. Worker constriction, collision, accident, damage	5.3.1.1. Falling to the rock bolt workbench with overloaded material
			5.3.1.2. Falling rock bolt lifting due to poor lifting equipment and work practices.
			5.3.1.3. Worker struck due to insufficient mounting during rock bolt withdrawal test.
			5.3.1.4. Falling of rock bolts due to self-weight when installing rock bolts and collision with workers below.
			5.3.1.5. Fall due to self-weight, fear of collision with lower workers
			5.3.1.6. Risk of falling workers by opening the charging car door when loading lock bolts in the charging car.

Work Type (WBS)	Risk Category	Risk	Risk Factor
5. Rock bolt	5.3. Safety	5.3.1. Worker constriction, collision, accident, damage	5.3.1.7. Construction of rock bolts at one time after excavating two layers to reduce construction time and risk of falling.
		5.3.2. Safety accident	5.3.2.1. Injuries during rock bolt insertion and grouting work
			5.3.2.2. Falling when inserting and grouting rock bolts at the top end
			5.3.2.3. Safety accidents due to plastic fragments when using resin
			5.3.2.4. Safety accident risk due to selection of rock bolt filling material.
			5.3.2.5. Difficulty in constructing rock bolts due to difficulty in constructing the design length
	5.4. Technology	5.4.1. Reduced rock bolt effect	5.4.1.1. Failure to consider the joint direction and insufficient adhesion to the pressure plate shotcrete surface.
		5.4.2. Poor installation of rock bolt	5.4.2.1. Fixing material dripping
			5.4.2.2. Difficulty in verifying settlement
			5.4.2.3. Number of drilled holes.
		5.4.3. Mortar quantity increase	5.4.3.1. Excessive joints
6. Lining	6.1. Construction	6.1.1. Occurrence of cracks in the ceiling	5.4.3.2. Fault zone emergence
			5.4.3.3. Groundwater seepage
			5.4.4. Lack of support
			5.4.4.1. Insufficient rock bolt length, off-specification
			6.1.1.1. Difficulty in injecting ready-mixed concrete at the top end
			6.1.1.2. Difficulty compacting concrete
			6.1.1.3. Poor mix
			6.1.1.4. Underground water occurs due to damage to the waterproofing membrane

Work Type (WBS)	Risk Category	Risk	Risk Factor
6. Lining	6.1. Construction	6.1.2. Leak	6.1.2.1. Underground water leakage due to water pressure in the lining due to damage to the waterproof membrane
			6.1.2.2. Split casting of joint wall
			6.1.2.3. Damage to the waterproof membrane when assembling rebar
			6.1.2.4. Poor construction of water stop plate.
		6.1.3. Longitudinal cracks, lining cracks, and breakage at the ceiling.	6.1.3.1. Lining back settlement.
			6.1.3.2. Long-term hydraulic load on the lining due to poor water pressure relief in the drainage tunnel.
			6.1.3.3. Cracks in mass concrete in cross-sectional tunnel section
			6.1.3.4. Reconstruction due to poor interface in subsequent processes (railway system, road disaster prevention).
			6.1.3.5. Shrinkage and non-compliance with installation spacing and depth of construction joints.
			6.1.3.6. Construction after convergence of main beam displacement.
			6.1.3.7. Insufficient vibration during casting and cold joint with insufficient spacing.
			6.1.3.8. Water pressure is generated in the lining.
		6.1.4. Additional cost incurred	6.1.4.1. Expenses incurred due to the use of vertical car lift and movement within the tunnel (mixer truck transport).
			6.1.4.2. Transportation costs cannot be applied when contracting for materials to be paid (mixer truck transportation)
			6.1.4.3. Surface replacement after construction of lead haunch section.
			6.1.4.4. Defects in the finished surface of the connection due to separate casting of the lining lead haunch area.
			6.1.4.5. Surface replacement costs due to defective lining surface and damage to lining foam.
		6.1.5. Elution water generation	6.1.5.1. Excessive water generation
			6.1.5.2. Problems with concrete adhesion, strength, and thickness.

Work Type (WBS)	Risk Category	Risk	Risk Factor
6. Lining	6.2. Safety	6.2.1. Occurrence of accidents, worker entrapment, and collision	6.2.1.1. Fall during setting work at the top of the form for lining casting work
			6.2.1.2. Risk of falling during work such as welding the top of the lining
			6.2.1.3. Falling during movement for assembling the outer foam of the opening lining
			6.2.1.4. Fall due to failure to install handrails on top of rebar between rebar sections.
			6.2.1.5. Falls while installing a workbench due to workers not wearing safety belts.
			6.2.1.6. Falls during work due to failure to install safety railings at the end of the rebar workbench
			6.2.1.7. Falling while going up and down the rebar workbench without lifting facilities
			6.2.1.8. Fall in the construction of each piece of lining foam.
			6.2.1.9. Fall material between the footplates inside the fall truck and the worker fell through the central passageway
			6.2.1.10. Rebar falling while being hoisted due to rebar hoist rope breakage.
			6.2.1.11. Collision and fall due to vibration of rebar during lifting work with 1 rope hanger
			6.2.1.12. Collision with a forklift moving backwards due to the lack of a warning light on the rear of the forklift
			6.2.1.13. During the lifting of the rebar, the rope fell off due to a release device not being installed on the hook, causing the rebar to fall.
			6.2.1.14. Electric shock while cutting and bending rebar with a rebar processor
			6.2.1.15. Neck and back pain caused by working toward the ceiling (musculoskeletal disease).
			6.2.1.16. Collapse during installation and dismantling of the rebar workbench due to unskilled work procedures.

Work Type (WBS)	Risk Category	Risk	Risk Factor
6. Lining	6.2. Safety	6.2.1. Occurrence of accidents, worker entrapment, and collision	6.2.1.17. Fall due to excessive gap between the truck work platform and the tunnel wall.
			6.2.1.18. Bronchial damage caused by flying dust during lining foam sanding work.
			6.2.1.19. Overturn/collapse due to the use of unsuitable materials in place of rebar.
			6.2.1.20. Workers fall and slip due to poor material arrangement of lining foam stairs and work scaffolding.
			6.2.1.21. Electric shock and fire during lining foam welding and fusing work.
			6.2.1.22. Danger due to fire and lack of oxygen during curing using a hot air fan in winter
			6.2.1.23. Nearby workers stabbed by rebar during rebar placement
			6.2.1.24. Optic nerve damage as rebar powder gets into the eye while assembling rebar toward the ceiling
			6.2.1.25. In an inclined shaft, a steel workbench rolls on a slope and collapses due to external collision.
			6.2.1.26. Traffic jam accident due to failure to place a signalman and failure to install a vehicle stopper while the concrete mixer truck is reversing
			6.2.1.27. Fall accident due to failure to install inspection window opening cover during lining placement
			6.2.1.28. Workers are trapped due to malfunction due to the foot switch cover not being installed during cutting work for rebar assembly
			6.2.1.29. Constriction accident due to insufficient fixation of the rebar between the slope section
			6.2.1.30. Risk of damage/collapse due to non-compliance with formwork bogie structural review and arbitrary changes
			6.2.1.31. Collapse due to non-compliance with scaffolding structural review and non-compliance with assembly order

Work Type (WBS)	Risk Category	Risk	Risk Factor
6. Lining	6.2. Safety	6.2.1. Occurrence of accidents, worker entrapment, and collision	6.2.1.33. Fall due to poor handling, unloading, or transportation of reinforcing bar materials
			6.2.1.34. Fall from workbench due to movement
			6.2.1.35. Concerns about collision and narrowing when the remicon vehicle reverses
		6.2.2. Fire	6.2.2.1. Risk of electric shock and fire during lining foam welding and fusing work
			6.2.2.2. Danger due to fire and lack of oxygen during curing using a hot air fan in winter
	6.3. Technology	6.3.1. Architectural limits	6.3.1. 1. Unsecured construction limits
7. Waterproof	7.1. Construction	7.1.1. Leak	7.1.1.1. Poor welding of waterproof paper at connection part
			7.1.1.2. Bad culvert connection at the bottom
			7.1.1.3. Reinforced sagging anchor connection, not air tested
			7.1.1.4. Connection part air test not conducted
			7.1.1.5. Damage to waterproof sheet during reinforcement work
			7.1.1.6. Poor welding of waterproof sheet joints
			7.1.1.7. Torn waterproof sheet during rebar work
			7.1.1.8. Poor connection between construction joint and waterproof sheet
			7.1.1.9. Leakage due to lack of waterproofing membrane overlapping
			7.1.1.10. Leakage due to poor connection of waterproof membrane longitudinal drainpipe
			7.1.1.11. Watertight plate damage due to rebar assembly and rebar fixed anchor construction
			7.1.1.12. NATM-Cut Tunnel Joint

Work Type (WBS)	Risk Category	Risk	Risk Factor
7. Waterproof	7.1. Construction	7.1.1. Leak	7.1.1.13. Influence of the surrounding groundwater level
		7.1.2. Accidents caused by insufficient surface clearance in the tunnel	7.1.2.1. Induced cracks in the lining due to damage to the waterproofing film.
		7.1.3. Sagging of waterproof paper	7.1.3.1. Installation of shotcrete waterproofing fixing device at regular intervals
		7.1.4. Dropped after installation of tunnel waterproofing membrane	7.1.4.1. Strong wind inflow into the tunnel, insufficient quantity of fixing facilities installed at end connections
		7.1.5. Waterproof membrane damage	7.1.5.1. Shotcrete surface has severe irregularities and damage during the construction of reinforcing bars and formwork
		7.1.6. Damaged waterproof sheet	7.1.6.1. Damage when installing lining foam
			7.1.6.2. When the lining is placed, the waterproof sheet is torn due to the weight of the concrete
			7.1.6.3. Tearing caused by shotcrete protrusions
			7.1.6.4. Torn due to lack of rock bolt head trim
	7.2. Safety	7.2.1. Accidents, damages (due to high-place work, excessive movement, working environment, working conditions, welding work, material falling, collision, etc.)	7.2.1.1. Pain in the neck and back because of working toward the ceiling
			7.2.1.2. Lower back pain due to excessive movements, such as moving up and down the waterproof seat cart
			7.2.1.3. Falling due to self-weight of sheet members, collision, etc., while lifting the waterproof sheet to the workbench
			7.2.1.4. Falling accidents due to slipping on stairs due to insufficient lighting when getting on and off the workbench
			7.2.1.5. Collision of the truck by a moving vehicle under the truck, such as an excavator
			7.2.1.6. A worker falling through the center aisle due to material falling between the inner scaffolding of the cart.

Work Type (WBS)	Risk Category	Risk	Risk Factor
7. Waterproof	7.2. Safety	7.2.1. Accidents, damages (due to high-place work, excessive movement, working environment, working conditions, welding work, material falling, collision, etc.)	7.2.1.7. Worker collision due to falling upper material while moving work cart
			7.2.1.8. When installing and dismantling the waterproof workbench, overturning and collapsing due to unlearned work order in advance
			7.2.1.9. Burns during fusion welding of waterproof sheet
			7.2.1.10. Concerns about loosening and deformation of member connections due to damage accumulated during movement of the work cart due to irregularities in the floor, poor inclination of the upper foot of the cart
			7.2.1.11. Fall/collapse due to the use of unqualified materials for the waterproof bogie
			7.2.1.12. Fall due to excessive gap between the platform and the tunnel wall
			7.2.1.13. Electric shock accident due to use of outdated welding machine and failure to install electric shock arrestor
			7.2.1.14. Fall while working on the waterproof sheet at the top of the waterproof cart
			7.2.1.15. Falls while installing a workbench due to workers not wearing seat belts
			7.2.1.16. Fall while lifting the waterproof sheet to the top of the workbench
			7.2.1.17. Stumble and fall while forcibly working at the end of the workbench
			7.2.1.18. Electric shock accident due to use of outdated welding machine and failure to install electric shock arrestor
		7.2.2. Fire	7.2.2.1. Fire occurred during welding/cutting work around the area after installing the waterproof sheet
			7.2.2.2. Waterproof sheet fire due to overheating of bonding equipment
	7.3. Management	7.3.1. Damaged waterproof sheet	7.3.1.1. Tear due to foreign (nail, fixed iron, etc.) matter in shotcrete

Work Type (WBS)	Risk Category	Risk	Risk Factor
7. Waterproof	7.3. Management	7.3.1. Damaged waterproof sheet	7.3.1.2. Difficulty in repairing damaged parts
			7.3.1.3. Difficulty in checking the joint condition of the damaged part
			7.3.1.4. Impossible to check micro-damage.
			7.3.1.5. Damage caused by fixtures when fixing the waterproof sheet
			7.3.1.6. Carelessness when assembling the lining steel formwork
	7.4. Technology	7.4.1. Poor connection of waterproof sheet	7.4.1.1. Inadequate adapter temperature.
			7.4.1.2. Difficulty in identifying defective joints
			7.4.1.3. Improper connection speed
			7.4.1.4. Insufficient joint spacing (insufficient joint length)
			7.4.1.5. Poor connection between waterproof sheet and non-woven fabric
8. Drain	8.1. Construction	8.1.1. Linkage of groundwater back into the lining	8.1.1.1. Insufficient induced drainage before waterproofing work
		8.1.2. Leakage at construction joints	8.1.2.1. Construction joints caused by separate casting
		8.1.3. Clogged drain	8.1.3.1. Inflow of foreign substances due to carelessness during construction
		8.1.4. Roadbed uplift	8.1.4.1. Generation of drainage water pressure
		8.1.5. Possibility of uplift on the pavement surface due to uplift	8.1.5.1. Water reservoir in the lower part of the geologic zone
		8.1.6. Subgrade separation and deformation	8.1.6.1. Poor drainage
		8.1.7. Ceiling settlement due to groundwater level drop	8.1.7.1. Over drain
	8.2. Management	8.2.1. Damaged packaging	8.2.1.1. Poor drainage of the bottom outlet
	8.3. Safety	8.3.1. Safety accident	8.3.1.1. Risk of falling when working at height
	8.4. Technology	8.4.1. Culvert depth difference caused by excavation type (ex., inverted section, normal section, etc.)	8.4.1.1. Design error

Work Type (WBS)	Risk Category	Risk	Risk Factor
9. Common area	9.1. Safety	9.1.1. Collapses when used as an emergency evacuation passage	9.1.1.1. Lack of rigidity
		9.1.2. Electric shock accident due to non-use of machinery and equipment.	9.1.2.1. Leakage current, grounding, earth leakage breaker not installed
		9.1.3. Risk of falling accidents due to poor installation of worker passageways	9.1.3.1. Floor irregularities
		9.1.4. Accidental cuts from high-speed rotating equipment.	9.1.4.1. Floor irregularities
		9.1.5. Eye damage from residual concrete fragments	9.1.5.1. Insufficient safety measures during sanding of utility tunnel formwork.
	9.2. Technology	9.2.1. Separation and level difference on the mounting surface of the common hole lid	9.2.1.1. Defect in the mounting surface of the joint lid.
	9.3. Construction	9.3.1. Poor settlement of new and old joint	9.3.1.1. Excessive split pouring of utility tunnel.
		9.3.2. Crack occurrence	9.3.2.1. No cutting
10. Shaft gates and retaining walls	10.1. Safety	10.1.1. Shaft gate-slope confined space disease occurrence	10.1.1.1. Lack of oxygen during asphalt coating work
	10.2. Construction	10.2.1. Shaft slope collapse and loss.	10.2.1.1. Neglecting the back slope after installing the gate.
			10.2.1.2. Insufficient shaft length or parapet height during heavy rains.
			10.2.1.3. Continuous loss of slopes in the shaft section due to rain
			10.2.1.4. Insufficient drainage on the side of the shaft
		10.2.2. Retaining wall sliding and ground settlement.	10.2.2.1. Insufficient backfilling of the retaining wall
		10.2.3. Retaining wall collapse.	10.2.3.1. Insufficient infiltrate treatment facility in retaining wall causes an increase in back earth pressure
			10.2.3.2. Increase in active soil pressure on retaining walls due to increase in volume of frozen soil in winter
			10.2.3.3. Increase in retaining wall dynamic earth pressure due to increased frozen soil volume in winter.

Work Type (WBS)	Risk Category	Risk	Risk Factor
10. Shaft gates and retaining walls	10.2. Construction	10.2.3. Retaining wall collapse	10.2.3.4. Insufficient compaction of the ground under the retaining wall
			10.2.3.5. Cracks and damages due to poor placement management
		10.2.4. Shaft top slope failure	10.2.4.1. Inappropriate shaft type or location
			10.2.4.2. Installation of shaft doors on steep slopes or large-cut slopes
			10.2.4.3. Inadequate shaft drainage.
		10.2.5. Shaft deformation	10.2.5.1. Shaft installed on a one-sided slope, causing lateral soil pressure
11. Shaft part (Shaft slope)	11.1. Construction	11.1.1. Falling rock and stones	11.1.1.1. Horizontal cuts in tunnel shaft rock in the direction of excavation
			11.1.1.2. Absence of shaft slope reinforcement
			11.1.1.3. Insufficient pumice clearance at the top of the tunnel slope.
		11.1.2. Shaft door damage	11.1.2.1. Rockfall, avalanche
		11.1.3. Shaft slope collapse	11.1.3.1. Poor slope protection
			11.1.3.2. Topography
			11.1.3.3. Failure to select an appropriate construction method to secure slope stability
			11.1.3.4. Discrepancy between the design shaft slope and the actual shaft slope on the ground.
			11.1.3.5. Slope reinforcement not implemented after shaft part installation.
			11.1.3.6. Stability not reviewed when cutting slope, inappropriate selection of reinforcement type according to review.
			11.1.3.7. Improper shaft location (valley, piled soil layer, etc.)
			11.1.3.8. Slope collapse due to a large amount of surface water inflow from concentrated lakes
			11.1.3.9. Slope collapse due to ground weakness caused by rainfall during the rainy season.

Work Type (WBS)	Risk Category	Risk	Risk Factor
11. Shaft part (Shaft slope)	11.1. Construction	11.1.4. Deformation and collapse	11.1.4.1. Measurement management standard exceeded
			11.1.4.2. Step-by-step over excavation
			11.1.4.3. Lack of earth anchor fixing effect
			11.1.4.4. Use of defective materials (deformed oral materials, etc.)
			11.1.4.5. Background tensile crack
12. Auxiliary facilities	12.1. Construction	12.1.1. Intensification of cracks due to gaps at the top of the lining	12.1.1.1. Impact of load on facilities (signboards, jet fans) at the top of the lining
	12.2. Management	12.2.1. Poor air quality inside the tunnel	12.2.1.1. Air supply system centered on dead surface
		12.2.2. Noise complaints and air delays	12.2.2.1. Impossible to install soundproof doors until a certain section is entered.
		12.2.3. Water pollution	12.2.3.1. After tunnel opening, it is necessary to reduce PH for a certain period, but it is reflected in the design only during the excavation period, and facilities that can reduce PH for other periods are not reflected in the design.
			12.2.3.2. Wall washing water generated during wall washing in the tunnel should be separated and treated separately, but in many cases, it is not reflected in the design
	12.3. Safety	12.3.1. Electric shock	12.3.1.1. Insufficient distribution box management during electrical construction
			12.3.1.2. Inadequate management of electrical facilities in tunnels
			12.3.1.3. Inadequate installation of ventilation facilities in tunnels
		12.3.2. Suffocation accident	12.3.2.1. Insufficient ventilation during use of chemicals in confined spaces
		12.3.3. Fire	12.3.3.1. Fire caused by sparks during welding/cutting work during miscellaneous steel construction
13. Instrumentation	13.1. Construction	13.1.1. Accidents such as cracks and collapses	13.1.1.1. Lack of reliability in measured values
			13.1.1.2. Displacement error occurs due to not conducting initial measurement after installation

Work Type (WBS)	Risk Category	Risk	Risk Factor
14. Auxiliary method (steel pipe multi-stage grouting)	14.1. Construction	14.1.1. Grouting Injection Pipe Crashes Due to Pipe Vibration	14.1.1.1. Grouting injection pipe clogged
		14.1.2. Wall collapse	14.1.2.1. Steel pipe insertion and caulking work not carried out immediately after drilling
			14.1.2.2. Groundwater leakage and soil leakage after drilling is completed
		14.1.3. Inadequate ground improvement (consolidation of joint surface) due to early hardening of grouting material	14.1.3.1. Poor mixing of hardener
		14.1.4. Drilling slime flows into the steel pipe	14.1.4.1. Damaged steel pipe inlet bending part
		14.1.5. Collapse	14.1.5.1. Cave-in due to overburden
	14.2. Management	14.2.1. Material fall	14.2.1.1. Concerns about worker collision in the event of sliding or falling due to self-weight after inserting multi-stage steel pipe materials
			14.2.1.2. Unbound wire and falls during the transportation of multi-stage steel pipes
			14.2.1.3. Defective hook release device
	14.3. Safety	14.3.1. Risk of electric shock	14.3.1.1. Mixer power supply, short circuit in distribution panel
		14.3.2. Safety accident	14.3.2.1. Fall, collision, and constrictions during steel pipe insertion
			14.3.2.2. When inserting a steel pipe after drilling, the pipe is pushed back, and there is a risk of hitting workers in the rear
	14.4. Technology	14.4.1. Collapse due to lack of support	14.4.1.1. Insufficient design length due to poor ground
			14.4.1.2. Grouting pressure not secured due to cracks, etc.
			14.4.1.3. Insufficient water and reinforcement
			14.4.1.4. Lack of required support
		14.4.2. Poor grouting strength	14.4.2.1. Poor grouting mixing ratio
	14.5. Design	14.5.1. Waiting cost due to scope change	14.5.1.1. Type change in the tunnel.
			14.5.1.2. Personnel and equipment input problems when changing the presence or absence of multi-stage steel pipes

Work Type (WBS)	Risk Category	Risk	Risk Factor
15. Ground investigation before construction	15.1. Technology	15.1.1. Construction cost increase and construction delay due to ground grade (pattern) change	15.1.1.1. Poor geotechnical survey results from underbidding by geotechnical survey company.
			15.1.1.2. Poor GPS reception in mountainous terrain makes it difficult to drill in the correct location.
			15.1.1.3. Estimation analysis after a geotechnical survey of adjacent sections due to inability to enter equipment in mountainous terrain.
		15.1.2. Increased construction cost and delayed construction due to additional ground investigation	15.1.2.1. Generation of additional factors reflecting physical exploration
			15.1.2.2. Failure to conduct an appropriate investigation according to tunnel type, etc.
			15.1.2.3. Inexperienced ground survey company.
			15.1.2.4. Possible existence of vulnerable ground in the survey omission section.
		15.1.2. Increased construction cost and delayed construction due to additional ground investigation	15.1.2.5. Due to poor ground investigation at the time of design, it differs from the ground investigation result before construction.
			15.1.2.6. Ground survey not conducted due to equipment inaccessibility.
		15.1.3. Increased construction cost and delayed construction due to soft ground	15.1.3.1. Impossible to reflect design when creating discontinuous soft ground.
			15.1.3.2. Different design and construction survey
		15.1.4. Groundwater level drop	15.1.4.1. Insufficient preliminary investigation of groundwater level
			15.1.4. Groundwater contamination due to insufficient treatment after boring investigation
		15.1.5. Ground disturbance	15.1.5.1. Excessive survey works due to ground volatility
			15.1.5.2. Groundwater contamination through boring holes
			15.1.5.3. Inflow of surface/groundwater in the tunnel due to poor quality of closed holes after completion of ground investigation
			15.1.5.4. Settlement due to ground disturbance by drilling water during the ground investigation of the soft layer

APPENDIX C: Validation Set and the Corresponding Generated Cypher Query

S/N	Natural language questions (Validation set)	Generated Cypher query
1	List the risk category associated with the tunnel project.	match (n:risk) return distinct n.riskcategory as response
2	What are the risk factors of "Lining" of the tunnel project?	match(n:wbs{name:"Lining"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response
3	How many risks are in the project?	match(n:risk) return count(n) As response
4	List risks associated with the "Geotechnical Investigation" of the tunnel project.	match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response
5	What are the risks of the "Excavation" of the tunnel project?	match(n:wbs{name:"Excavation"})-[:has_risk]->(m:risk) return m.name As response

APPENDIX D: Questions (Test set)

S/ N	Natural language questions (Test set)	Generated Cypher queries	Remark
1	What are the risks associated with the geotechnical investigation of the tunnel project?	match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk) return "Risk: "+ m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
2	List the works in the tunnel project.	match(n:Tunnel)-[:has_work_type]->(m:wbs) return count(m) As response	Correct
3	What are the works in the tunnel project?	match(n:Tunnel)-[:has_work_type]->(m:wbs) return m.name as response	Correct
4	List the risk factors of Excavation in the tunnel project.	match(n:wbs{name:"Excavation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: "+ m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
5	Count the number of risks in the project.	match(n:risk) return count(n) As response	Correct
6	What are the risk categories associated with the tunnel project?	match (n:risk) return distinct n.riskcategory as response	Correct
7	How many risk category is associated with the tunnel project?	match (n:risk) return count(distinct n.riskcategory) as response	Correct
8	What are the risks associated with the Rock bolting of the tunnel project?	match(n:wbs{name:"Rock bolting"})-[:has_risk]->(m:risk) return "Risk: "+ m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
9	How many risks are associated with the Geotechnical Investigation of the tunnel project?	match(n:wbs{name:"Geotechnical Investigation"})-[:has_risk]->(m:risk) return count(m) As response	Correct
10	What are the risk factors of Drain in the tunnel project?	match(n:wbs{name:"Drain"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: "+ m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
11	List risks associated with the water roofing of the tunnel project.	match(n:wbs{name:"Waterproofing"})-[:has_risk]->(m:risk) return "Risk: "+ m.name + " - "+ "Risk Category: " + m.riskcategory As response	Incorrect
12	What are the risks associated with the Water Proofing of the tunnel project?	match(n:wbs{name:"Water Proofing"})-[:has_risk]->(m:risk) return "Risk: "+ m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
13	How many risks are associated with the waterproofing of the tunnel project?	match(n:wbs{name:"Water Proofing"})-[:has_risk]->(m:risk) return count(m) as response	Correct
14	List the risk factors of Formwork in the tunnel project.	match (n:wbs{name:"Formwork"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: "+ m.name + " - Risk Factor: " + l.Riskfactor As response	Correct

S/N	Natural language questions (Test set)	Generated Cypher queries	Remark
15	What are the risks of Formwork in the tunnel project?	match (n:wbs{name:"Formwork"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
16	What are the risks of Formwork in the tunnel project	Error	Incorrect
17	Count the risks of Formwork in the tunnel project	match(n:wbs{name:"Formwork"})-[:has_risk]->(m:risk) return count(m) As response	Correct
18	What are the risk factors of Auxiliary Facilities in the tunnel project?	match(n:wbs{name:"Auxiliary Facilities"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
19	List the risks of auxiliary facilities in the tunnel project.	match (n:wbs{name:"Auxiliary Facilities"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
20	What are the risks of Instrumentation in the tunnel project?	match(n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
21	Count the risks of Instrumentation in the tunnel project.	match(n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk) return count(m) As response	Correct
22	Identify the risk factors of Steel Pipe Multi-stage Grouting in the tunnel project.	match(n:wbs{name:"Steel Pipe Multi-stage Grouting"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
23	Identify the risks of Steel Pipe Multi-stage Grouting in the tunnel project.	match (n:wbs{name:"Steel Pipe Multi-stage Grouting"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - "+ "Risk Category: " + m.riskcategory As response	Correct
24	Count the risks of Steel Pipe Multi-stage Grouting in the tunnel project.	match (n:wbs{name:"Steel Pipe Multi-stage Grouting"})-[:has_risk]->(m:risk) return count(m) as response	Correct
25	List risks associated with the Gang Gates and Retaining Walls of the tunnel project.	match (n:wbs{name:"Steel Pipe Multi-stage Grouting"})-[:has_risk]->(m:risk) return count(m) as response	Correct
26	How many risk factors are s associated with the Gang Gates and Retaining Walls of the Tunnel project.	match (n:wbs{name:"Gang Gates and Retaining Walls"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) as response	Correct
27	What are the risk factors associated with the Gang Gates and Retaining Walls of the Tunnel project?	match (n:wbs{name:"Gang Gates and Retaining Walls"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct

S/N	Natural language questions (Test set)	Generated Cypher queries	Remark
28	How many risks of Drain are in the project?	match (n:wbs{name:"Drain"})-[:has_risk]->(m:risk) return count(m) As response	Correct
29	Count the risks of Drain in the tunnel project.	match(n:wbs{name:"Drain"})-[:has_risk]->(m:risk) return count(m) As response	Correct
30	List the risks of Drain in the tunnel project.	match (n:wbs{name:"Drain"})-[:has_risk]->(m:risk) return m.name + " - " + " Risk Category: " + m.riskcategory as response	Correct
31	Count the risks of Common areas in the tunnel project.	match(n:wbs{name:"Common area"})-[:has_risk]->(m:risk) return count(m) As response	Correct
32	What are the risk factors of Instrumentation in the tunnel project?	match (n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
33	Mention the risk factors of Drain in the tunnel project.	match(n:wbs{name:"Rock bolting"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) As response	Correct
34	What are the risk factors of Rock bolting in the tunnel project?	match(n:wbs{name:"Rock bolting"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
35	List risks associated with the formwork of the tunnel project.	match (n:wbs{name:"Formwork"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response	Correct
36	List risks associated with the Shotcrete Work of the tunnel project.	match(n:wbs{name:"Shotcrete Work"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response	Correct
37	What are the risk factors in the Shotcrete Work of the tunnel project?	match(n:wbs{name:"Shotcrete Work"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
38	What are the risk factors in the Shotcrete Work?	match(n:wbs{name:"Shotcrete Work"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
39	How many risk factors are in the Shotcrete Work?	match (n:wbs{name:"Shotcrete Work"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) As response	Correct
40	How many risks are in the Shotcrete Work?	match(n:wbs{name:"Shotcrete Work"})-[:has_risk]->(m:risk) return count(m) As response	Correct

S/N	Natural language questions (Test set)	Generated Cypher queries	Remark
41	Count the risks in Support of Treatment.	match(n:wbs{name:"Support Treatment"})-[:has_risk]->(m:risk) return count(m) As response	Correct
42	Count the risk factors in Support Treatment.	match (n:wbs{name:"Support Treatment"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) as response	Correct
43	What are the risk factors in supporting treatment?	match (n:wbs {name:"Support Treatment"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
44	What are the risks in Support Treatment?	match(n:wbs{name:"Support Treatment"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response	Correct
45	What are the risks in Support Treatment of the project?	match(n:wbs{name:"Support Treatment"})-[:has_risk]->(m:risk) return "Risk: " + m.name + " - " + "Risk Category: " + m.riskcategory As response	Correct
46	What are the risks in the instrumentation of the project?	match (n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk) return m.name + " - " + "Risk Category: " + m.riskcategory as response	Correct
47	List the risk factors in the instrumentation of the project.	match(n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return "Risk: " + m.name + " - Risk Factor: " + l.Riskfactor As response	Correct
48	Count the risk factors in the instrumentation of the project.	match(n:wbs{name:"Instrumentation"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) As response	Correct
49	How many risk factors are in the Steel Pipe Multi-Stage Grouting of the project?	match(n:wbs{name:"Steel Pipe Multi-stage Grouting"})-[:has_risk]->(m:risk)-[:has_riskfactor]->(l:riskfactor) return count(l) As response	Correct
50	How many risks are there in the project?	match (n:risk) return count(n) as response	Correct

APPENDIX E: Answers Returned by QAR-System for Questions Q1-Q10

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

Ask Question

Generated Cypher Query

```
match(n:Tunnel)-[:has_work_type]->(m:wbs) return m.name As response
```

1. Geotechnical Investigation
2. Steel Pipe Multi-stage Grouting
3. Instrumentation
4. Auxiliary Facilities
5. Shaft part Work
6. Gang Gates and Retaining Walls
7. Common area
8. Drain
9. Water Proofing
10. Lining
11. Rock bolting
12. Formwork
13. Shotcrete Work
14. Support Treatment
15. Excavation

Figure E-1: Answer to question Q1 using QAR-System

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

List the risk categories in tunnel project.

Ask Question

Generated Cypher Query

```
match (n:risk) return distinct n.riskcategory as response
```

1. Construction

2. Management

3. Safety

4. Technical

5. Environment

6. Design

Figure E-2: Answer to question Q2 using QAR-System

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

List all the risks under construction risk category?

Ask Question

Generated Cypher Query

```
match (n:risk) where n.riskcategory = "Construction" return n.name as response
```

1. Increased vertical/horizontal strain
2. Collapsing(tunnel collapse subsidence)
3. Collapse of the posterior membrane
4. Economic loss due to under-excavation and over-excavation
5. Cost increase due to drilling alignment error
6. Economic loss due to deterioration of blasting efficiency
7. Waste and environmental issues
8. Increase in economic loss and cost (division, waste disposal)
9. Additional yard storage and increased yard storage costs
10. Arch formation difficulty
11. Cost increase due to excessive use of shotcrete material
12. Insufficient secondary support
13. Defective shotcrete attachment

Figure E-3: Answer to question Q3 using QAR-System

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

Ask Question

Generated Cypher Query

```
match (n:risk{riskcategory:"Design"}) return n.name as response
```

1. Project delay due to design change
2. Sunk cost
3. Suspension of construction due to the need for reinforced steel support
4. Waiting cost due to type change

Figure E-4: Answer to question Q4 using QAR-System

A Question-Answering System for Identifying Risks in Construction Projects

Ask Question

```
match (n:risk{riskcategory:"Technical"}) return n.name as response
```

- Figure E-5: Answer to question Q5 using QAR-System**

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

What are the risks associated with safety risk category?

Ask Question

Generated Cypher Query

```
match (n:risk[riskcategory:"Safety"]) return n.name + " - Risk Category: " + n.riskcategory as response
```

1. Worker electric shock - Risk Category: Safety
2. Worker trapping, conflict - Risk Category: Safety
3. Occupational accidents related to workers' diseases - Risk Category: Safety
4. Fire - Risk Category: Safety
5. Explosion - Risk Category: Safety
6. Charging car fall - Risk Category: Safety
7. Car overturn accident - Risk Category: Safety
8. Safety Accident - Risk Category: Safety
9. Collapse - Risk Category: Safety
10. Occupational accidents of workers - Risk Category: Safety
11. Defective shotcrete - Risk Category: Safety
12. Insufficient shotcrete strength - Risk Category: Safety
13. Increased shotcrete disposal cost - Risk Category: Safety
14. Formwork Falling Accident - Risk Category: Safety
15. Worker safety accident - Risk Category: Safety
16. Strong support conduction occurs - Risk Category: Safety

Figure E-7: Answer to question Q7 using QAR-System

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

List the risks associated with Management risk category of the tunnel project.

Ask Question

Generated Cypher Query

```
match(n:risk) where n.riskcategory = "Management" return n.name + " - " + " Risk Category: " + n.riskcategory as response
```

1. Damage to surrounding structures (roads, buildings, underground pipes, utility poles, etc.) - Risk Category: Management
2. Damage to surrounding structures (roads, buildings, underground pipes, ground poles, etc.), sinkholes - Risk Category: Management
3. Possibility of civil complaints for damage to fixed-temperature facilities such as neighboring houses and barns, delay in construction - Risk Category: Management
4. Risk of falling after excavation and before support installation - Risk Category: Management
5. Delayed construction period - Risk Category: Management
6. Damaged waterproof sheet - Risk Category: Management
7. Damaged waterproof sheet - Risk Category: Management
8. Damaged packaging - Risk Category: Management
9. Poor air quality inside the tunnel - Risk Category: Management
10. Noise complaints and air delays - Risk Category: Management
11. Water pollution - Risk Category: Management
12. Process disruptions and air delays due to inefficient responses - Risk Category: Management
13. Material Fall - Risk Category: Management

Figure E-8: Answer to question Q8 using QASRisk

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

Generated Cypher Query

```
match (n:risk{riskcategory:"Environment"}) return n.name + " - " + " Risk Category: " + n.riskcategory as response
```

1. Environmental problems - Risk Category: Environment

Figure E-9: Answer to question Q9 using QASRisk

QAR-SYSTEM

A Question-Answering System for Identifying Risks in Construction Projects

Generated Cypher Query

```
match (n:risk{riskcategory:"Environment"}) return count(n) as response
```

1. 1

Figure E-10: Answer to question Q10 using QASRisk

APPENDIX F: Answers Returned by ChatGPT Plus for Questions Q1-Q10

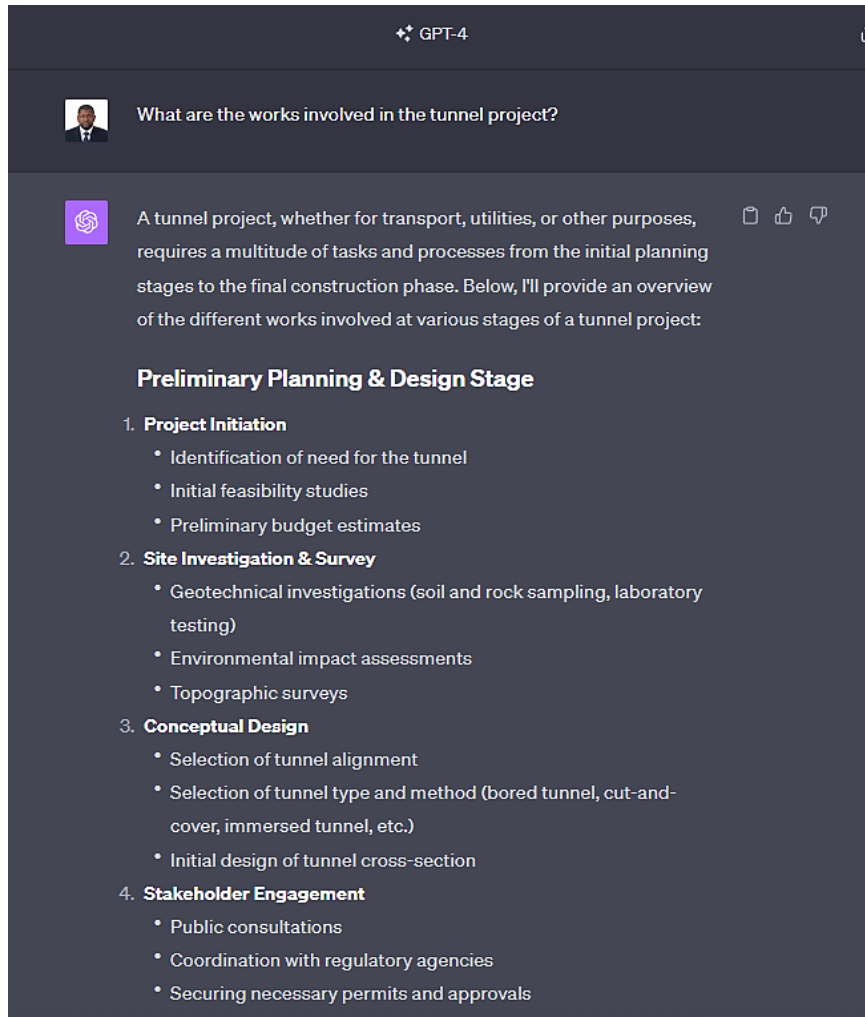


Figure F-1: Answer to question Q1 ChatGPT Plus

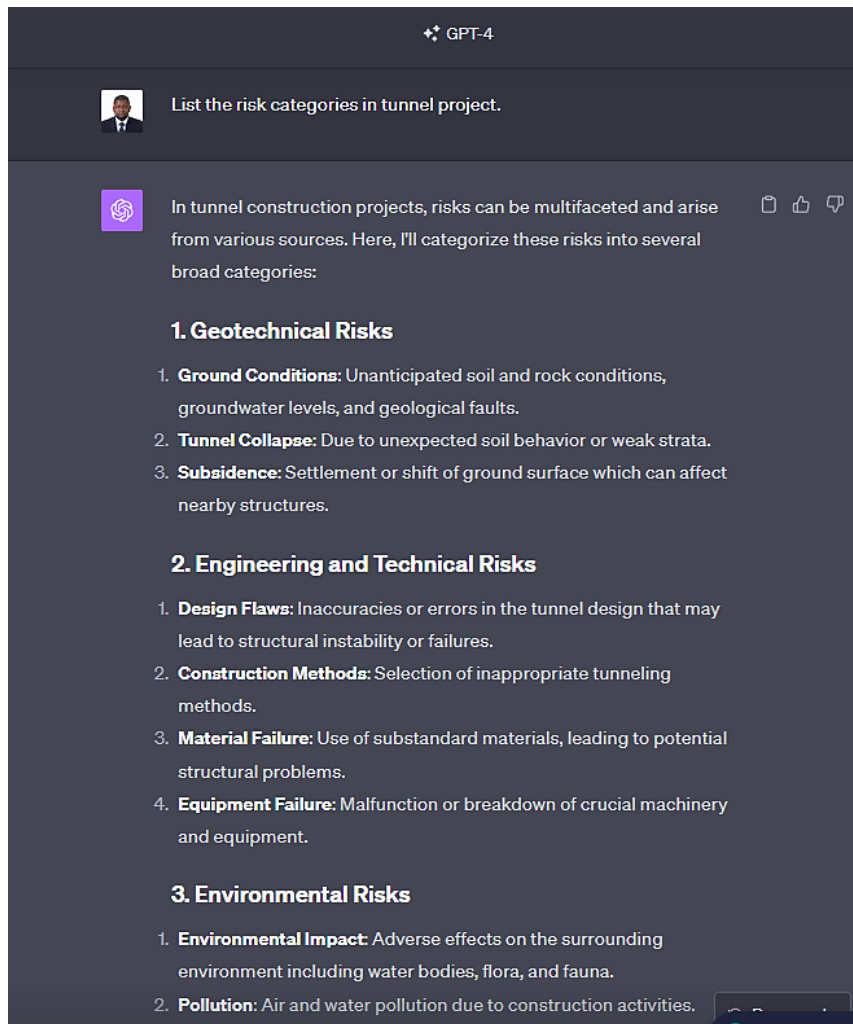


Figure F-2: Answer to question Q2 ChatGPT Plus

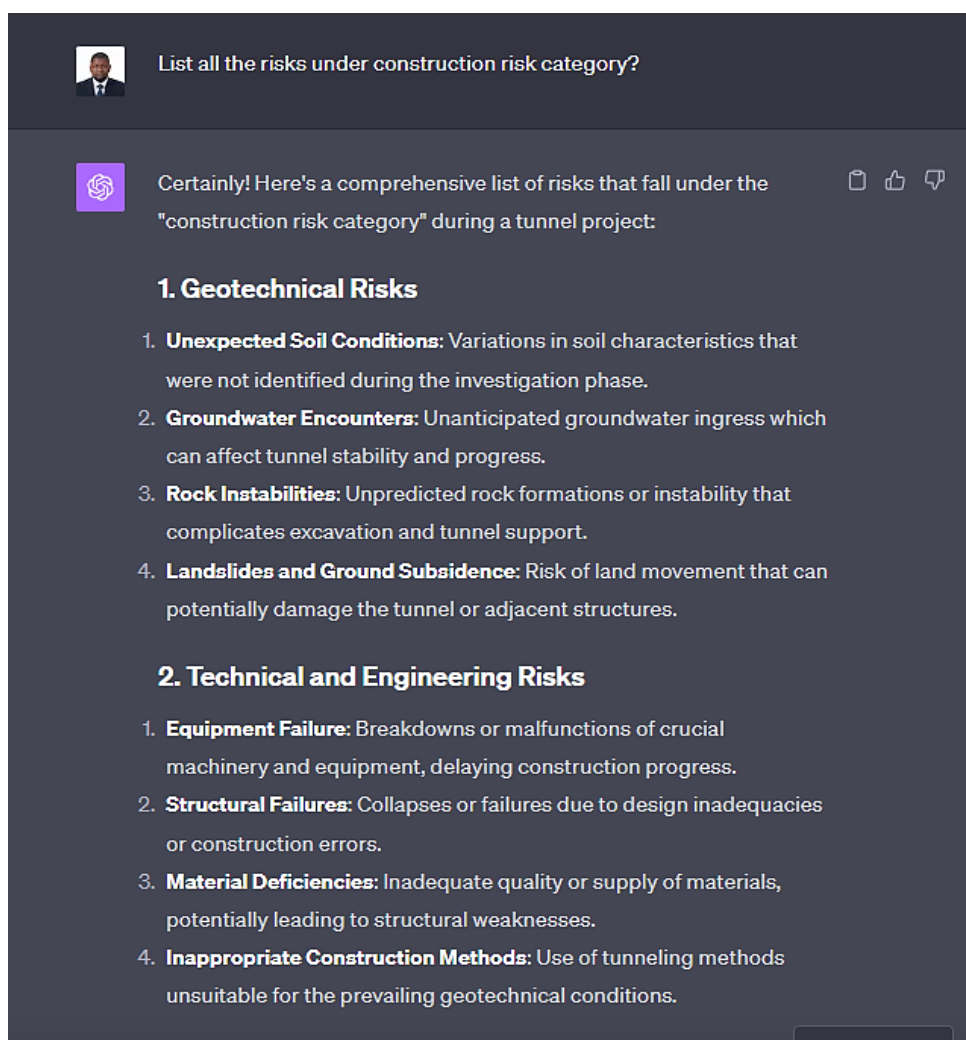


Figure F-3: Answer to question Q3 ChatGPT Plus

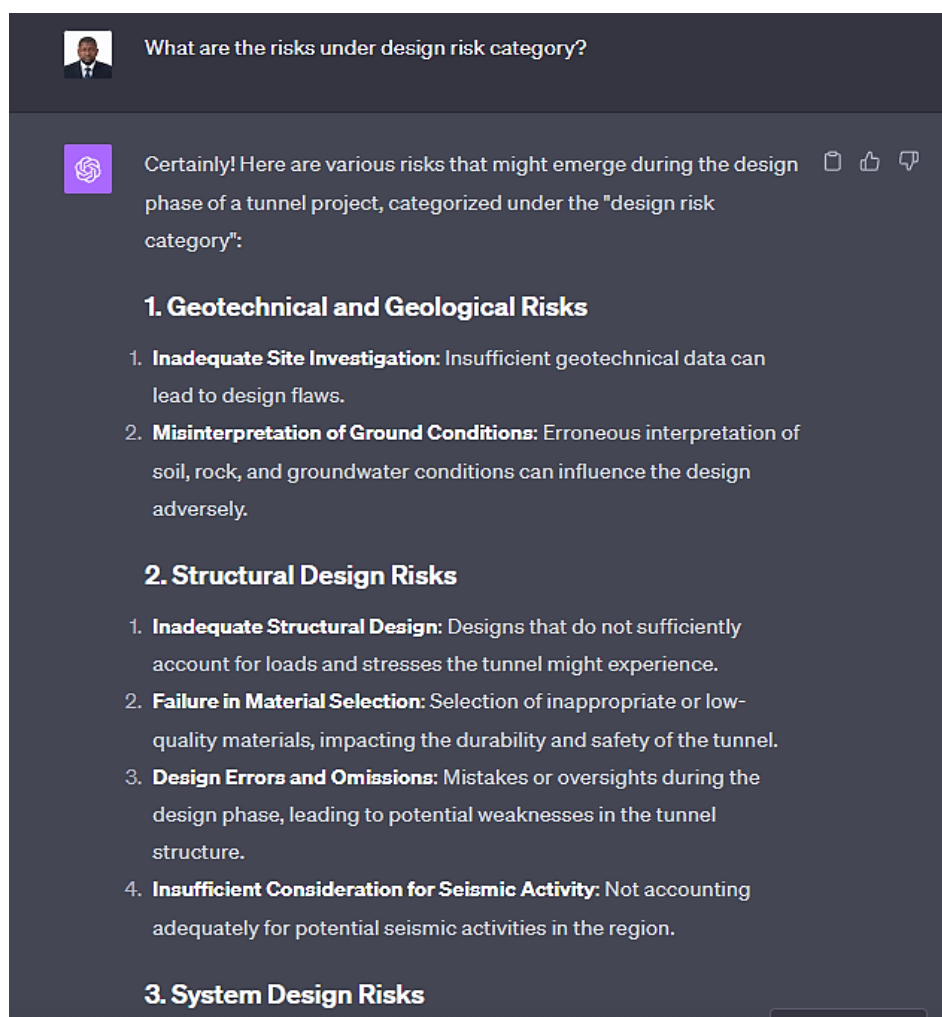


Figure F-4: Answer to question Q4 ChatGPT Plus

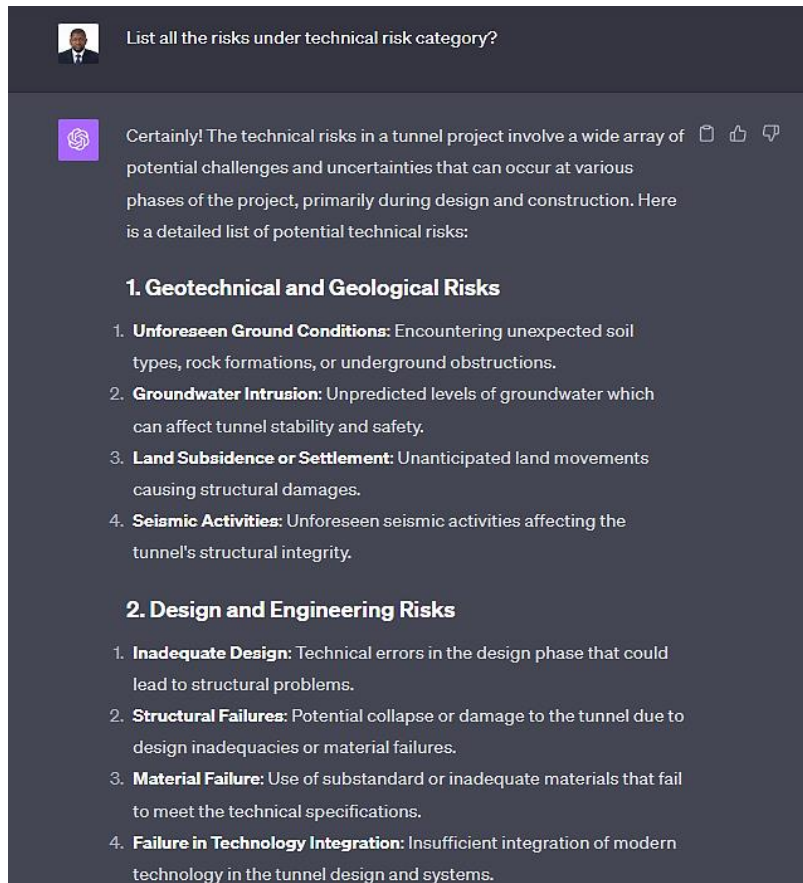


Figure F-5: Answer to question Q5 ChatGPT Plus

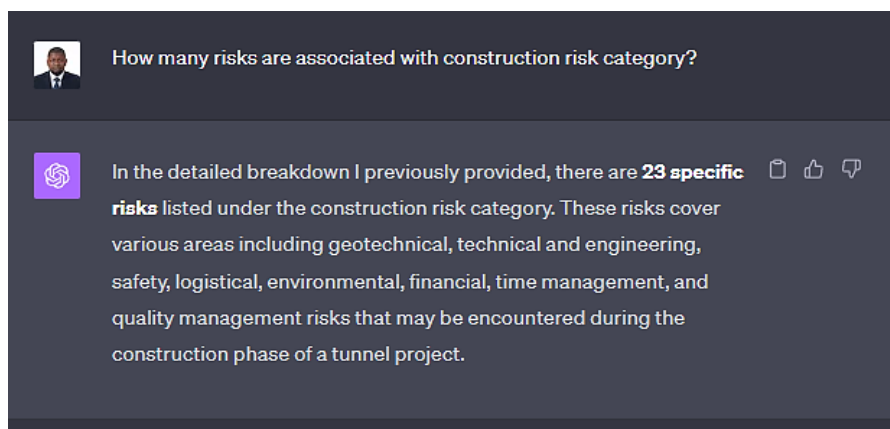


Figure F-6: Answer to question Q6 ChatGPT Plus

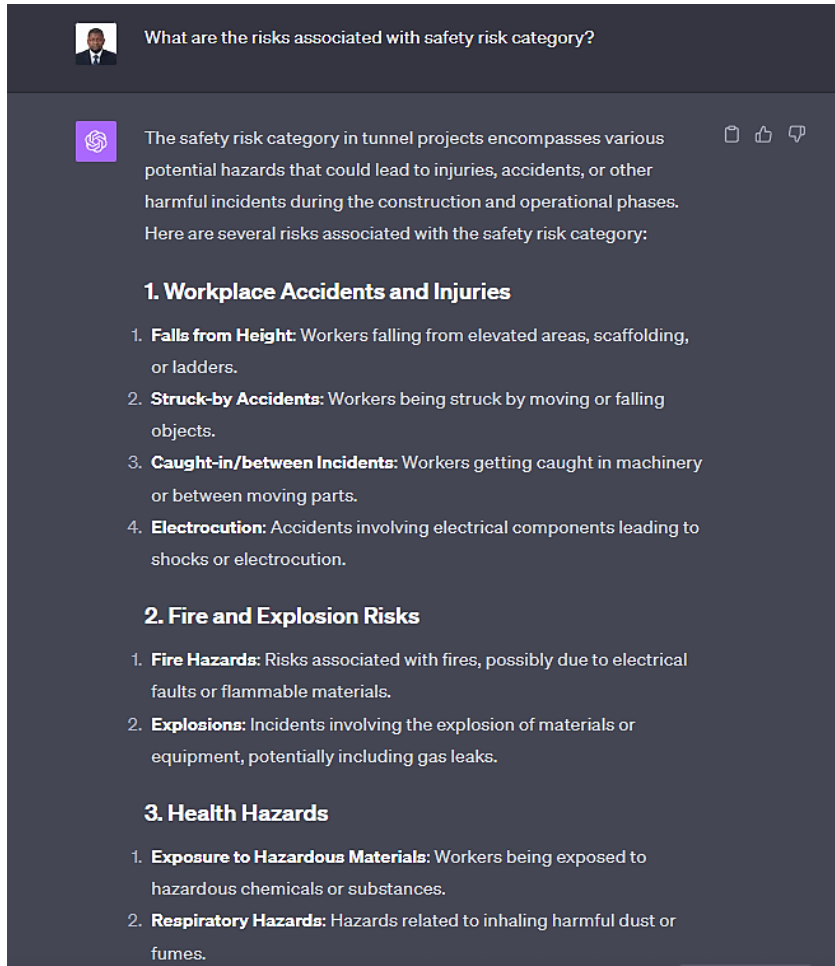


Figure F-7: Answer to question Q7 ChatGPT Plus



List the risks associated with Management risk category of the tunnel project.



Certainly! In tunnel projects, the "management risk category" often encompasses a variety of risks associated with the administrative, organizational, and strategic aspects of managing the project. Here is a detailed list of potential management risks:



1. Project Management Risks

- Scope Creep:** Uncontrolled changes or continuous growth in the project's scope, which can lead to delays and budget overruns.
- Poor Project Planning:** Insufficient or incorrect planning that can result in inefficiencies and failures.
- Inadequate Resource Allocation:** Misallocation of resources including personnel, machinery, and materials, leading to delays and increased costs.

2. Financial Management Risks

- Budget Overruns:** Exceeding the allocated budget due to unforeseen expenses or poor financial management.
- Inaccurate Cost Estimation:** Failure to accurately estimate the costs associated with the project, leading to financial strains.
- Fluctuating Currency Rates:** Risks associated with fluctuations in currency rates affecting the project's financial stability.

Figure F-8: Answer to question Q8 ChatGPT Plus



What are the risks associated with Environment risk category of the tunnel project?



Environmental risks in a tunnel project pertain to potential negative impacts on the surrounding environment and ecosystems during different stages of the project. These risks require proper assessment and management to prevent or mitigate adverse effects. Here is a list of some potential environmental risks associated with tunnel projects:



1. Land and Soil Impact

1. **Soil Contamination:** Release of hazardous substances that could contaminate the soil.
2. **Land Degradation:** Negative impact on the land quality due to excavation and construction activities.
3. **Ground Subsidence:** Changes in the ground level that might result from tunneling activities, possibly affecting nearby structures and ecosystems.

2. Water Resources and Quality

1. **Water Contamination:** Potential contamination of nearby water bodies due to spills or discharges.
2. **Alteration of Water Tables:** Changes in groundwater levels or flow patterns due to tunnel construction.
3. **Erosion and Sedimentation:** Increased erosion and sedimentation in water bodies due to construction activities.

3. Air Quality

1. **Air Pollution:** Release of pollutants into the air from construction activities, vehicles, and machinery.
2. **Dust Emissions:** Generation of dust during construction, affecting air quality and nearby communities.
3. **Greenhouse Gas Emissions:** Emission of greenhouse gases due to the use of fossil fuels during construction and operation.

Figure F-9: Answer to question Q9 ChatGPT Plus

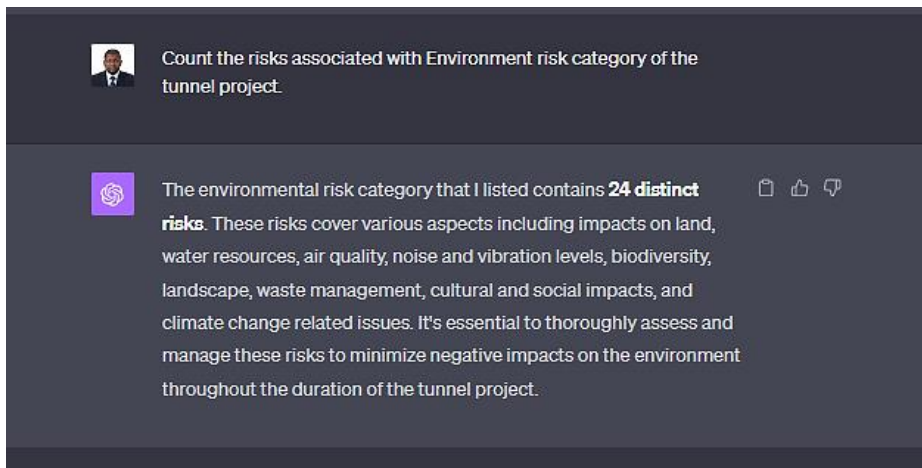


Figure F-10: Answer to question Q10 ChatGPT Plus

APPENDIX G: Answers Returned by Bing Chat for Questions Q1-Q10

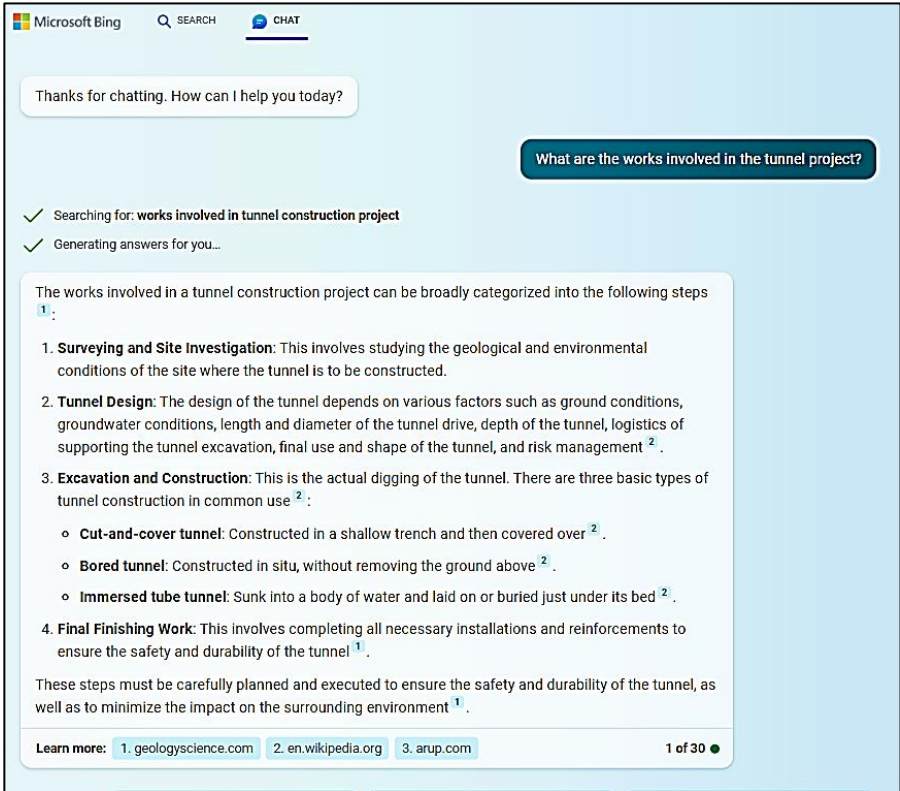


Figure G-1: Answer to question Q1 Bing Chat

Microsoft Bing SEARCH CHAT

well as to minimize the impact on the surrounding environment ¹.

Learn more: [1. geologyscience.com](#) [2. en.wikipedia.org](#) [3. arup.com](#) 1 of 30 ●

List the risk categories in tunnel project.

✓ Searching for: **risk categories in tunnel construction project**

✓ Generating answers for you...

The risk categories in a tunnel construction project can be broadly divided into the following categories ¹ ² ³:

1. **Technical Risks:** These are risks related to the technical aspects of the project. The occurrence of a technical risk usually carries cost and schedule consequences ¹.
2. **Schedule Risks:** These risks impact the timeline of the project, typically affecting cost escalation and project overhead ¹.
3. **Condition of Newly Built Tunnels:** Factors such as tunnel radius and net distance can influence the risk of the project ³.
4. **Existing Tunnel Structure Condition:** Conditions such as water seepage, structural cracking, and material deterioration in an existing tunnel structure can pose risks when constructing a new tunnel underneath ³.
5. **Hydrogeological Condition:** The water and geological conditions of the site can significantly impact the risk profile of the project ³.

These risks are often interrelated and must be carefully considered in the context of the specific set of project conditions and circumstances ¹.

Learn more: [1. pubs.aip.org](#) [2. researchgate.net](#) [3. hindawi.com](#) [4. doi.org](#) [5. doi.org](#) 2 of 30 ●

Figure G-2: Answer to question Q2 Bing Chat

Microsoft Bing SEARCH CHAT

List all the risks under construction risk category?

✓ Searching for: risks under construction category in tunnel project

✓ Generating answers for you...

The risks under the construction category in a tunnel project can include ¹ ²:

1. **Worker Availability:** The availability of skilled workers can pose a risk, especially during times of crisis such as the COVID-19 pandemic ¹.
2. **Site Accessibility:** The accessibility of the construction site can be a significant risk factor. This can be influenced by factors such as geographical location, weather conditions, and local infrastructure ¹.
3. **Shortage of Construction Materials:** A shortage of necessary construction materials can delay the project and increase costs ¹.
4. **Inadequate Epidemic Prevention Materials:** During a pandemic, inadequate supplies of epidemic prevention materials can pose a risk to the health and safety of the workers ¹.
5. **Social Environment:** The social environment, including factors such as public opinion and local regulations, can impact the progress of the project ¹.
6. **Political Epidemic Prevention Pressure:** Political pressures related to epidemic prevention can also pose a risk, especially during a pandemic ¹.
7. **Closed Construction and Poor Ventilation:** Tunnel construction work is under greater risk of epidemic spread and infection due to its features of closed construction and poor ventilation ¹.

These risks need to be carefully managed to ensure the successful completion of the tunnel construction project ¹ ².

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Figure G-3: Answer to question Q3 Bing Chat

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epidemic spread and infection due to its features of closed construction and poor ventilation...

These risks need to be carefully managed to ensure the successful completion of the tunnel construction project ¹ ².

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What are the risks under design risk category?

✓ Searching for: risks under design category in tunnel project

✓ Generating answers for you...

The risks under the design category in a tunnel project can include ¹ ²:

- 1. Geotechnical Risks:** These are risks related to the geological conditions of the site. Incorrect assumptions about the geotechnical conditions can lead to design flaws ¹.
- 2. Hydrological Risks:** These are risks related to the water conditions of the site. Incorrect assumptions about the hydrological conditions can also lead to design flaws ¹.
- 3. Design Errors:** Errors in the design process can lead to issues during construction and operation of the tunnel ².
- 4. Inadequate Design:** The design may not adequately account for all possible scenarios, leading to potential issues during construction and operation ².
- 5. Changes in Design Standards or Regulations:** Changes in design standards or regulations after the design phase has been completed can lead to non-compliance issues ².

These risks need to be carefully managed through thorough site investigation, careful design, and regular review and update of the design ¹ ².

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Figure G-4: Answer to question Q4 Bing Chat

List all the risks under technical risk category?

✓ Searching for: risks under technical category in tunnel project

✓ Generating answers for you...

The risks under the technical category in a tunnel project can include ¹ ² :

1. **Geotechnical Risks:** These are risks related to the geological conditions of the site. Incorrect assumptions about the geotechnical conditions can lead to technical issues ¹.
2. **Hydrological Risks:** These are risks related to the water conditions of the site. Incorrect assumptions about the hydrological conditions can also lead to technical issues ¹.
3. **Technical Errors:** Errors in the technical aspects of the project can lead to issues during construction and operation of the tunnel ².
4. **Inadequate Technical Planning:** The technical planning may not adequately account for all possible scenarios, leading to potential issues during construction and operation ².
5. **Changes in Technical Standards or Regulations:** Changes in technical standards or regulations after the technical planning phase has been completed can lead to non-compliance issues ².

These risks need to be carefully managed through thorough site investigation, careful technical planning, and regular review and update of the technical aspects ¹ ².

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Figure G-5: Answer to question Q5 Bing Chat

How many risks are associated with construction risk category?

✓ Generating answers for you...

There are seven risks associated with the construction risk category in a tunnel project:

1. Worker Availability
2. Site Accessibility
3. Shortage of Construction Materials
4. Inadequate Epidemic Prevention Materials
5. Social Environment
6. Political Epidemic Prevention Pressure
7. Closed Construction and Poor Ventilation

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Figure G-6: Answer to question Q6 Bing Chat

What are the risks associated with safety risk category?

✓ Searching for: risks under safety category in tunnel project

✓ Generating answers for you...

The risks associated with the safety category in a tunnel project can include ¹ ² ³ :

- Tunnel Stability:** Risks related to rock or earth falls and rock bursts ³ .
- Changing Ground Conditions:** Risks related to strata and stress fluctuations ³ .
- Limited Space and Access:** Risks related to confined spaces ³ .
- Health and Safety:** The construction industry is well known for being one of the UK's highest-risk sectors. Tunnels are unique; their design and construction are different from other structures. They have the same risks as surface construction does but within a confined space underground ¹ .
- Epidemic Spread and Infection:** With the features of closed construction and poor ventilation, tunnel construction work is under greater risk of epidemic spread and infection, compared to other construction projects ² .

These risks need to be carefully managed through thorough site investigation, careful technical planning, and regular review and update of the technical aspects ¹ ² .

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[2. hindawi.com](#)
[3. bing.com](#)
[4. safetyrisk.net](#)
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Figure G-7: Answer to question Q7 Bing Chat

List the risks associated with Management risk category of the tunnel project?

✓ Searching for: risks under management category in tunnel project

✓ Generating answers for you...

The risks associated with the management category in a tunnel project can include ¹ ² :

- Project Management Risks:** These are risks related to the management of the project, including planning, coordination, and control of the project ¹ .
- Supply Chain Risks:** These are risks related to the supply chain of the project, including the availability and delivery of materials and equipment ¹ .
- Contractual Risks:** These are risks related to the contracts of the project, including issues with contractors and subcontractors ² .
- Financial Risks:** These are risks related to the financial aspects of the project, including cost overruns and funding issues ² .
- Regulatory Risks:** These are risks related to regulatory compliance, including changes in laws and regulations ² .

These risks need to be carefully managed through effective project management practices ¹ ² .

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Figure G-8: Answer to question Q8 Bing Chat

What are the risks associated with Environment risk category of the tunnel project?

✓ Searching for: risks under environment category in tunnel project
 ✓ Generating answers for you...

The risks associated with the environment category in a tunnel project can include ¹ ²:

- 1. Environmental Impact:** The construction of tunnels can have significant impacts on the surrounding environment, including noise, dust, vibration, and disruption to local ecosystems ¹.
- 2. Weather Conditions:** Adverse weather conditions can pose risks to the construction process, including heavy rain, snow, and extreme temperatures ².
- 3. Natural Disasters:** Natural disasters such as earthquakes, landslides, and floods can pose significant risks to tunnel construction ².
- 4. Ground Conditions:** The ground conditions at the construction site can pose risks, including the presence of hazardous materials in the soil or groundwater ².
- 5. Air Quality:** The quality of air inside the tunnel during construction can pose health risks to workers ².

These risks need to be carefully managed through thorough site investigation, careful planning, and implementation of appropriate safety measures ¹ ².

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Figure G-9: Answer to question Q9 Bing Chat

Count the risks associated with Environment risk category of the tunnel project.

✓ Generating answers for you...

There are five risks associated with the environment risk category in a tunnel project:

1. Environmental Impact
2. Weather Conditions
3. Natural Disasters
4. Ground Conditions
5. Air Quality

10 of 30 ●

Figure G-10: Answer to question Q10 Bing Chat

APPENDIX H: Answers Returned by HuggingChat for Questions Q1-Q10

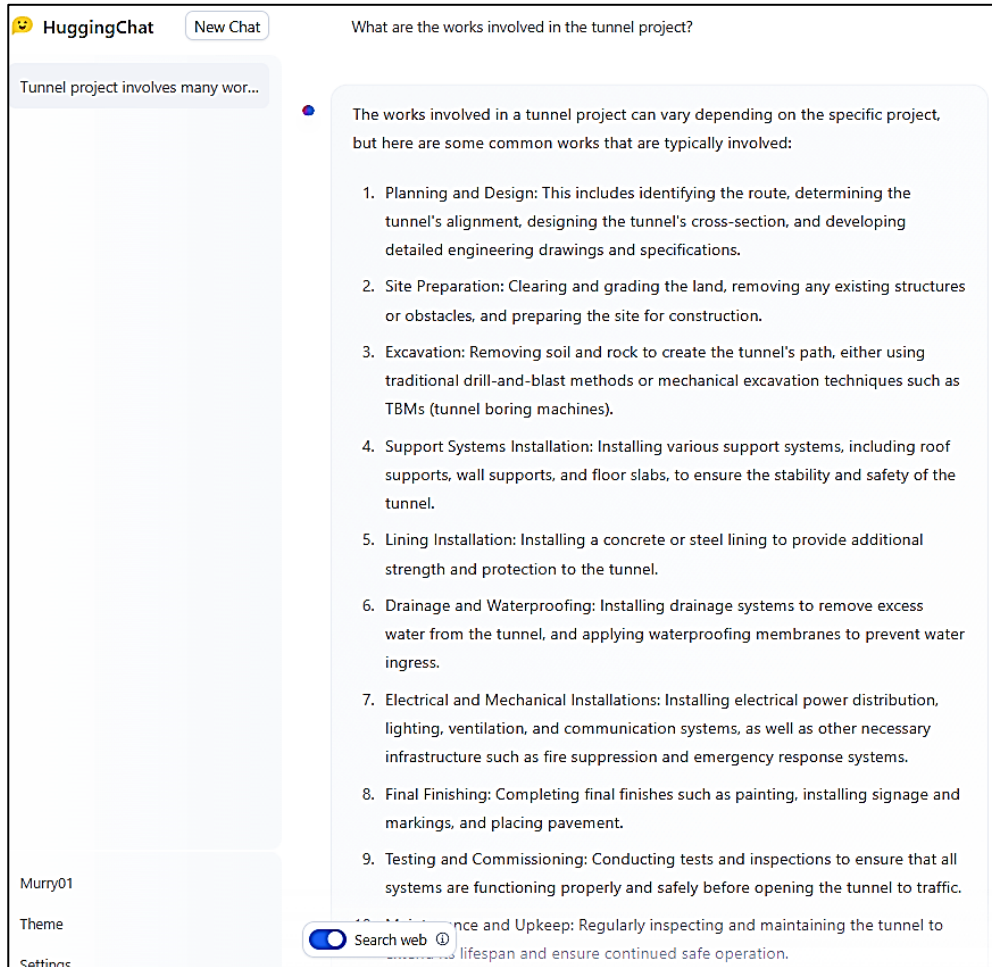


Figure H-1: Answer to question Q1 HuggingChat

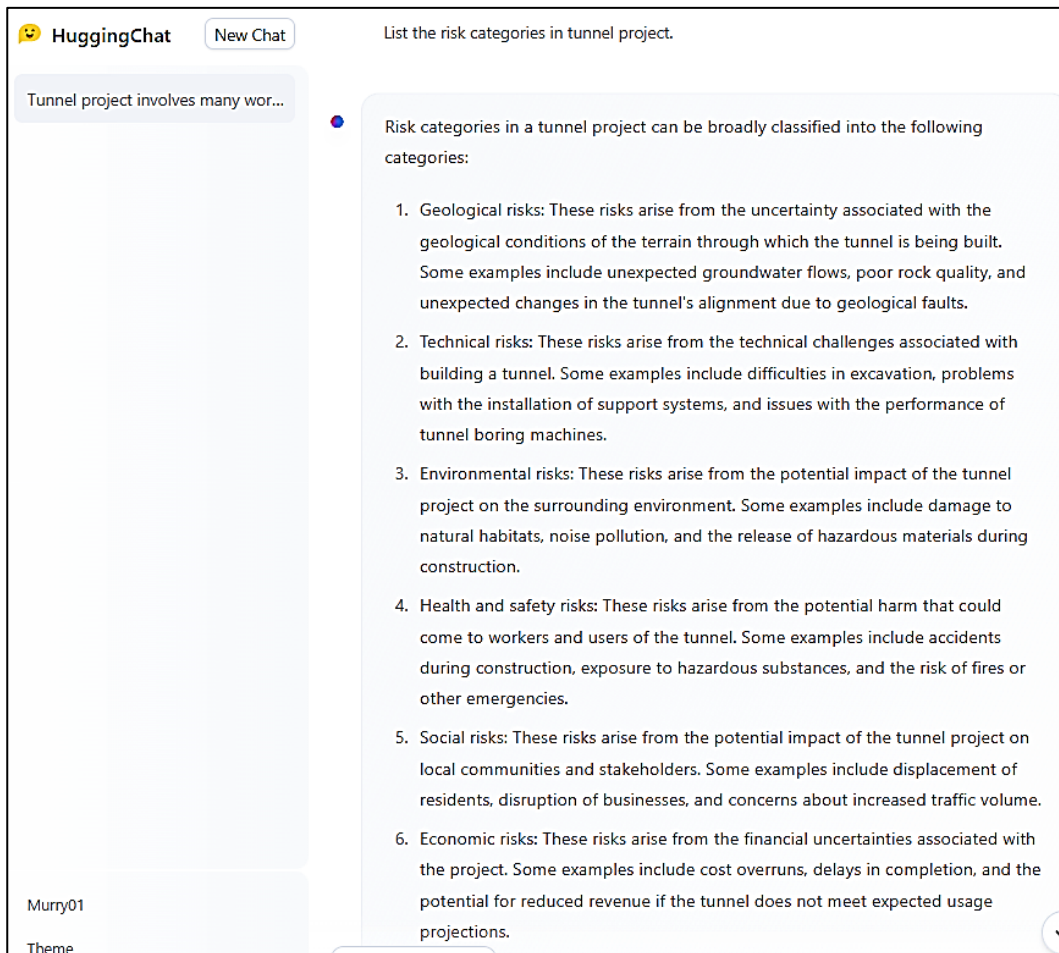


Figure H-2: Answer to question Q2 HuggingChat

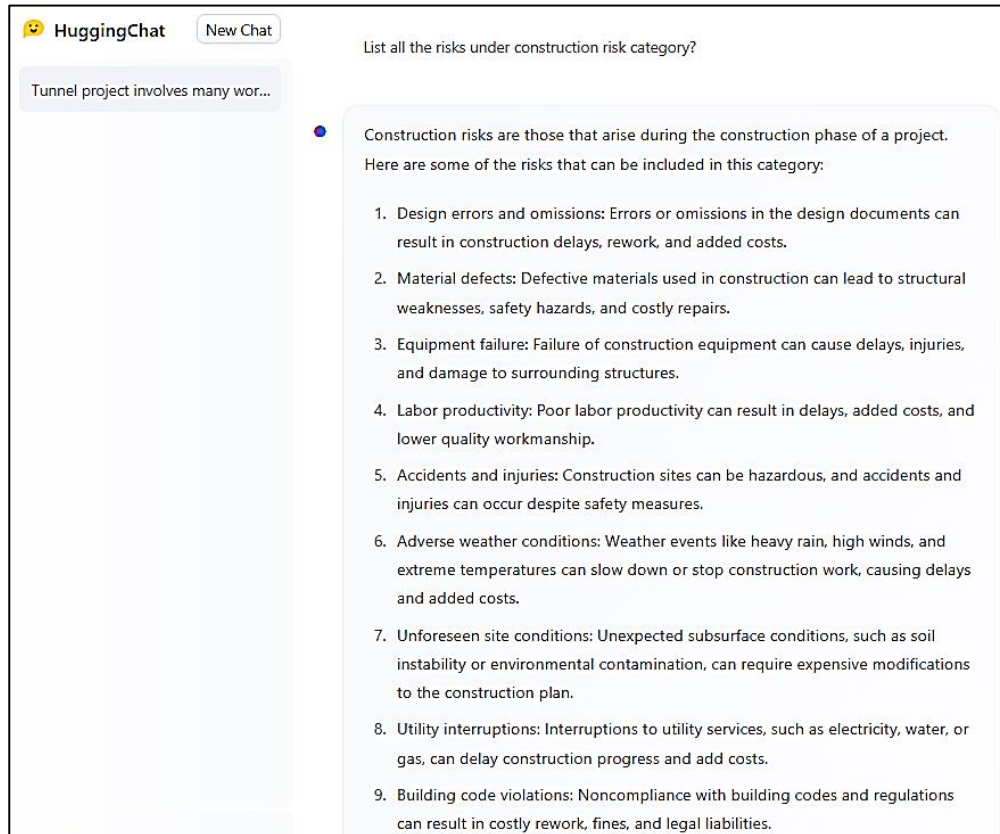


Figure H-3: Answer to question Q3 HuggingChat

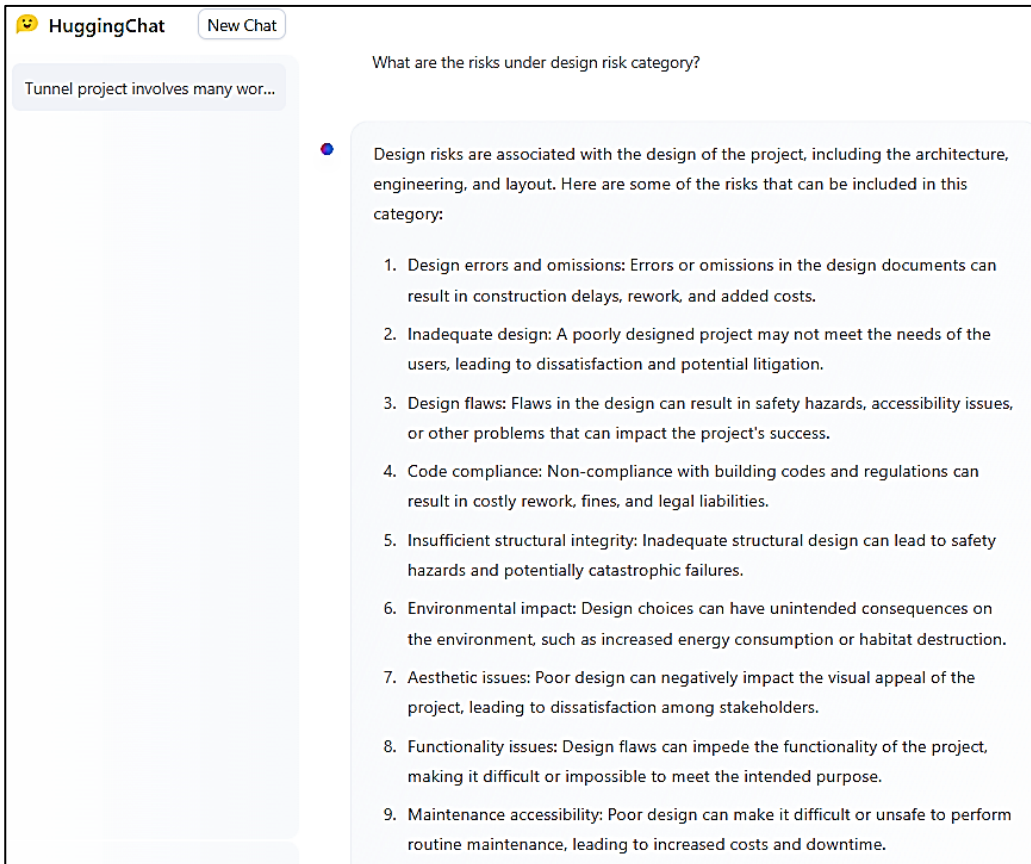


Figure H-4: Answer to question Q4 HuggingChat

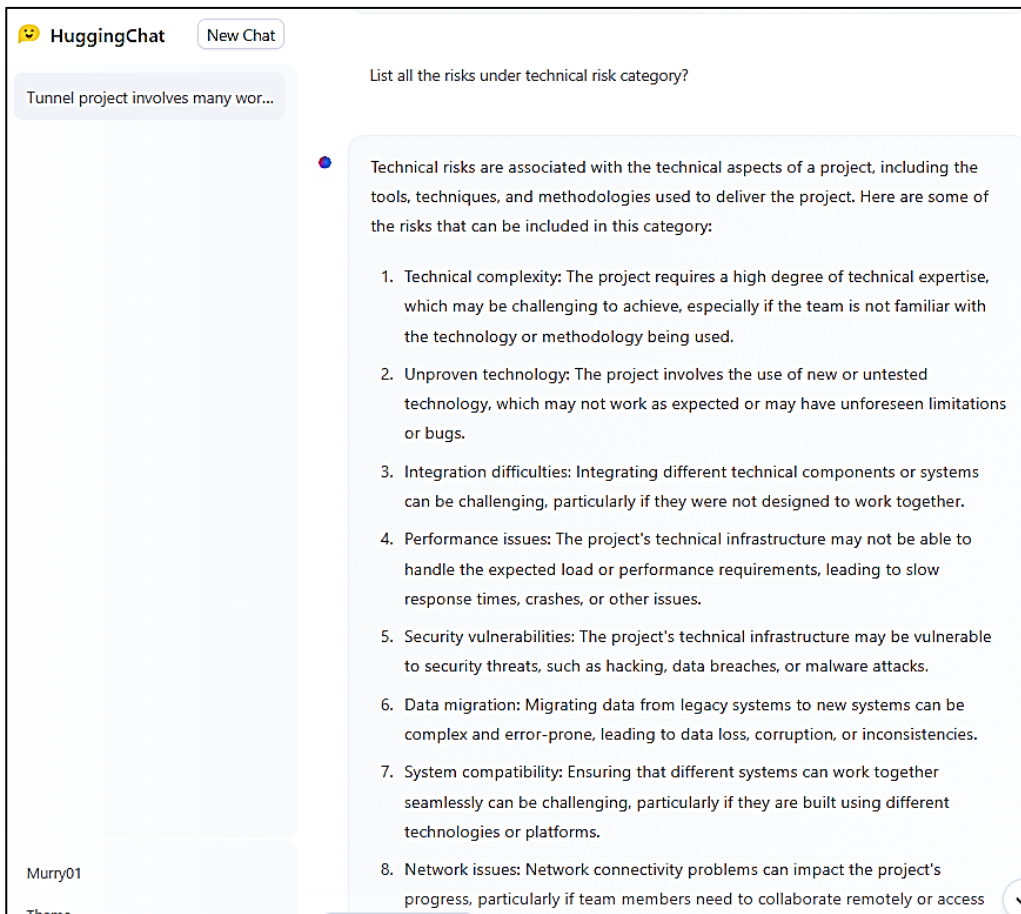


Figure H-5: Answer to question Q5 HuggingChat

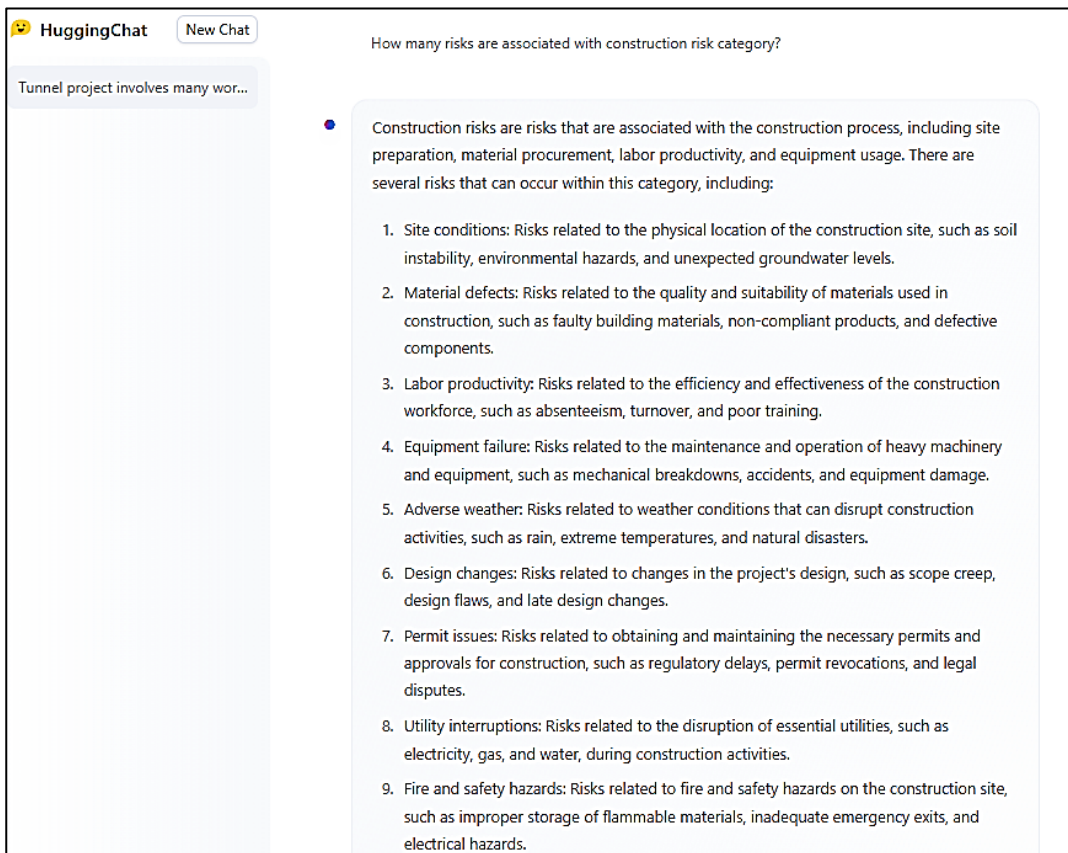


Figure H-6: Answer to question Q6 HuggingChat

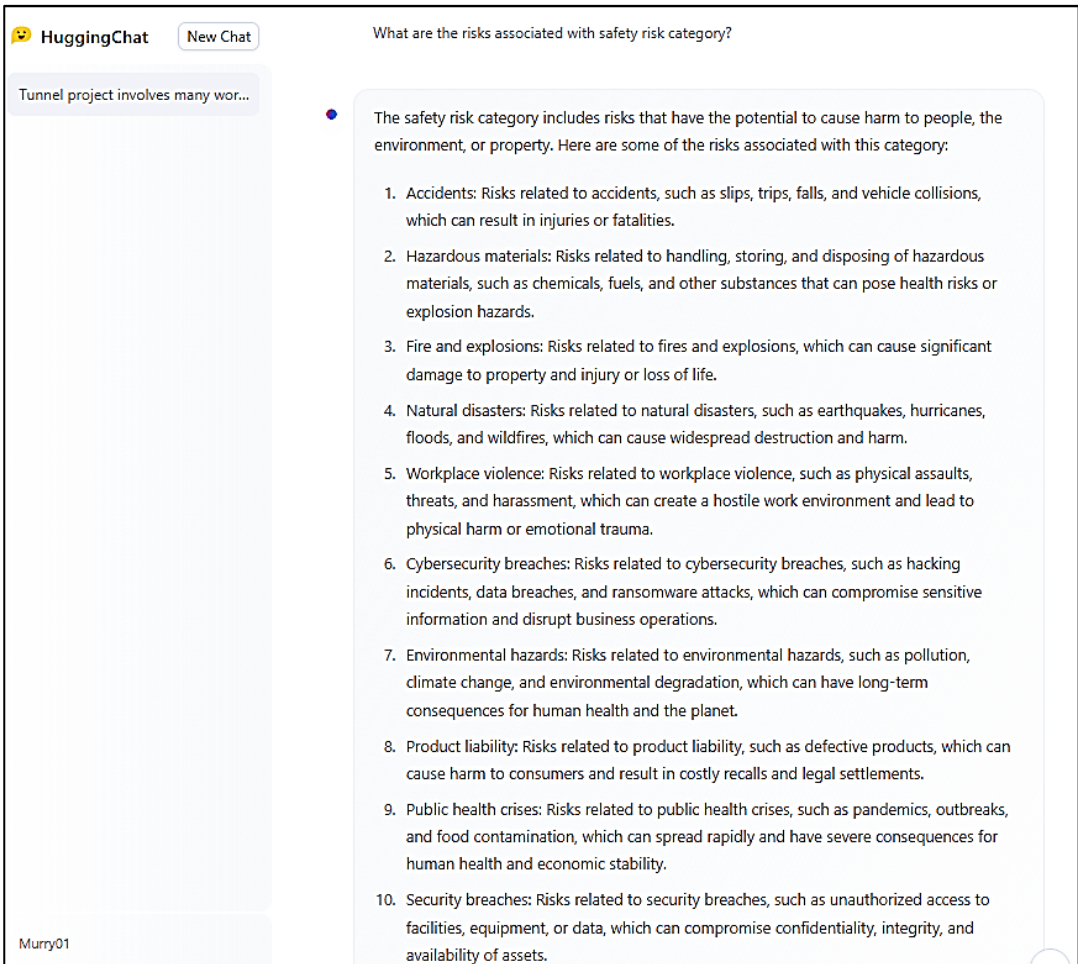


Figure H-7: Answer to question Q7 HuggingChat

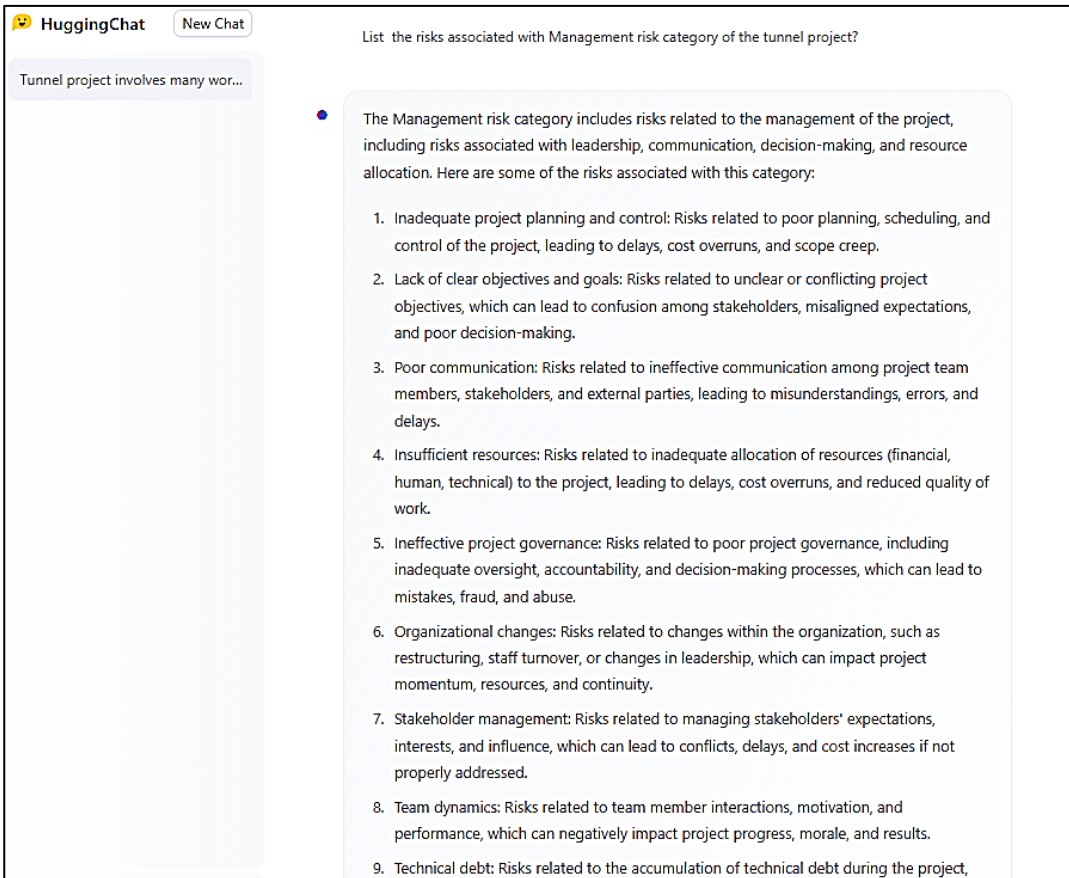


Figure H-8: Answer to question Q8 HuggingChat

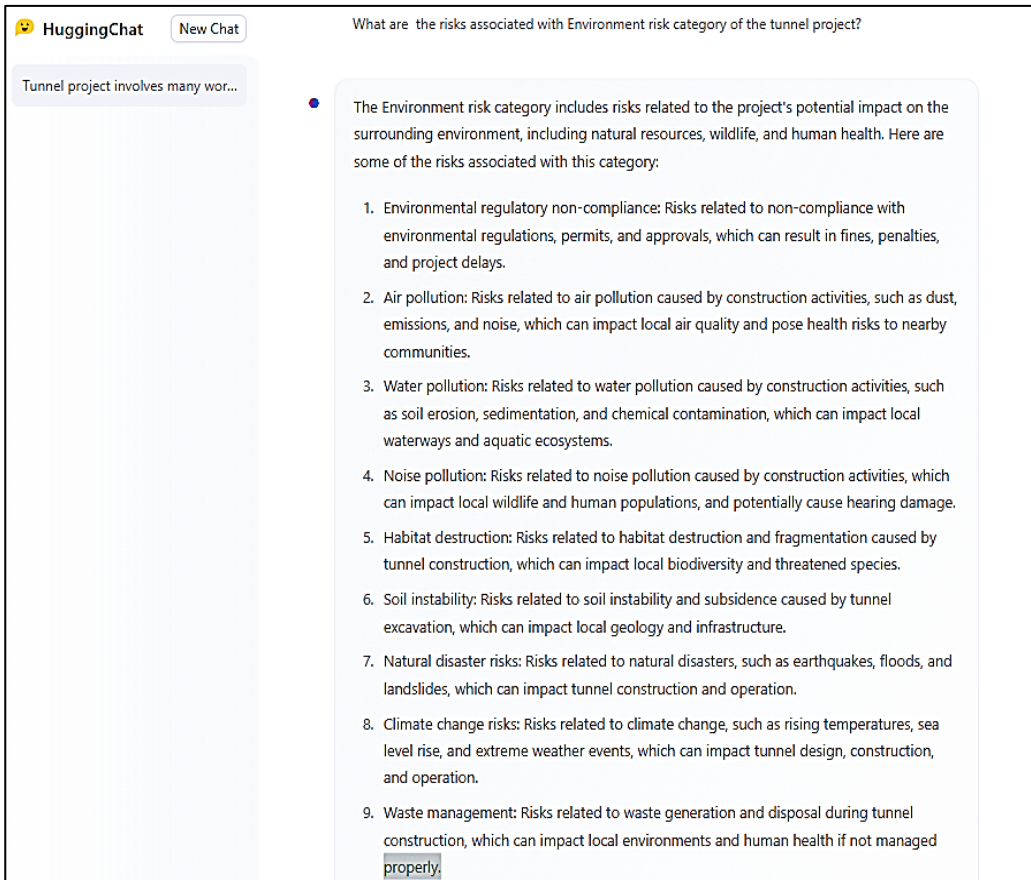


Figure H-9: Answer to question Q9 HuggingChat

Count the risks associated with Environment risk category of the tunnel project.

Sure! There are 15 risks associated with the Environment risk category of the tunnel project:

1. Environmental regulatory non-compliance
2. Air pollution
3. Water pollution
4. Noise pollution
5. Habitat destruction
6. Soil instability
7. Natural disaster risks
8. Climate change risks
9. Waste management
10. Community health risks
11. Emergency response plan
12. Security risks
13. Land acquisition risks
14. Cultural heritage risks
15. Public perception risks

I hope this helps! Let me know if you have any further questions.

Figure H-10: Answer to question Q10 HuggingChat

APPENDIX I: Comparison of answers to Q2 & Q6 by the four models

S/N	Question	ChatGPT4	Bing Chat	HuggingChat	QAR-System
1	List the risk categories in tunnel project.	Geotechnical, Engineering & Technical, and Environment Risks	Technical risks, Schedule risks, Condition of Newly Built Tunnels, Existing Tunnel Structure Condition, and Hydrogeological Condition	Geological, Technical, Environmental, Health & Safety, Social and Economic risks	Construction, Management, Safety, Technical, Environmental and Design Risks
2	How many risks are associated with construction risk category?	23	7	9	63